

Review

Radiation track, DNA damage and response—a review

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Received 11 May 2015, revised 1 February 2016

Accepted for publication 25 May 2016

Published 21 September 2016



Corresponding Editor Bob Austin

Abstract

The purpose of this paper has been to review the current status and progress of the field of radiation biophysics, and draw attention to the fact that physics, in general, and radiation physics in particular, with the aid of mathematical modeling, can help elucidate biological mechanisms and cancer therapies. We hypothesize that concepts of condensed-matter physics along with the new genomic knowledge and technologies and mechanistic mathematical modeling in conjunction with advances in experimental DNA (Deoxyrinonucleic acid molecule) repair and cell signaling have now provided us with unprecedented opportunities in radiation biophysics to address problems in targeted cancer therapy, and genetic risk estimation in humans. Obviously, one is not dealing with ‘low-hanging fruit’, but it will be a major scientific achievement if it becomes possible to state, in another decade or so, that we can link mechanistically the stages between the initial radiation-induced DNA damage; in particular, at doses of radiation less than 2 Gy and with structural changes in genomic DNA as a precursor to cell inactivation and/or mutations leading to genetic diseases.

The paper presents recent development in the physics of radiation track structure contained in the computer code system KURBUC, in particular for low-energy electrons in the condensed phase of water for which we provide a comprehensive discussion of the dielectric response function approach. The state-of-the-art in the simulation of proton and carbon ion tracks in the Bragg peak region is also presented. The paper presents a critical discussion of the models used for elastic scattering, and the validity of the trajectory approach in low-electron transport. Brief discussions of mechanistic and quantitative aspects of microdosimetry, DNA damage and DNA repair are also included as developed by the authors’ work.

Keywords: track structure, radiation biophysics, DNA damage, DNA repair, inelastic scattering, elastic scattering, microdosimetry

(Some figures may appear in colour only in the online journal)

1. Introduction—state-of-the-art in biophysical studies of radiation track

In radiation protection, the ultimate aim is to estimate the risks associated with radiation exposure (cancer and genetics) and reduce it to be as low as possible⁷. In radiation therapy, we aim to kill the tumour cells, but avoid causing secondary cancer—it is a tweedledum tweedledee situation. On the one hand, ionizing radiation⁸ is used in cancer therapy as a healing agent, while at the same time we need to protect ourselves from its harmful effects. Ionizing radiation is used in many medical and industrial applications, while in the environment it is perceived as one of the most toxic agents. For over a century the radiation research community has striven to increase the knowledge of physical properties and biological effects which follow exposure of human cells, tissue, organs, and the whole body, to be able to enhance its use in medicine or estimate the risks involved. Ionizing radiation appears in nature, emitted from decaying radionuclides or it is man made when atomic particles (electrons, and heavy atomic particles) are accelerated. Ionizing radiation also exists in space, namely cosmic rays, which originate from outside the solar system in the form of galactic cosmic rays (GCRs) and solar particle events (SPE) emitted from the Sun. Therefore, the protection of astronauts and electronics in satellites and spacecraft from UV (ultra-violet light) and ionizing radiation forms a major activity in space travel and exploration. In a previous era, medical radiation physics (MRP) used to be an area of activity exclusive to the physics domain and physicists, but in the 21st Century, we need to understand the properties of radiation track in the context of radiation biophysics, radiation biology and medicine.

Some questions in the field of radiation research embracing medical radiation physics and biology, and radiation protection include: What are the important events defining radiation tracks? Is ‘dose’ the appropriate unit to measure radiation quantity? How well can we estimate the genetic and cancer risks from exposure to low doses of radiation, including those received in radiation therapy and diagnostics in humans? Are there qualitative and quantitative differences between the DNA damage produced in human cells by x-rays versus energetic electrons, or from those induced by heavy particles such as iron ions? Is there a correlation between DNA damage and cell death? Are cancer and genetic risks below the radiation doses detectable from the atomic bomb studies in Japan important?

⁷ In radiation protection, low doses are considered to be about 0.1 Sv (NRC, BEIR VII Report, 2006).

⁸ Photons and neutrons *per se* are considered non-ionizing radiation, but once the particle imparts its energy to the medium it generates secondary electrons. It is these electrons that are the source of damage and of interest in radiation biology. Therefore, the important matter is to be able to describe quantitatively the spectrum of electrons produced by photon or neutron interaction with the target.

In this paper, we describe the properties of radiation tracks of electrons and heavy particles and the recent development in the field of radiation biophysics. In chapter 2, we present the state-of-the art in the interaction of electrons with condensed matter and, in particular, low-energy electrons as these constitute most of the energy deposited by the track. The discussion is limited to water as it is the most abundant biological matter in the human body and the cells. There are two principle modes of interaction of radiation with matter: (a) inelastic interactions, which determine the size of the energy loss, and (b) elastic interactions, which govern the scattering or the shape of radiation track. Much knowledge is available for the former, but not a lot for the elastic interactions and, in particular, for the condensed phase of water and at low-energy electrons (see chapter 3). Chapter 4 is devoted to inelastic interactions of protons and carbon ion tracks. Such particles are used in modern radiation therapy. Energetic particles move in straight lines without much deviation from their path. Scattering of particles at low energies and, in particular, electrons are of immense interest as these are the main sources of DNA damage in human cells. A short discussion of DNA damage and repair follows in chapter 5.

In radiation therapy, we are primarily interested in ideally killing the tumour cells, without affecting the healthy tissue surrounding the tumor. In radiation protection, we are primarily interested in knowing what is the risk involved if we are exposed to environmental, medical or accidental radiation. By ‘risk’, we mean the genetics and the cancer risks after exposures to low doses of ionizing radiation. Other issues related to radiation risk, which have not been discussed in this paper, are to do with the bystander effect, genomic instability, inflammation and immune responses. For the latter, evidence supports that the DNA damage response and repair and immune response (ImmR) signaling networks work together favoring the correct function of multi-cellular organisms, supported by recent bioinformatics data that underlines this close relationship [229]. The ImmR is a new field of research and to our knowledge, mechanistic and mathematical models have not yet been proposed for this effect.

Figure 1 is a schematic presentation of the stages of how radiation damage in human cells is broadly processed in five stages. For a complete mechanistic description of the damage produced in the human cell nucleus, we need to provide a mathematical description linking one stage to the next. Some of the questions that arise at this point include: To what extent has the field of radiation research, medical radiation physics, and biological sciences succeeded in characterizing the principle events? To what extent can the principal stages be linked together? Do we have sufficient experimental data for a description of the processes at each stage from exposure of the cell to mutation and cell death? Undoubtedly, this was the task

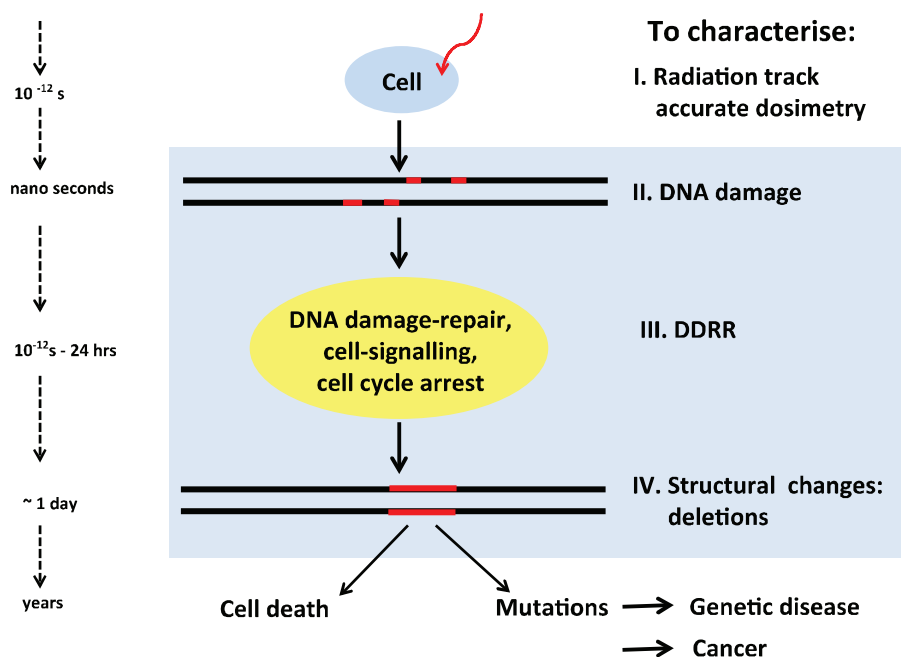


Figure 1. A schematic presentation of the principal stages after human cells are exposed to ionizing radiation. The approximate time course of each stage is indicated on the LHS. The major task is to characterize each stage to be able to make a mechanistic description of the radiation effect in the human cell or tissue. Such a model facilitates selecting optimal radiation therapy modalities. Moreover, it paves the way for the estimation of genetic and cancer risk factors in humans.

Douglas Lea initiated in his pioneering work and eloquently described in his book ‘Actions of radiations on living cells’ published in 1946 [1].

The ultimate challenge we seek is to be able to bridge the gap from irradiation of the cell to how mutations or cell death will follow as a consequence of DNA damage response and repair.

In recent decades, considerable knowledge has been accumulated on the multi-step nature of cancers, the ‘hallmark’ capabilities acquired during the process of carcinogenesis (that enable tumor growth and metastatic dissemination [2]) and the molecular aspects of some, albeit not all of the underlying mechanisms. This knowledge has provided the basis for, among others, cancer therapies that target one or more of the hallmark capabilities with the aim of impairing tumor growth and progression. Furthermore, the discovery of key proteins involved in the DNA damage response (DDR) pathways helped the formulation of cancer therapy strategies to target those proteins that increase the sensitivity of cancer cells to ionizing radiation or chemotherapy agents [3–5]. Although some therapies against specific signaling proteins or pathways have been successful at treating certain cancers, considered overall, success with these efforts thus far has been limited, i.e. the clinical responses have largely been transitory, followed by almost inevitable relapses. Several possible reasons for these observations have been suggested, one of them being that each of the hallmark capabilities is regulated by partially redundant signaling pathways, allowing some cancer cells to survive with residual function and adapt to the selective pressure of the therapy and re-establish the functional capability of renewed tumor growth [2].

Basic studies have taught us that (a) eukaryotic cells respond to radiation or chemical insults by activating DNA-damage

response pathways, which encompass signal transduction processes, which alert the cell to the presence of DNA damage and trigger various downstream events: cell-cycle arrest, repair and apoptosis; (b) the adverse biological consequences of exposure to DNA damaging agents such as mutation, chromosomal aberrations and cell death arise from mis-repaired or unrepaired DNA damage, especially DNA double-strand breaks (DSBs), and (c) cells mutated in one repair pathway may recruit alternative/backup pathways to restore the functions of repair machinery and cell survival [6].

A successful targeted anti-tumor regimen requires thorough understanding of the underlying mechanisms involved, starting from the exposure of cells and tissue, DNA damage, repair of DNA damage, and of protein interactions or malfunctions that lead to cancer development. Due to the environmental dependency (tumor versus normal tissue), the nature of cellular response, high degree of pathway choices, intricate nature of the signaling network, and lack of sufficient quantitative data on the dynamics of DNA repair, cell signaling, cell-cycle progression and apoptosis, potentially very effective anticancer strategies, have failed to go from concept to clinical practice (an example is the failure of the synthetic lethality by poly ADP ribose polymerase (PARP) inhibitors). Such deficiency makes it increasingly evident that we need a more comprehensive approach, theoretical in parallel with experiments to uncover mechanism(s) of cancer development, and make predictions about the clinical outcome and risk. A mechanistic mathematical approach includes modeling of the radiation exposures, and a biological process in DDR can pave the way to advance the field and suggest more optimal treatment schedules and tumor-cell inactivation pathways such as senescence by selecting radiation modalities like light ions and suitable adjuvant agents.

Table 1. MCTS simulations code KURBUC for electrons and heavy ions.

Code	Particle	Energy range	Date created	Reference
Kurbuc_Electron ^a	Electron	10 eV–10 MeV	1993	[18]
Kurbuc_Pits99 ^b	All ions	≥ 0.3 MeV u^{-1}	1999	[19]
Kurbuc_Proton ^c	Protons	1 keV–1 MeV	2001	[20]
Kurbuc_Alpha ^d	^4He ion	1 keV u^{-1} –2 MeV u^{-1}	2002	[21]
Kurbuc_Chem ^e	Chemistry	$\geq 10^{-12}$ s	2006	[22]
Kurbuc_Neutron ^f	Neutron	Thermal–100 MeV	2007	—
Kurbuc_e _{liq} ^g	Electron	10 eV–10 keV	2008	[23]
Kurbuc_P ^h	Protons	1 keV–300 MeV	2010	[24]
Kurbuc_Carbon ⁱ	Carbon ions	1 keV u^{-1} –10 MeV u^{-1}	2013	[25]

^a A detailed description of the Kurbuc_Electron code, including total and differential cross sections based on model calculations for water vapor and direct experimental data, has been given in previous publications [7, 18, 31, 32].

^b Kurbuc_Pits99 is an MCTS code for the simulation of track segments of protons and heavier ions with initial energy ≥ 0.3 MeV u^{-1} . A detailed description and input to the code have been published [19]. The code was designed in a modular form to make it easy to replace the electron simulation part with another one. The data base of the cross section for the code contains water vapor, liquid water, amorphous ice, and DNA.

^c The Kurbuc_proton [20] was the first MCTS code to produce the full-slowng-down of proton track, 1 keV u^{-1} –1 MeV u^{-1} , with the unique feature charge-exchange model for H^+ , H^0 and H^- in the Bragg peak.

^d Kurbuc_Alpha [21], similar to the Kurbuc_proton, produces the full-slowng-down of alpha particles with energies 1 keV u^{-1} –2 MeV u^{-1} including the interactions for low-energy He^{2+} , He^+ , and He^0 in water.

^e Kurbuc_Chem [22] describes the prechemical evolution of radiolysis of water product in an infinite water medium to stable radical OH, H and e_{aq} .

^f Kurbuc_Neutron (unpublished) produces the spectrum of secondary particle, thermal—100 MeV.

^g Kurbuc_e_{liq} [23] is the version of Kurbuc_electron with liquid water cross-section input (see chapter 2).

^h Kurbuc_P [24] is the version of the code extended to 300 MeV to simulate proton-therapy beam energies.

ⁱ Kurbuc_Carbon [25] is the first example to produce the full slowing down of carbon ion, 1 keV u^{-1} –10 MeV u^{-1} , including charge exchange using the three-body CTMC model.

In our approach, we use mathematical models and computer simulations in conjunction with experimental evidence in anticipation of offering solution(s) to cancer problems. We assume most biological experiments and radiation therapy regimens are based on single-radiation track effects.

1.1. Biophysical modeling studies

1.1.1. Simulation of radiation track for monoenergetic electrons and ions. *State-of-the-art: The physics of interaction of ionizing radiation with matter has been the subject of research and investigation since the discovery of x-rays, but at low electron energies of less than 1 keV it is the subject of much uncertainty and raises many questions. In this context, Monte Carlo track structure (MCTS) studies and the computer codes we have developed (see table 1) contain state-of-the-art physics knowledge and provide a rich source of experience by making it possible to estimate those parameters of radiation effects and express them in numbers, though we may not yet be able to measure them directly (see chapters 2 and 3; Nikjoo et al 2012, 2006) [7].*

There are many ways to simulate radiation track from one to four dimensions, including space and time as the most sophisticated type. These include: (a) single parameter track (linear energy transfer (LET)), z_D , y_D , Q , W_R and relative biological effectiveness (RBE)) [8], Howard-Flanders [9]; (b) the 1D deterministic description of ion transport based on the solution of the Boltzmann equation [10]; amorphous-track description based on radial dose profile around heavy charged particle trajectories, but ignoring the stochastic structure of the track [11]; (c) condensed history method codes (for example MCNP [12], FLUKA [13], GEANT [14], PHITS [15], SHEILD-HIT [16], MCEP [17]; (d) 3D MCTS codes

with stochastic (for example KURBUC codes (see table 1) and Partrac code [26]), and (d) space and time evolution by taking into account the creation and diffusion of radical species in water (KURBUC codes (see table 1)). Historically, the major breakthrough in simulation of photon and electron transport should be credited to the work of Berger in 1963 on the code ETRAN [27], which laid the foundation for what is known today as the condensed-history method and subsequently led to the development of full MCTS codes (Oak Ridge code Turner *et al* (1985) [28]; GSF code MOCA7, Paretzke, 1987 [29]; Toulouse code, Terrissol and Beaudre, 1990 [30]; KURBUC code, Uehara *et al* (1993) [18]).

The bulk of this article is devoted to current progress in the description of the physics input in simulations of electrons and heavy charged particle tracks. Our experience and work have been translated, and is contained in the MCTS code system KURBUC, a suite of MCTS codes for the simulation of electrons and ions, summarized in table 1.

SUMMARY

- Many general-purpose Monte Carlo codes [12–17], and MCTS codes [18–32] are now available, in the public domain or through collaboration, for simulations of photons, electrons, neutrons and heavy ion tracks and dosimetry.
- Physics and simulation of electron tracks below 100 eV in condensed phase remain an area of investigation and improvement.

1.1.2. Interaction of radiation with matter. *State of the art: In chapters 2–4 we describe the latest development in interactions of electrons and heavy charged particles in water as a biological surrogate.*

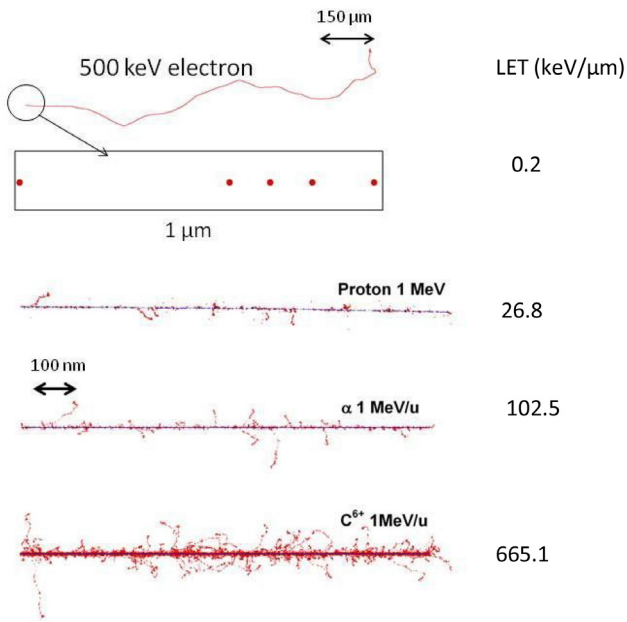


Figure 2. Comparing tracks of electrons, protons, He and carbon ions. For the electron shown, a full-slowing-down track, while for the others only 1 μm track segments. The most obvious observation is that the number of inelastic interactions increases with increasing of LET. The box for the 500 keV shows the number of primary electron interactions in a 1 μm length. The scale for ions is 100 nanometer (nm) as indicated.

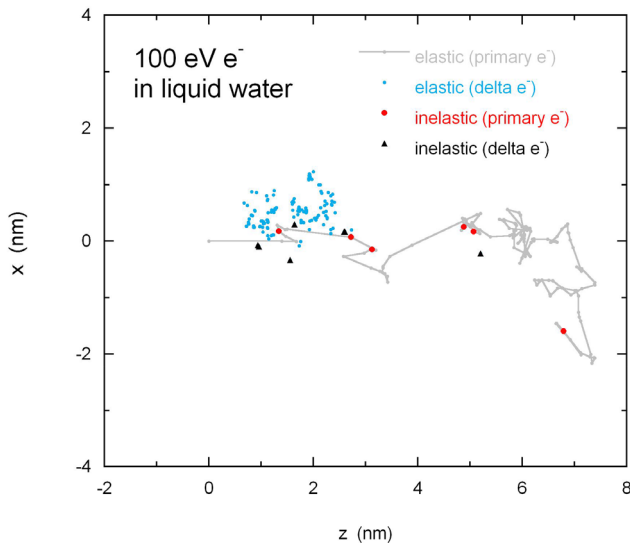


Figure 3. Projection of a track on the (z, x) plane of a 100 eV electron started at the origin (0,0,0) in liquid water. The gray line represents the full path of the primary electron including elastic scattering, on which the energy deposition is due to ionization and excitation, is represented by the red dots. The interaction points for elastic and inelastic scattering of the secondary electrons are represented by the blue dots and the black triangles, respectively.

In principle, the interaction of radiation with matter and biological cells involves the transfer of energy from the particulate radiation to atoms or molecules of the medium. This comes about through various modes of atomic and nuclear interactions in the form of inelastic and elastic scatterings. The

inelastic atomic interactions are the events producing ionizations and excitations (see chapters 2 and 3; [6]). The patterns of energy depositions may differ because of the aggregate of the medium, types of radiation and the kinetic energy of the particle (see figure 2). Elastic scattering is a nuclear interaction, causing scattering of the particles. Elastic scattering determines the shape of the track (see chapter 4).

Human cells are composed mainly (>70%) of water. Therefore, the interaction of radiation can be categorized by physical interaction and physicochemical interactions leading to early biological effect. The basic concept of radiation track is demonstrated in a number of examples.

Example 1: comparing track segments of electron and heavy ion in figure 1.

Example 2: simulation of a 100 eV electron track in liquid water. The 100 eV electron track shown in figure 3 was generated by the Kurbuc_eliq code [23]. The electron was set in motion at the origin (0,0,0) in the direction of the z -axis. In almost all radiation track presentations, elastic interactions are absent (not shown). This is because it causes the picture to be very complicated and difficult to trace. The 100 eV electron was chosen for convenience, as it does not create a busy picture. The basic elements of the track simulation are the elastic and inelastic interactions by the primary and delta electrons, and similarly for elastic scatterings by the primary and delta electrons. A critical discussion of models used for the elastic scattering of low-energy electrons is presented in section 3. Because of their stochastic nature, no two 100 eV electron tracks look the same. However, by simulating a number of electron tracks we obtain the average macroscopic quantities that may be compared to the experimental data.

Example 3: comparing macroscopic quantities of electrons of different energies. In table 2 we compare some track parameters for electrons of various energies according to the model presently used in KURBUC. These include the number of ionization and elastic scatterings per track, average mean-free-paths, and thermalization distances. Data have been averaged for five simulated tracks generated by the KURBUC codes (see table 1).

Example 4: simulation of a photon of Al_K is shown in figure 4.

Example 5: comparing depth-dose distribution for electrons generated by ^{60}Co γ -photons with full slowing down of 200 MeV protons.

Both radiations, ^{60}Co γ -photons and protons, are used in radiation therapy. The most obvious observation is that proton track deposits more of its energy at the end of the track, while ^{60}Co γ -photons have their peak at about 5 mm. The proton dose at the entrance window is much less than that of γ -photons. The proton distributions were generated by the code Kurbuc_Proton and for ^{60}Co γ -photons by the Kurbuc_Photon [17] (see figure 5).

Table 2. Some average quantities of electron tracks in vapor and liquid water.

Electron (eV) in liquid water	Average MFP (elas + inel) (cm)	Average MFP (inel) (cm)	Thermalizat length (cm)	No. of elastic per track	No. of ionization per track	No. of excitation per track	No. of sub excit. per track
11	1.93×10^{-8}	3.96×10^{-8}	6.55×10^{-8}	48.2	0.0	1.0	1.0
50	2.21×10^{-8}	3.23×10^{-8}	8.83×10^{-8}	19.2	2.0	1.0	3.0
100	2.06×10^{-8}	2.10×10^{-8}	6.16×10^{-8}	105.0	3.4	4.2	4.4
1000	2.71×10^{-8}	6.53×10^{-8}	6.61×10^{-8}	780.8	41.8	26.6	42.8
Electron (eV) in water vapor.							
11	1.66×10^{-8}	1.62×10^{-8}	4.24×10^{-8}	76.0	0.0	1.0	1.0
50	2.06×10^{-8}	2.15×10^{-8}	8.27×10^{-8}	49.6	1.0	2.4	2.0
100	1.94×10^{-8}	2.60×10^{-8}	5.60×10^{-8}	89.8	3.0	3.8	4.0
1000	2.45×10^{-8}	7.03×10^{-8}	6.37×10^{-8}	923.0	34.4	29.0	35.4

Example 6: a track of $2 \text{ MeV u}^{-1} {}^4\text{He}_2$ ion in water in figure 6.

Summary: The technology for the simulation of radiation tracks at molecular level is now highly advanced. One major question to explore is the detailed understanding of the physics of interactions of low-energy electrons below 100 eV in the condensed phase of matter, in particular, that of water. Currently, there is only one accurate measurement of the energy loss function of liquid water [4] in the region 10–150 eV. The latter experiment has allowed for a full model description of energy loss and momentum transfer leading to the calculations of cross sections [90].

1.2. Microdosimetry

State-of-the-art microdosimetry is a better way of characterizing radiation track. It was invented by Rossi et al [33] to overcome the problems of dose measurements in small-sized detectors and volumes, with diameters $\leq 1 \mu\text{m}$, similar to biological targets. Figure 7 articulates the conceptual idea. The MCTS method is the ideal way to estimate microdosimetry parameters with a high level of accuracy. Nikjoo et al have computed microdosimetry parameters in cylinders and spheres of diameters 0.001–2.0 μm for soft x-rays, monoenergetic electrons and heavy ions (see table B2 appendix B)

Microdosimetry was a revolutionary idea in radiation biophysics and set the course on how to link radiation physics to biological observations for the next three decades. Barendsen (1968) eloquently summarized the importance of microdosimetry for radiation biology and radiation protection. He discussed the relation between biological effects produced by ionizing radiations and spatial distributions of energy depositions, in general, and quantitative aspects and implications of differences between biological effects of different types of radiation—it is generally assumed that the properties of ionized and excited molecules produced by radiation to a first approximation are independent of the velocity, mass and charge of the ionizing particles. Consequently, differences between effects of various ionizing radiation must be due to the differences between spatial distributions of ionizations in the irradiated material [34].

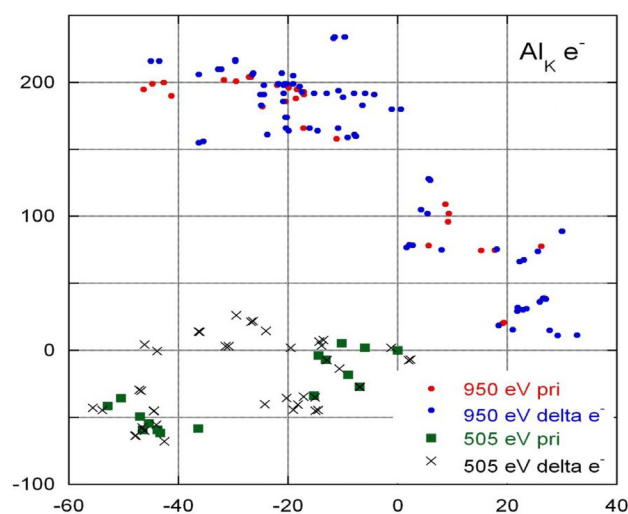


Figure 4. Tracks of two electrons generated from the interaction of a single photon of ultrasoft x-ray Al_K in DNA medium. The most probable photon interaction generates a primary photo-electron (950 eV) and an Auger electron (505 eV) set in motion in random directions. The full detailed calculations describing the spectrum of electrons released have been published [57]. The ordinate and abscissa are in units of angstrom.

In the 1960s, the radiation research community was concerned with the importance of the hazards of exposure of humans to low dose and dose rates of ionizing radiation of relevance to health physics problems. It is intriguing that similar issues still dominate the work of radiation-protection funding bodies in Europe, USA and Japan. Here are some statements of the problems discussed at the time: quantitative aspects of differences between biological effects produced by different radiation; To be able to make an extrapolation from high doses and dose rates to low doses and dose rates of less than 1 mrad (10 μGy) per hour, we need to have mechanistic knowledge and models of dose-effect relationship. The relevant relative biological effectiveness (RBE) values on which the choice of quality factor (Q) values have to be based are frequently unknown (see table 3). Is there a synergetic effect when irradiating cells by high and low LET radiation simultaneously? Thirty-five years later we still see many published papers and meeting presentations still concerned with

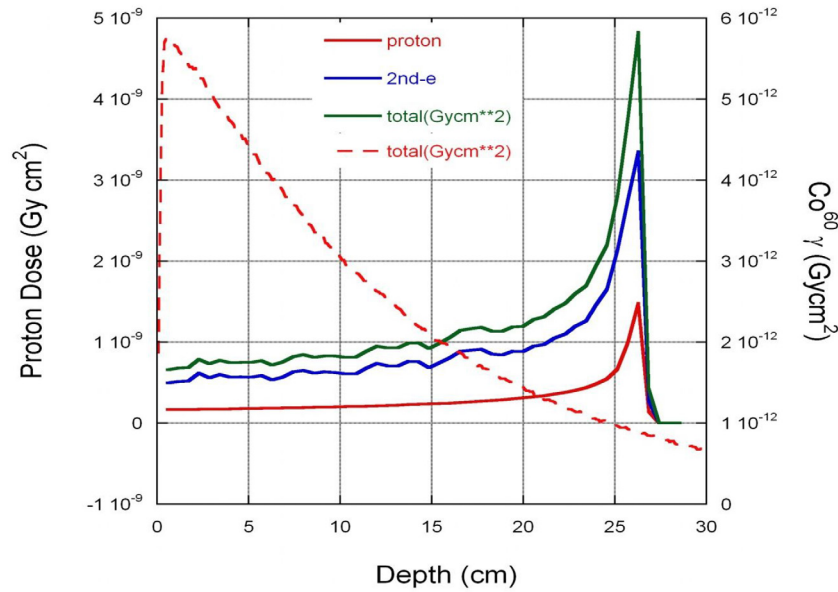


Figure 5. Comparing the dose deposited by one track of 200 MeV protons with that of ^{60}Co γ -photons (red dashed line).

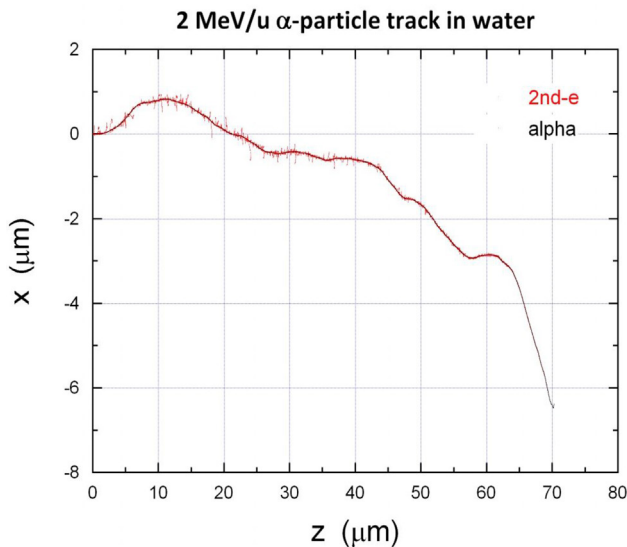


Figure 6. Full slowing down of a $2 \text{ MeV u}^{-1} {}^4\text{He}_2$ ion track simulated by the computer code Kurbuc_Alpha (table 1). The number of events is presented in appendix A.

the same issues that Rossi and Barendsen discussed in 1968 [33, 34]. It should be noted that all RBE values in the clinic are estimations that have been derived from cellular studies and calculations. The term RBE is a ‘black box’ depending on many physical and biological factors. As such, the term RBE does not define a mechanism of biological effect.

1.2.1. Microdosimetry calculations from the first principle. *The problem: What is the absolute probability of energy deposition by a single track of radiation under test in a small-sized volume ($1 \text{ nm} \leq \text{diameter} \leq 1 \text{ } \mu\text{m}$)?*⁹

Analytical method is one way of solving the problem, but the solution provides an average macroscopic value. The

⁹ E in this notation is equivalent to energy imparted, ϵ , total energy deposited in the target volume.

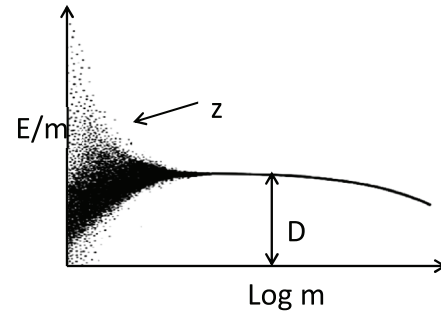


Figure 7. Specific energy as a function of the mass of the sensitive volume of a detector for which dose is determined. The flat part covers the region in which the absorbed dose can be established in a single measurement. As the size of the detector is reduced the uncertainty in the measurement increases and tends to infinity at nanometer levels. Rossi, in 1968 [33], used a similar picture to review and discuss the basic concepts related to the local or microscopic distribution of radiation energy deposition and techniques for the measurements of dose in small volumes and relate these to radiobiological mechanisms. Therefore, microdosimetry, in principle, is concerned with the characterization of radiation field at microscopic level.

MCTS method is the only way to solve the problem. David Charlton first provided the solution to the problem in 1985, in a paper presented at the 10th Symposium on Microdosimetry [39]. We published our more recent results in 2014 [40].

In figure 8, we demonstrate the above statement by the following examples for the calculation of absolute frequencies of energy deposition by radiation under test as a function of specific energy z , the energy deposited in a volume of defined dimensions. The significance of the parameter $f(>E)$ is that it describes the probability of a track capable of depositing an amount of energy in a specified volume. All other microdosimetry parameters can be derived from $f(>E)$ distributions. Therefore, $f(>E)$ allows the comparison of the physical effectiveness of radiation tracks of different qualities. In figure 8, we compare the probabilities of three types

Table 3. Weighting factors in radiation protection and radiation therapy [35].

Radiation under test	^a RBE _{clinic} = ^b W _{isoE}	RBE ^b values derived from LQ model for radiation protection	Reference
⁶⁰ Co γ photons/x-rays	1.25 ^c	1.2 ^g	[36]
⁶⁰ Co γ photons/protons in SOBP at 5 cm	1.1 ^d	1.1 ^g	[36]
⁶⁰ Co γ photons/C ion SOBP	2.4 ^e	2.9 ^g	[37]
⁶⁰ Co γ photons/neutrons	3.1 ^f	3.2 ^g	[38]

^a RBE calculated; clinical RBE is assumed to be equal to W_{isoE} [36].

^b W_{isoE}: when the reference radiation is ⁶⁰Co- γ photons and the absorbed dose of the reference source given in fractions of 2 Gy in 5 d a week, the weighting factor assumed equal to W_{isoE} [35, 36].

^d IAEA [36].

^e Matsufuji *et al* [37].

^f Batterman *et al* [38].

^{c, g} Lindborg *et al* [35].

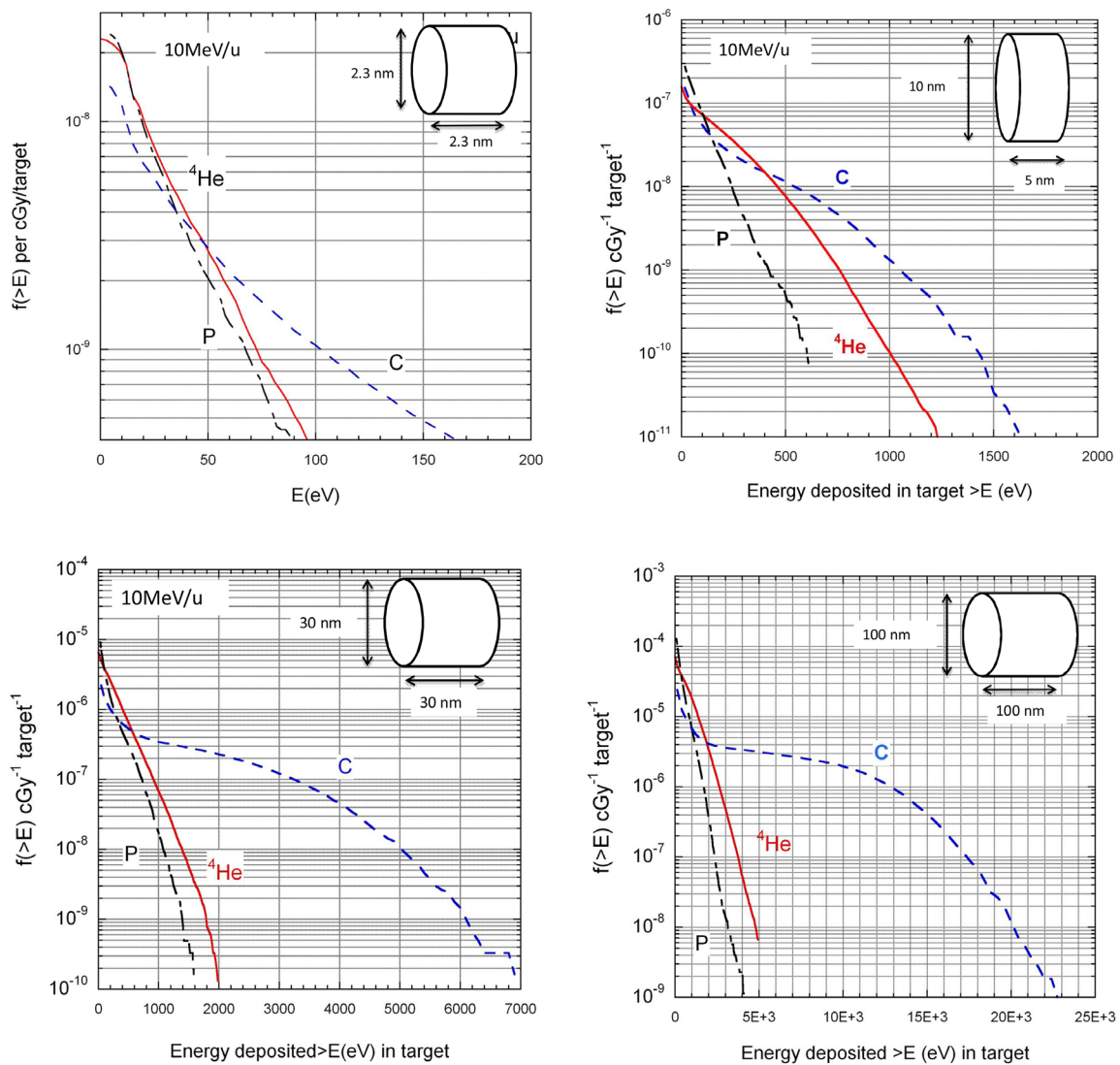


Figure 8. Absolute frequencies of energy depositions $f(>E)$ per unit dose per target versus the energy deposited ($>E$ eV) in small cylindrical volumes for the three types of radiation of 10 MeV u^{-1} proton, helium, and carbon. We can distinguish approximately two distinct regions. As an example in panel A, the crossover for C is at about 50 eV with He, and at 35 eV with P. The LET values of the radiation ranked carbon, helium and proton from high to low. The value of $f(>E)$ is a measure of the effectiveness of the radiation. Therefore, below 35 eV carbon ions are the least effective, while the >35 eV carbon ions with higher LET values are capable of depositing more energy in the same volume (data from Monograph 2010b listed in table B3 of appendix B).

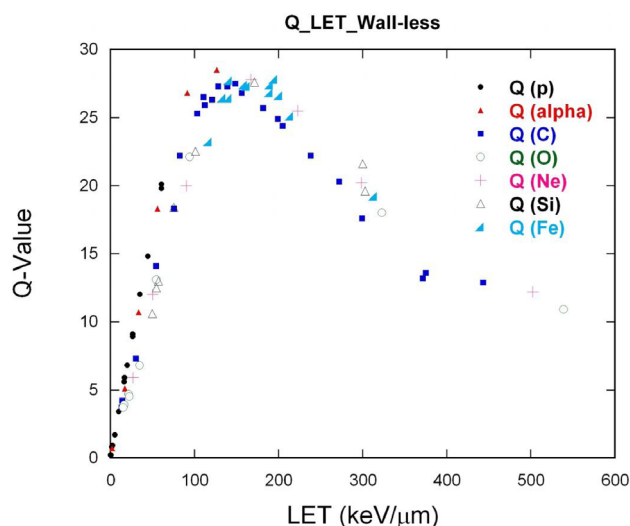


Figure 9. The calculated quality factor (Q) in table B2 (appendix B) for a wall-less proportional counter of $1\ \mu\text{m}$ diameter. The detailed method of calculation is given by Nikjoo *et al* [41].

of radiation, proton, helium ion and carbon ion of 10MeV u^{-1} energy, depositing energy in cylinders of width and height $2.3 \times 2.3\text{ nm}$; $10 \times 5\text{ nm}$; $30 \times 30\text{ nm}$, and $100 \times 100\text{ nm}$. What does the observation tell us? Panel A: Carbon ion is more effective than proton and helium when the energy deposition by a single track E is about $>50\text{ eV}$, but less effective below that. Similar arguments are applicable to data seen in panels B, C and D for larger target sizes. In table B1 of appendix B, we show some selected microdosimetry parameters for the various types of radiation in the cylinders. In table B2 of appendix B, microdosimetry parameters are presented for selected ions in spherical targets with sensitive volume (SV) = $1\ \mu\text{m}$. For more detailed information and data, we refer the reader to the published monographs (see table B3 of appendix B for references). In figure 9, we show the distribution of quality factors Q for all radiation types tested in table B2 of appendix B in a wall-less counter of $1\ \mu\text{m}$ diameter.

Summary:

- Low-energy electrons play a crucial role in characterizing the radiation field
- Radiation interacts with matter stochastically
- Stochastic quantities are needed for a detailed description of the track
- Microdosimetry is a better way of characterizing radiation track
- Microdosimetric quantities vary with the size of the target volume
- Microdosimetry is a recommended method for characterizing the radiation quality factor when biological effectiveness is not well known, which is true for radiotherapy beams in the Bragg peak
- References to the data base of microdosimetry parameters calculated from first principle for soft x-rays, electrons and heavy ions using MCTS codes are given in table B3 (appendix B).

1.3. Modeling and simulation of a human cell nucleus including organization of genomic DNA

State-of-the-art: The most advanced model of the cell nucleus for use with biophysical study of DNA damage induced by ionizing radiation has been published by Nikjoo and Girard [42]. The model is based on a canonical B-DNA to build the nucleosome and the 30 nm solenoid chromatin structures. In turn, the chromatin structure was used to form the loops of factories in chromosome domains. The model of the cell nucleus contains a complete copy of the human genome (NCBI) [43].

The organization of genomic DNA in mammalian cell nucleus is shown in figure 10. The model of chromosomal territories is based on the conceptual model that was proposed by Cook [44]. In the model, the nucleus consists of ‘domains’, each containing one of the two ‘arms’ of a chromosome. Each arm contains ‘factories’, and each factory has ‘scaffolds’ to which loops of DNA are anchored. The region around a factory occupied by most of the attached fiber is termed a ‘cloud’.

The size of each domain and amount of DNA it contains are taken from ‘Parameters of the Human Genome’ [45]. Other quantities, such as the average number of base-pairs in a loop, and the range of possible loop lengths, are wherever possible compatible with those suggested by Cook [44] and summarized in table 4. In the present implementation of the model, the cell nucleus is represented by a sphere of diameter $10\ \mu\text{m}$, though the size of the spherical nucleus can be changed according to the cell type and size. Each chromosome is modeled as a pair of adjacent spheres, one for each arm. To represent the human cell nucleus 92 spheres must, therefore, be fitted into the outer sphere. The volumes of the spheres are proportional to the amounts of DNA that they need to contain. They are fitted so that they can touch in an efficient manner, but not overlap. The sizes of the chromosomal domains are based on the parameters of the human genome given by Morton [45]. The domains are considered spherical, each containing the DNA belonging to one chromosome arm. The placement of the spherical domains can be chosen in several ways: entirely at random, or partly according to the size of the domain, thereby selecting smaller or larger domains to be at the inner or outer regions.

The chromatin structure was assumed to take the form of 30 nm solenoidal fiber. The length of the loops of fiber is variable between 7.5–175 kilobase pairs (kbp). Factories are modeled as spheres of diameter 50 nm. At the attachment points, the DNA is assumed locally unwound, so the fiber as a whole does not have to be curved smoothly through an attachment point. The number of loops per factory is in the region of 10–20, and the number of factories can be between 100–200. On average, a loop contains 86.5 kbp of DNA, and may extend to a distance of about 250 nm from the factory center. As an example, taking chromosome 1, arm 1, its domain when fitted into the nucleus with all other domains contains 128 megabase pairs (Mbp) and has a radius of $0.993\ \mu\text{m}$. The volume of this spherical domain is $4.1\ \mu\text{m}^3$ and the ‘density’ (percentage space occupancy) of the fiber in the domain can be estimated by considering the volume occupied by a 30 nm

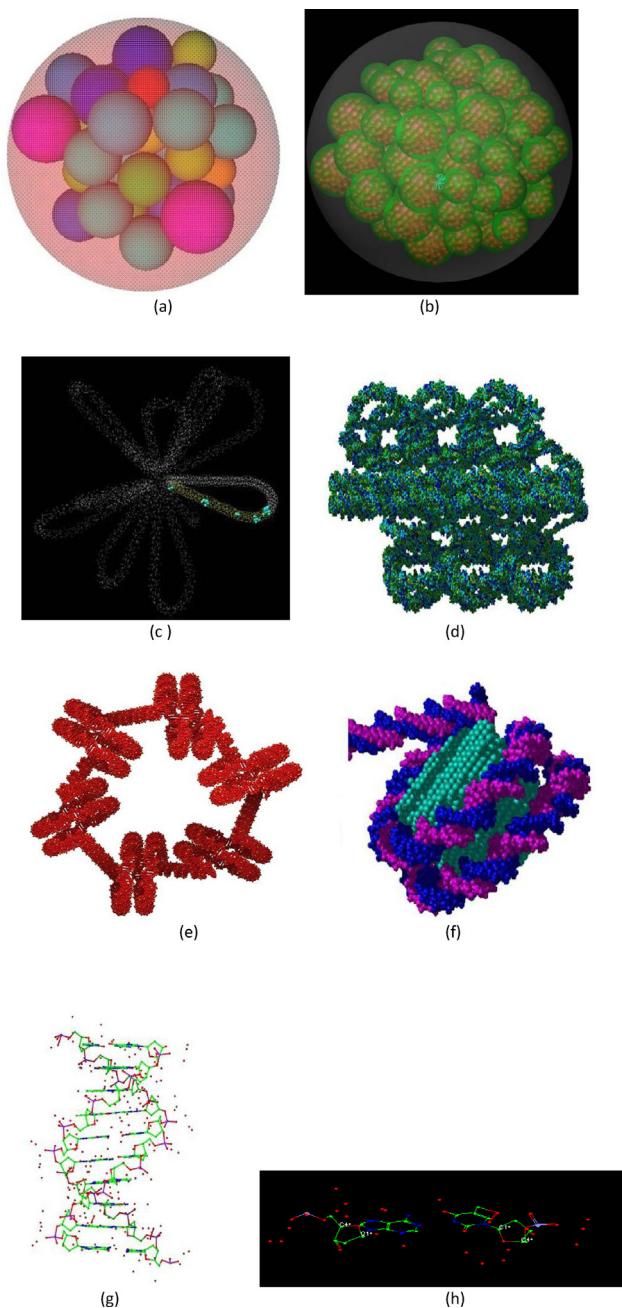


Figure 10. Organization of DNA in mammalian cells [7]: (a) A model of a cell nucleus with spherical chromosome domains each representing an arm of a chromosome. (b) A chromosome domain with spherical factories. (c) One factory with 11 chromosome loops of in X chromosome domain. In this instance, one loop contains the HPRT gene (48 kbp). Cyan shows the exons in between the introns. (d) A segment of a chromatin fiber. The loops of chromatin consist of this basic element. (e) A cross section of a chromatin loop. (f) A nucleosome, an element of the chromatin. (g) A 10bp DNA double helix, a basic unit used to construct all higher order DNA structures [228]. (h) A single base-pair to construct a double helix (Watson and Crick base pairing). The free red dots represent first hydration shell water. The bonded dots are oxygen. There are ~ 20 water molecules per base-pair [228].

fiber, 128 Mbp long. This gives a volume of fiber of $0.76 \mu\text{m}^3$ and a density of 19%. A similar calculation based on a 10 nm fiber with 17 bp per nm would give a volume of $0.59 \mu\text{m}^3$ and a density of about 14%.

The following simplifying assumptions were made to reduce the amount of data and data processing. The number of loops attached to each factory is taken to be 11. The amount of chromatin fiber that a factory can accommodate is assumed to be fixed, and equal to the number of loops times the mean length of 86.5 kbp. This is about 952 kbp per factory. The number of factories per domain adjusts itself to be able to handle all the chromatin in the domain. On this basis, the longest arms may contain as many as 165 factories. Figure 9 shows the organization of DNA in a mammalian cell.

1.4. Modeling and simulation of DNA damage by radiation track

State-of-the-art. Experimental techniques in DNA damage measurements are not yet capable of producing detailed resolution at a single base-pair. In fact, the usual technique of measuring DNA damage and, in particular, the DNA DBS, pulse-field-gel-electrophoresis (PFGE), has a resolution of about 1 Mbp. Currently, MCTS is the only method capable of simulating quantitatively the DNA damage at a single base-pair including DSB, single-strand break (SSB), base damage (BD), and the source of damage from direct and indirect effects. We assume an energy deposition of $\geq 17.5 \text{ eV}$ by a single track in the volume of sugar-phosphate can cause an SSB with a probability of one. A similar assumption is considered for the creation of a BD. A DSB is considered to be induced if there were two SSBs on the opposite strands in one turn of DNA. Use of the MCTS method has enabled us to predict the frequencies of complex types of DNA damage. The experimental data for the induction of SSB and DSB for radiation of different quality are nearly constant (independent of radiation type, LET) while frequencies of complex DSB increase with increasing LET. The yield of DSB is a linear function of radiation dose (46, 31). Recent ideas and progress in the DNA damage field have been reviewed by Georgakilas [218] providing up-to-date data on the induction and repair of clustered DNA lesions. In addition, Stewart provides very useful data and computer code for MCTS calculations, which are used by many laboratories [219]. The latter authors discuss the idea of how bistranded clustered DNA damage can be converted to unistranded and then mutations [220], in particular, for high-LET radiation.

1.4.1. Chemical interaction of radiation with water. Energy depositions by radiation track in water cause ionization and excitation of water molecules leading to reactive oxygen species (ROS). These ROS, in turn, diffuse in the environment and may cause damage, for example, in the cellular environment. Here, we provide some important references to the topic, as it has been well researched and explored. The important time-frame here is the prechemical stage $< 10^{-12} \text{ s}$ after which the radicals are stable and fully formed. The development of chemical modules simulating the prechemical and chemical stages of charged particle tracks in pure liquid water calculations are usually based on the physical track structure for electrons and ions.

Table 4. Approximate structural features of the human cell nucleus model.

Cell nucleus diameter	Variable, 10 μm diameter is usually considered
Chromatin fiber diameter	30 μm
Chromatin fiber structure	Solenoidal
Nucleosome height	10 nm
Nucleosome per fiber element	6
DNA structure	Canonical
DNA cross section	2.3 nm
One element of DNA (cylindrical)	10 base-pair
Length of DNA in the genome	$\sim 10^9$ bp
Base-pair per fiber element	1191
Fiber element per chromatin loop	Randomized subject to average range
Loops per cell nucleus	72 600
Factories per cell nucleus	6600
Average number of loops per factory	11 (range from 10–20)
Average bp per loop	86.5 kbp
Number of bp per loop	7.5–175 kbp
Diameter of factory	50 nm
Factories per chromosome arm domain	Up to 165

The chemistry code enables us to simulate the non-homogenous chemistry that pertains to electrons, and ions of various LET. Time-dependent yields of chemical species produced by electrons and ions of different energies can then be calculated and compared with the G values obtained from experiments. The references to the published papers provide the general and more specific detail to the subject [22, 221–227].

1.4.2. Energetics of DNA damage. When a radiation track crosses a human cell, it causes DNA damage. To short-cut the huge set of biophysical and biochemical reactions involved in causing a strand break or a BD, we considered the question: Does DNA damage arise from ‘ionization’ or from ‘energy depositions’? In our modeling of DNA damage, we assumed that ‘energy deposition’ in the volume of ‘sugar-phosphate’ (S-P) is responsible for causing strand break or BD. However, if the track event is in the water in the cell environment, we assume that ionization causes hydrolysis of water creating hydroxyl radical and other species. Therefore, DNA damage arises from direct energy deposition and the reaction of diffusing water radicals with DNA. The quantity 17.5 eV energy deposition in the S-P volume was first suggested by Charlton and Humm [47] in a simulation of an experiment by Martin and Haseltine [48] and remains the only value to date for the energetics of DNA damage based on biological experiment and evidence. The original simulation was repeated to confirm the finding by Nikjoo *et al* [49], and Pomplum *et al* [50]. For the reaction of the OH radical with the S-P moiety, we consider a probability of 0.13 [51] to cause an SSB. A DSB was assumed to be formed if two SSBs were produced on opposite strands within 10 bp. Although other published models use different criteria, the total yield of SSBs and DSBs for most models is nearly the same and is consistent with experimental data. Figure 11 shows the classification of DNA damage [39, 52]. An ‘X’ represents a strand break or a BD produced either by a direct energy deposition or from the reaction of an OH radical.

1.5. Modeling and simulation of DNA repair

State-of-the-art: The field of experimental DNA repair is very advanced and much knowledge has been gained in understanding the role of DNA repair proteins and of different repair pathways in mammalian cells. There remains a major question in the field: What is the cause of slow repair of DNA DSBs? Two ideas have been put forward: Decondensation of chromatin fiber, and the other is the repair of complex DSBs. Recently, Taleei and Nikjoo [53–55] produced, for the first time, a mechanistic model of DSB repair and suggested that complex damage was responsible for the slow repair of DSB. Similarly, Rahmanian *et al* [56] suggested a similar mechanism for the slow repair of non-DSB DNA damage [46] (see section 5).

In section 5, we present a detailed discussion of the repair of DSB. To articulate the issue, some examples of repair times of different types of DSB are presented in table 5 and figure 12. The average times for the repair of simple DSB, DSB in the heterochromatin, and complex DSB for the incident ions and electrons are listed in table 5. The simple DSBs are repaired fast in about 29 min, the simple DSBs in the heterochromatin region are repaired in about 232 min, and the complex DSBs are repaired in a slow process in about 700 min. The average repair time for simple DSB, complex DSB, and DSB in the heterochromatin does not vary with the radiation quality, however, the complexity ratio changes with the radiation quality. The complex DSBs are mainly responsible for slow repair DSB kinetics. Figure 12 lists the repair kinetics for three examples of simple and complex DSB. As predicted by the model independent of incident radiation quality, the repair kinetics for simple DSB is orders of magnitude faster than complex DSB. The current experimental technology is not capable of verifying the repair kinetics for individual DSB (complex and simple) induced by ionizing radiation. However, in recently published literature there has been inference that the slow kinetics of DSB repair is the result of chromatin decondensation as an alternative interpretation to complex DSB [230–232], but remains as another speculation.

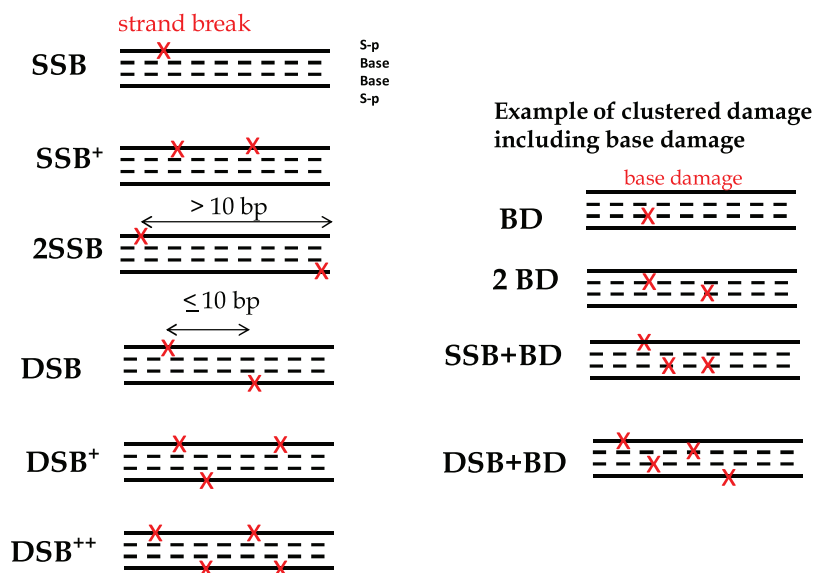


Figure 11. Model of DNA damage induced by direct energy deposition and reaction of OH radicals. For ease of observation, the double helix is shown as four untwisted linear lines. The solid lines at the top and bottom represent the S-P backbone; the two dash lines represent the bases. An 'X' represents an SSB, or a BD in one turn of double helix DNA. If two Xs are on opposite strands within 10bp of each other, it will be considered a DSB. If two SSBs are more than 10bp apart, it is denoted by 2SSB, and if two SSBs are within 10bp apart, but on the same strand of DNA, it is denoted by SSB⁺. Other configurations such as DSB⁺, and DSB⁺⁺ are forms of complex DSBs. Non-DSB breaks include an SSB, SSB⁺, 2SSB, BD, 2BD, and SSB+BD [46].

When a radiation track crosses a human cell, it causes DNA damage. To short-cut the huge set of biophysical and biochemical reactions involved in causing a strand break or a BD, we considered the question: Does DNA damage arise from 'ionization' or from 'energy depositions'? In our modeling of DNA damage, we assumed that 'energy deposition' in the volume of 'S-P' is responsible for causing strand break or BD. However, if the track event is in the water in the cell environment, we assume that ionization causes hydrolysis of water creating hydroxyl radical and other species. Therefore, DNA damage arises from direct energy deposition and the reaction of diffusing water radicals with DNA. The quantity 17.5eV energy deposition in the S-P volume was first suggested by Charlton and Humm [47] in a simulation of an experiment by Martin and Haseltine [48] and remains the only value to date for the energetics of DNA damage based on biological experiment and evidence. The original simulations were repeated to confirm the finding by Nikjoo *et al* [49], and Pomplum *et al* [50]. For the reaction of the OH radical with the S-P moiety, we consider a probability of 0.13 [51] to cause an SSB. A DSB was assumed to be formed if two SSBs were produced on opposite strands within 10bp. Although other published models use different criteria, the total yield of SSBs and DSBs for most model calculations produce similar results, as expected.

2. Inelastic scattering of low-energy electrons in condensed media

2.1. The dielectric response function

Inelastic collisions leading to electronic excitations (including ionizations) represent the main energy-loss mechanism of non-relativistic electrons in matter. Exact calculations of

Table 5. Average repair time for simple DSB, DSB in heterochromatin and complex DSB is listed for 1000 DSB induced by the simulated particles.

Particles	Simple DSB (min)	DSB in heterochromatin (min)	Complex DSB (min)
Electrons			
0.3 keV	28	229	679
1.5 keV	28	230	689
4.5 keV	28	230	689
Protons			
0.3 MeV	28	246	691
0.5 MeV	31	249	684
0.75 MeV	31	213	711
1.0 MeV	31	213	711
2.0 MeV	28	229	679
4.0 MeV	29	231	698
α particles			
2.0 MeV	28	254	699
3.0 MeV	30	241	710
4.0 MeV	30	247	706
6.0 MeV	28	218	700
8.0 MeV	28	220	715
10.0 MeV	28	229	736

the cross sections for inelastic scattering in condensed media are not feasible due to the many-body nature of the problem. The plane-wave Born approximation (PWBA) provides the standard theoretical framework for calculating such cross sections in both the gas and the condensed phase [58].

We should point out that for the condensed phase there is a change in nomenclature, that is, the conventional cross section (σ), in units of $\frac{\text{area}}{\text{atom}}$ or $\frac{\text{area}}{\text{molecule}}$, commonly used for the gas phase, is replaced by a macroscopic cross section (Λ) which has units of $\frac{1}{\text{length}}$. The reason for this change is that in the

SP ...D.....	Example 1
BD	Classification: a simple DSB
B ...I.....	Repair time: 27.8 minutes
SP D.....	
SP ..D.....	Example 2
BD	Classification: a simple DSB
B I...I.....I	Repair Time: 32.1 minutes
SP ...D.....D.	
SPD..	Example 3
BDI.I	Classification: a simple DSB (in the Heterochromatin)
BD...	Repair Time: 275.6 minutes
SPI...	
SPD.....	Example 4
B I....D.....I.....	Classification: a complex DSB
B II...II.....II.....DI...I..	Repair Time: 826.8 minutes
SPI.....D..D..D.....	
SPDD..D....	Example 5
BDI	Classification: a complex DSB
B I.....	Repair Time: 442.8 minutes
SP ...DD.D....D..	
SPD.....	Example 6
B I.....I..D.....D.I.....	Classification: a complex DSB
BI.I.....I..	Repair Time: 908.4 minutes
SPD..D...D..D	

Figure 12. The repair time for three examples of simple and complex DSB is computed. Each example contains the linear representation of the DNA model. The DNA model is shown with four dotted horizontal lines that represent S-P backbone and bases (B) of a double helical DNA. The damage is noted with I for indirect damage and D for direct damage. (a) Examples of simple DSB repair times. (b) Examples of complex DSB repair times.

condensed phase the interaction involves a large number of target electrons that belong to different atoms or molecules; thus, it is conceptually more appropriate to associate the cross section with the likelihood of interaction per unit distance traveled in the medium. The two cross sections are related through $\Lambda = N\sigma$, where N is the number of atoms or molecules per unit volume.

In the condensed phase and within the PWBA, the (complex) macroscopic dielectric response function of the material, $\varepsilon(E, q)$, with E and q being the energy- and momentum-transfer, respectively, is the key quantity for calculating inelastic cross sections based on the so-called linear-response theory (under equilibrium or quasi-equilibrium conditions). These calculations are most valid over macroscopic spatial variations of rather weak external fields (i.e. external perturbations). These conditions are better satisfied for small- q values (long wavelength limit) than for large- q values. Inelastic cross-section calculations for liquid water (and other bulk amorphous solids) are generally carried out within the above approximations. However, these approximations may not be well satisfied for nanostructures (e.g. DNA) which may exhibit a certain degree of order at the nanoscale. In such cases, one may have to start from the momentum-vector-dependent microscopic dielectric matrix. However, dielectric matrix calculations are, in general, much more complicated than dielectric function calculations, because they require elaborate electronic-structure calculations.

In the PWBA, the doubly-differential inelastic cross section in energy- and momentum-transfer is

$$\frac{d^2\Lambda(E, q; T)}{dE dq} = \frac{1}{\pi a_0 T} \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] \frac{1}{q}, \quad (1)$$

where a_0 is the Bohr radius, T is the electron kinetic energy and $\text{Im}[\dots]$ denotes the imaginary part of the argument. Note that for electron energies (T) below about 50 eV the wavelength of the electron may become comparable to inter-atomic (or inter-molecular) distances giving rise to coherent scattering effects not considered here (see section 3). Equation (1) may be recast in terms of the scattering angle θ using the relation

$$q = \sqrt{2m} [2T - E - 2\sqrt{T(T-E)} \cos \theta]^{1/2}. \quad (2)$$

It follows from equations (1) and (2) that $\text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right]$ is proportional to the electron energy-loss spectrum at a given incident electron energy and scattering angle in a transmission experiment. By virtue of this property, $\text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right]$ is called the energy-loss-function (ELF) of the material. Note that the ELF is solely a material property, whereas the electron-energy-loss-spectrum obtained in the transmission electron microscope depends upon the incident electron energy. All practical calculations related to inelastic scattering, essentially start from equation (1). For example, the singly-differential cross section in energy transfer is obtained by integrating equation (1) over q :

$$\frac{d\Lambda(E; T)}{dE} = \int_{q_{\min}}^{q_{\max}} \frac{d^2\Lambda(E, q; T)}{dE dq} dq = \frac{1}{\pi a_0 T} \int_{q_{\min}}^{q_{\max}} \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] \frac{1}{q} dq, \quad (3)$$

where the integration limits are

$$q_{\min}(E; T) = \sqrt{2m}(\sqrt{T} - \sqrt{T-E}) \quad (4a)$$

$$q_{\max}(E; T) = \sqrt{2m}(\sqrt{T} + \sqrt{T-E}). \quad (4b)$$

with m the electron rest mass. The $d\Lambda/dE$ is also sometimes called the differential inverse inelastic mean free path. The total inelastic cross section is obtained by integrating further equation (3) over E :

$$\Lambda(T) = \int_{E_{\min}}^{E_{\max}} \frac{d\Lambda(E; T)}{dE} dE = \frac{1}{\pi a_0 T} \int_{E_{\min}}^{E_{\max}} dE \int_{q_{\min}}^{q_{\max}} \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] \frac{1}{q} dq, \quad (5)$$

where the limits of integration are generally set at $E_{\max} = \infty$ and $E_{\min} = 0$, but are further specified depending on the type of excitation, e.g. for an ionization event in the n th shell $E_{\max} = \frac{1}{2}(T + B_n)$ and $E_{\min} = B_n$, with B_n being the binding energy of that shell (the factor by $\frac{1}{2}$ in $E_{\max}$ accounts for the indistinguishability between the primary and secondary electron). Note that Λ^{-1} is the inelastic mean free path. Finally, the electronic (or collision) stopping power is obtained from

$$S(T) = -\frac{dT}{dx} = \int_{E_{\min}}^{E_{\max}} E \frac{d\Lambda(E; T)}{dE} dE = \frac{1}{\pi a_0 T} \int_{E_{\min}}^{E_{\max}} E dE \int_{q_{\min}}^{q_{\max}} \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] \frac{1}{q} dq. \quad (6)$$

It follows from the above that, within the PWBA, the dielectric function, $\varepsilon(E, q)$, is the only non-trivial material property that is needed to calculate cross sections for inelastic collisions. To emphasize the importance of this fact, the 3D plot of the ELF of a material over the energy-momentum plane is given the special name ‘Bethe surface’ of the material. Note that both the dielectric function, $\varepsilon(E, q)$, and the ELF, $\text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right]$, are dimensionless. The Bethe surface is essentially a map of the full electronic excitation spectrum of the material in both energy and momentum space. Further insight into the excitation properties of the medium may be obtained by expressing the ELF as

$$\text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] = \sum_n \frac{\text{Im} [\varepsilon_n(E, q)]}{|\varepsilon(E, q)|^2}, \quad (7)$$

where n runs over all the electronic excitations of the system. The real part of the dielectric function, $\text{Re}(\varepsilon)$, describes the screening properties (i.e. the polarizability) of the medium, whereas the imaginary part, $\text{Im}(\varepsilon)$, describes the absorption properties. In general, the maxima of $\text{Im} [\varepsilon_n(E, q)]$ correspond to single-electron excitations (e.g. electron-hole pairs, inter-band excitations, inner-shell ionizations), whereas the minima of $|\varepsilon(E, q)|$ correspond to collective many-electron excitations (the so-called plasmons). The minima of $|\varepsilon(E, q)|$ take place when $\text{Im} [\varepsilon(E, q)] \ll 1$ and $\text{Re} [\varepsilon(E, q)] \approx 0$. Collective excitations are unique to the condensed phase due to the absence of screening in the gas phase, where $|\varepsilon(E, q)| = 1$. From equation (7) and the above discussion, it follows that the excitation spectrum in the gas phase is solely described by $\text{Im} [\varepsilon_n(E, q)]$. Therefore, the factor $|\varepsilon(E, q)|$ in the denominator

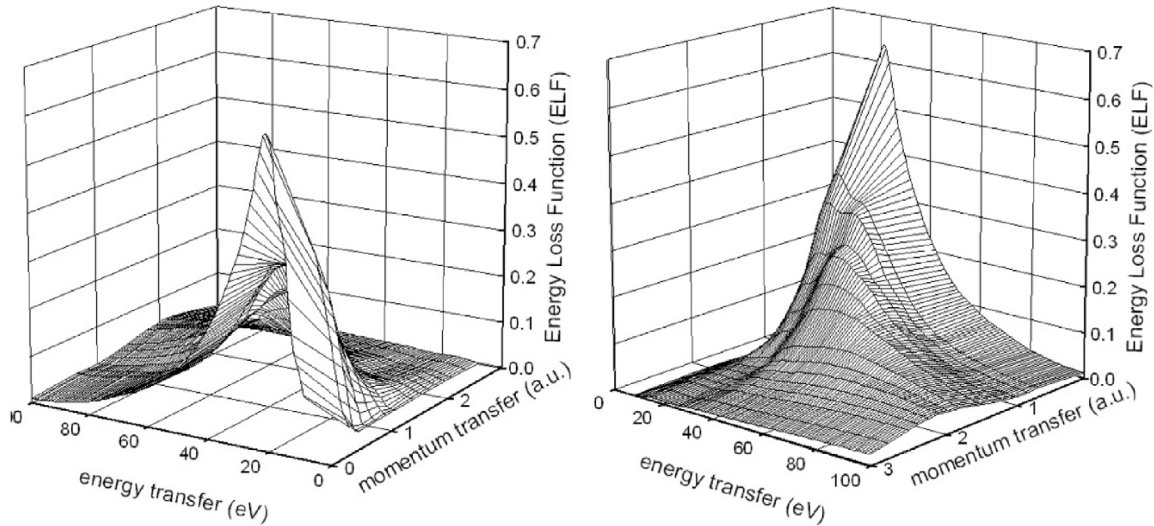


Figure 13. The Bethe surface of liquid water measured by inelastic x-ray scattering spectroscopy [59–61]. The left panel emphasizes the region of small- q excitations, whereas the right panel emphasizes the region of large- q excitations (q is in atomic units; multiplication by the factor 1.99×10^{-24} Å converts to SI units).

of equation (7) is the principal modulator of the excitation spectrum during condensation. In general, one observes that due to $|\varepsilon(E, q)|$ the excitation peaks of the gas phase appear at slightly higher energies in $\text{Im}[\frac{-1}{\varepsilon(E, q)}]$ of the condensed phase and the difference between the two diminishes as the excitation energy increases. As a result, inner-shell ionizations remain, to a good approximation, the same in the gas and condensed phase. The $\text{Im}[\varepsilon_n(E, q)]$ of the gas phase is directly related to the atomic or molecular generalized-oscillator-strength (GOS) through

$$\text{Im}[\varepsilon_n(E, q)] = 8\pi^2 a_0^3 R y^2 N (f_n(q)/E_n), \quad (8)$$

where $f_n(q)$ is the GOS for the n th electronic excitation of the atom or molecule (in the gas phase). Thus, equation (8) offers a reasonable approximation for inner-shell ionizations. By combining equations (7) and (8) the ELF of condensed media is often expressed as

$$\begin{aligned} \text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right] &= \text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right]_{\text{outer}} + \text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right]_{\text{inner}} \\ &= \underbrace{\sum_i \frac{\text{Im}[\varepsilon_i(E, q)]}{|\varepsilon(E, q)|^2}}_{\text{outer-shell electrons}} + \underbrace{\frac{8\pi^2 a_0^3 R y^2 N}{E} \sum_j \frac{df_j(E, q)}{dE}}_{\text{inner-shell electrons}}, \end{aligned} \quad (9)$$

with the outer-shell term being dependent upon the dielectric function, which includes all condensed-phase effects, and the inner-shell term, assumed to be unperturbed from the state of condensation, being calculated from the generalized oscillator strength of isolated atoms.

The q -dependence of the dielectric function originates from the fact that the behavior of electronic excitations may change with q . The dispersion relation $E_q = f(q; E_0)$, with E_0 the excitation energy for $q = 0$, is in general, unknown and varies across the excitation spectrum. In general, the excitation energy E_q increases with q (positive dispersion), because excitations with smaller wavelength (larger momentum transfer)

have a higher energy cost. In fact, for high enough q so that $\frac{q^2}{2m}$ is much larger than the binding energies of the system, the peaks of the ELF that correspond to dispersive excitations (e.g. plasmon-like excitations, e-h pairs) merge along the so-called Bethe ridge, which is defined by the free-electron dispersion line $E \approx \frac{q^2}{2m}$. In the Bethe ridge, the most probable energy transfer is (almost) equal to the recoil energy of a free and stationary electron. For a two-body collision, where the struck electron is free and stationary, the Bethe ridge is a delta function, i.e. $\text{Im}[\frac{-1}{\varepsilon(E, q)}] \propto \delta(E - \frac{q^2}{2m})$. In real systems, the Bethe ridge has a non-zero width (i.e. it is broadened) due to band-structure and exchange-correlation (XC) effects. Note also that, in real materials, many inter-band excitations are non-dispersive (i.e. they are fixed across all the q space) or they have a rather complicated evolution with q that deviates from the free-electron dispersion.

The Bethe surface of liquid water measured by inelastic x-ray scattering spectroscopy is shown in figure 13. At small q (see panel a) the Bethe surface has structures which originate from the various excitation levels and binding energies of liquid water according to equation (9). With increasing q (see panel b) the excitation peaks shift to higher energies (according to their dispersion rate), become broader, and their amplitude decreases. Any realistic model of the dielectric function of liquid water should be able to reproduce these general features.

A theoretical description of the Bethe surface rests on our knowledge of the wave functions of the material before and after an inelastic collision; i.e. the ground- and excited-state wavefunctions. Despite various attempts to determine $\varepsilon(E, q)$ of condensed biological media, and liquid water in particular, by *ab initio* techniques [62–66], the high computational cost needed to carry out electronic structure calculations for disordered many-electron systems (e.g. liquid water) over the whole excitation range and across all momenta, has severely compromised the accuracy of the final results. Thus, even within the simplified framework of the PWBA, the calculation

of inelastic scattering in condensed media from first-principles is far from being practical.

In the next sections, we will examine various models that aim in a tractable, yet realistic, description of the dielectric function (or the energy loss function) of condensed media. Such efforts are facilitated by the fact that the real and imaginary parts of the dielectric function are not independent, but are related through the Kramers–Krönig relations

$$\text{Re}[\varepsilon(E, q)] = 1 + \frac{1}{\pi} P \int_{-\infty}^{\infty} \text{Im}[\varepsilon(E', q)] \frac{dE'}{E' - E}, \quad (10a)$$

$$\text{Im}[\varepsilon(E, q)] = -\frac{1}{\pi} P \int_{-\infty}^{\infty} [\text{Re}[\varepsilon(E', q)] - 1] \frac{dE'}{E' - E}, \quad (10b)$$

where P stands for the Cauchy principal value. Therefore, if $\text{Re}[\varepsilon(E, q)]$ or $\text{Im}[\varepsilon(E, q)]$ is known, then the other can be determined through the Kramers–Krönig relations. On the other hand, a self-consistent model for the dielectric function must fulfill various so-called sum rules, which reflect causality. In the present context, the most important sum rules are the conductivity sum rule

$$\frac{2}{\pi} E_{\text{pl}}^{-2} \int_0^{\infty} E \text{Im}[\varepsilon(E, q)] dE = 1, \quad (11)$$

the Thomas–Reich–Kuhn sum rule (also known as the f -sum rule)

$$\frac{2}{\pi} E_{\text{pl}}^{-2} \int_0^{\infty} E \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] dE = 1, \quad (12)$$

and the Kramers–Krönig sum rule

$$\frac{2}{\pi} \int_0^{\infty} \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] \frac{dE}{E} + \text{Re} \left[\frac{1}{\varepsilon(E=0, q)} \right] = 1. \quad (13)$$

Conductors $\text{Re}[1/\varepsilon(0, 0)] = 0$ and equation (13) become (for $q = 0$) the so-called perfect-screening sum rule.

2.2. Electron gas models

As noted above, the dielectric function of real materials cannot be known exactly. Some very useful models originate from the homogeneous electron gas (HEG) theory, or so-called jellium model of the solid. In the HEG theory, one assumes that the material comprises free (i.e. unbound) electrons that move in a uniform and isotropic background of (positively charged) nuclei, which provide charge neutrality. The HEG theory depends on a single material parameter, the electron density. The HEG can exhibit only intra-band excitations in the form of single-electron excitations (i.e. e-h pairs) as well as plasmon excitations. The latter formally emerge as a solution of the equation

$$\begin{aligned} \varepsilon(E, q) = 0 &\Rightarrow \{\text{Re}[\varepsilon(E_{\text{pl}}(q), q)] = 0 \\ &\wedge \text{Im}[\varepsilon(\Delta E_{\text{pl}}(q), q)] = 0\}, \end{aligned} \quad (14)$$

where E_{pl} and ΔE_{pl} represent the plasmon energy and linewidth (or half-width at half-maximum), respectively. Their q -dependence defines the so-called *plasmon dispersion relations*, $E_{\text{pl}}(q) = f(q; E_{\text{pl}})$ and $\Delta E_{\text{pl}}(q) = f(q; \Delta E_{\text{pl}})$. Note that

the linewidth of an excitation (ΔE) has an inverse relation to its lifetime (τ), i.e. $\Delta E \propto \tau^{-1}$. An in-depth discussion of plasmons can be found in the classical book of Pines [67]. On the other hand, single-electron excitations emerge, when $\varepsilon_2(E, q) \neq 0$, and disperse according to the free-electron relation, $E \approx \frac{q^2}{2m}$, with a zero linewidth.

Interestingly, even for a model as simple as the HEG, the dielectric function is intractable, unless further approximations are made. The plasmon energy of the HEG at the optical limit ($q \rightarrow 0$), however, is known exactly

$$E_{\text{pl}} = \hbar \omega_{\text{pl}} = \sqrt{4\pi \hbar^2 e^2 N Z / m}, \quad (15)$$

where ω_{pl} denotes the oscillation frequency of the free-electron density and Z the number of electrons per atom or molecule. At $q \rightarrow 0$ the plasmon exhausts the f -sum-rule because single-electron excitations are not possible in the HEG at this limit.

Electronic excitations in condensed media have mixed-properties that range between those of single-electron and many-electron excitations. Moreover, excitations in real materials are both intra- and inter-band. The latter may be non-dispersive or show little dispersion with q . Thus, the HEG works best for the excitation properties of the conduction electrons of metals. However, it has also been found very useful for the weakly-bound (outer-shell) electrons of materials other than simple metals, as long as their ELF at $q \approx 0$ exhibits a well-defined peak at about the nominal plasmon energy given by equation (15). Interestingly, many biological materials (e.g. liquid water, DNA) fulfill the above condition despite the fact that they are wide band-gap insulators. A simple explanation is that for these materials, the condition $E_{\text{pl}} \gg E_{\text{gap}}$ holds and, consequently, many of their electronic excitation properties can be reasonably well understood based on the HEG theory.

In what follows, we will present various approximations to the dielectric function of the HEG that are useful in constructing a realistic model of the Bethe surface of arbitrary condensed media (e.g. biologically interesting materials like liquid water and polymers).

2.2.1. Drude model. The Drude model for metals assumes that the target electrons are set in a harmonic oscillation of frequency $\omega_{\text{pl}} = E_{\text{pl}}/\hbar$ under the influence of a spatially uniform electric field of frequency $\omega = E/\hbar$ induced by an external perturbation (e.g. incoming charged projectile). Assuming a damped oscillation due to the various dissipative processes that can take place in real materials (e.g. interaction of the excited electrons with imperfections or impurities) the dielectric function of the Drude model reads

$$\varepsilon_{\text{D}}(E; E_{\text{pl}}, \gamma) = 1 - \frac{E_{\text{pl}}^2}{E^2 + iE\gamma}, \quad (16)$$

where γ is the damping constant. The ELF of the damped Drude model is

$$\text{Im} \left[\frac{-1}{\varepsilon_{\text{D}}(E; E_{\text{pl}}, \gamma)} \right] = \frac{E_{\text{pl}}^2 \gamma E}{(E_{\text{pl}}^2 - E^2)^2 + (E\gamma)^2}, \quad (17)$$

with γ being approximately equal to the full-width-at-half-maximum of the plasmon peak. For small γ the peak in the ELF of equation (17) appears approximately at the energy E_{pl} . In fact, in the limit of zero damping the ELF of the Drude model is

$$\lim_{\gamma \rightarrow 0} \text{Im} \left[\frac{-1}{\varepsilon_{\text{D}}(E; E_{\text{pl}}, \gamma)} \right] = \frac{\pi}{2} E_{\text{pl}} \delta(E - E_{\text{pl}}). \quad (18)$$

which yields a plasmon peak of zero width due to the absence of any dissipative processes.

The Drude model can also be applied to insulators (and semiconductors) by assuming that a harmonic binding force of frequency ω_b acts upon the oscillatory motion of the excited electrons. In this case, the dielectric function of equation (16) takes the form

$$\varepsilon_{\text{D}}(E; E_{\text{pl}}, \gamma) = 1 + \frac{E_{\text{pl}}^2}{E_b^2 - E^2 - iE\gamma}, \quad (19)$$

where $E_b = \hbar\omega_b$. At high enough excitation energies ($E \gg E_b$), equation (19) reduces to the dielectric function, equation (16), of the standard Drude model for metals. The ELF of the dielectric function of equation (19) does not have the simple analytic form of equation (17), but must be calculated from $\text{Im}(-\frac{1}{\varepsilon})$ with ε being the $\varepsilon_{\text{D}}(E; E_{\text{pl}}, \gamma)$ of equation (19). The peak in the ELF of equation (19) does not appear at E_{pl} but (for small γ) it is shifted to higher energies according to $E_{\text{peak}} \approx \sqrt{E_{\text{pl}}^2 + E_b^2}$.

It can be shown that the Drude model exactly satisfies the sum rules, equations (11)–(13), independently of the value of its model parameters (E_{pl} , γ , E_b). In the present context, an important limitation of the Drude model is that it is not dispersive, i.e. it neglects the q -dependence of the dielectric function. Nonetheless, the Drude model coincides with the exact dielectric function of the HEG at $q = 0$ (where only plasmon excitations are possible) under the so-called random-phase approximation (RPA).

2.2.2. Lindhard model. An exact solution for the dielectric function of the HEG at $q \neq 0$ is only possible within the RPA, which assumes that each screening electron interacts with the *average* field generated by all other screening electrons. The dielectric function of the HEG within the RPA was first derived by Lindhard [68]

$$\varepsilon_{\text{L}}(E, q; E_{\text{pl}}) = 1 + (\chi^2/z^2)[f_1(x, z) + i f_2(x, z)]. \quad (20)$$

where

$$f_1(x, z) = \frac{1}{2} + \frac{1}{8z} \{1 - (z - x/4z)^2\} \ln \left| \frac{z - x/4z + 1}{z - x/4z - 1} \right| + \frac{1}{8z} \{1 - (z + x/4z)^2\} \ln \left| \frac{z + x/4z + 1}{z + x/4z - 1} \right| \quad (21a)$$

$$f_2(x, z) = \begin{cases} \pi x/8z, & 0 < x < 4z(1 - z) \\ (\pi/8z)\{1 - (z - x/4z)^2\}, & 4z(z - 1) < x < 4z(z + 1) \text{ and } x > 4z(1 - z) \\ \approx 0, & \text{otherwise} \end{cases} \quad (21b)$$

with $z = q/2q_{\text{F}}$, $x = E/E_{\text{F}}$, and $\chi^2 = e^2/\pi\hbar v_{\text{F}}$; v_{F} , q_{F} , E_{F} being the Fermi velocity, momentum, and energy, respectively. It can be shown that the Lindhard dielectric function also fulfills the sum rules at all q . From equation (14) the plasmon dispersion $E_{\text{pl}}^{\text{RPA}}(q)$ of the Lindhard model can be found as a solution of $\varepsilon_{\text{L}}(E, q; E_{\text{pl}}) = 0$. An *analytic* solution for $E_{\text{pl}}^{\text{RPA}}(q)$ is possible only for small q

$$E_{\text{pl}}^{\text{RPA}}(q) = E_{\text{pl}} + \alpha_{\text{RPA}} \frac{q^2}{2m} + O(q^4), \quad (22)$$

where the (dimensionless) dispersion coefficient is given by

$$\alpha_{\text{RPA}} = \frac{6}{5} \frac{E_{\text{F}}}{E_{\text{pl}}}. \quad (23)$$

At the optical limit, the Lindhard model reduces to the *dampless* Drude model

$$\lim_{q \rightarrow 0} \text{Im} \left[\frac{-1}{\varepsilon_{\text{L}}(E, q; E_{\text{pl}})} \right] = \lim_{\gamma \rightarrow 0} \text{Im} \left[\frac{-1}{\varepsilon_{\text{D}}(E; E_{\text{pl}}, \gamma)} \right] = \frac{\pi}{2} E_{\text{pl}} \delta(E - E_{\text{pl}}). \quad (24)$$

Thus, the Lindhard model provides an automatic extension for the Drude model to $q \neq 0$ (in a manner fully consistent with the sum rules) valid within the RPA. Note that whereas at $q = 0$ only plasmon excitations are possible (as also predicted by the Drude model), both single-electron and plasmon excitations are permitted in the Lindhard model at $q \neq 0$. An important outcome of the Lindhard model is that although plasmons are dampless (i.e. have infinite lifetime), there is a critical momentum transfer (q_c) above which they suddenly become strongly damped (practically disappear) by decaying to single-electron excitations. This is because for $q > q_c$ the plasmon energy falls within the energy spectrum of single-electron excitations and, therefore, energy-momentum conservation allows all the plasmon energy to be transferred to an excited electron. This is the so-called Landau damping mechanism.

2.2.3. Mermin model. A shortcoming of the Lindhard model is the absence of any plasmon damping mechanism other than the Landau damping effective at $q > q_c$. A dampless plasmon manifests as a delta-like peak in the Bethe surface. Clearly, an infinite plasmon lifetime is an oversimplification. Real materials exhibit plasmon peaks of finite width at any q due to various dissipation processes that cause plasmon decay, such as, for example, collisions among the excited electrons or collisions of the excited electrons with the nuclei. A damping-inclusive dielectric function within the RPA has been advanced by Mermin [69]. The Mermin dielectric function involves a combination of terms of the Lindhard dielectric function of complex energy ($\tilde{E} = E + i\gamma$) in the following manner

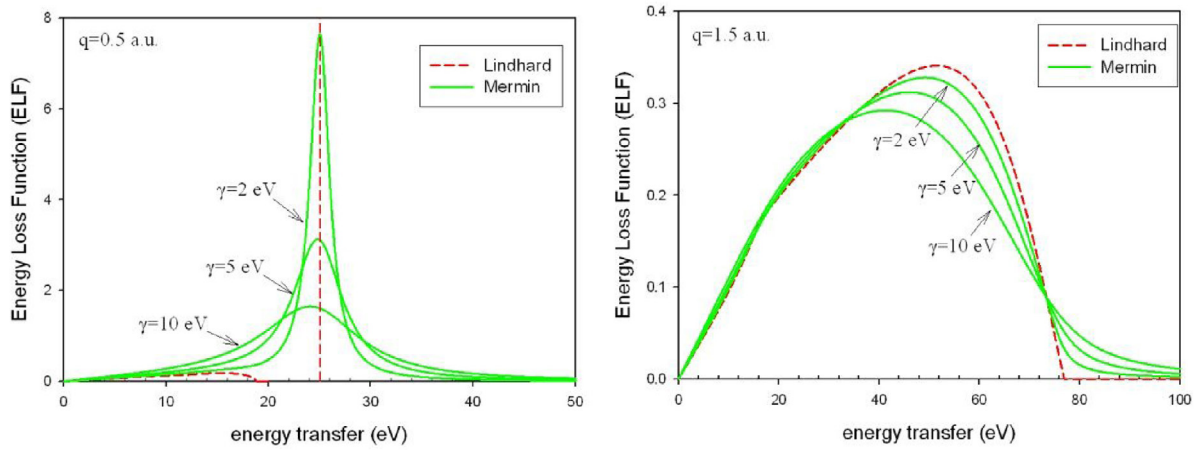


Figure 14. The ELF of the Lindhard (equation (21)) and Mermin (equation (25)) models of the dielectric response function of an electron gas with density equal to that of liquid water ($3.34 \text{ \AA}^{-3} \sim 10^{23} \text{ e}^- \text{ cm}^{-3}$). Left panel: The momentum transfer ($q = 0.5 \text{ a.u.}$) is below the Landau cut-off; Right panel: The momentum transfer ($q = 1.5 \text{ a.u.}$) is above the Landau cut-off. In both panels, the Mermin model is shown for three different values of the damping coefficient γ .

$$\begin{aligned} \epsilon_M(E, q; E_{pl}, \gamma) \\ = 1 + \frac{(1 + i\gamma/E)[\epsilon_L(\tilde{E}, q; E_{pl}) - 1]}{1 + (i\gamma/E)[\epsilon_L(\tilde{E}, q; E_{pl}) - 1]/[\epsilon_L(E = 0, q; E_{pl}) - 1]}, \end{aligned} \quad (25)$$

where γ is a damping constant, the value of which is not given by the model (as is also the case in the Drude model) and has to be evaluated by other means. Note that although the damping constant (γ) is fixed across the momentum space, the analytic properties of the Mermin dielectric function ensure that the width of the plasmon peak increases with q (i.e. it predicts linewidth dispersion) yielding a momentum broadening effect in ELF. By construction, the Mermin dielectric function fulfils the sum rules at all q and for all values of γ . The following limiting properties of the Mermin dielectric function are of interest in the present context. First, at the limit of zero damping ($\gamma \rightarrow 0$) the Mermin model recovers the Lindhard model

$$\lim_{\gamma \rightarrow 0} \epsilon_M(E, q; E_{pl}, \gamma) = \epsilon_L(E, q; E_{pl}). \quad (26)$$

Second, the Mermin model reduces to the damped Drude model at the optical limit ($q \rightarrow 0$):

$$\begin{aligned} \lim_{q \rightarrow 0} \text{Im} \left[\frac{-1}{\epsilon_M(E, q; E_{pl}, \gamma)} \right] &= \text{Im} \left[\frac{-1}{\epsilon_D(E; E_{pl}, \gamma)} \right] \\ &= \frac{E_{pl}^2 \gamma E}{(E_{pl}^2 - E^2)^2 + (\gamma E)^2}. \end{aligned} \quad (27)$$

A comparison of the Lindhard and Mermin models for the ELF of the HEG with an electronic density equal to liquid water is presented in figure 14. The ELF is plotted for two different values of q that fall below (panel a) and above (panel b) the Landau cut-off. The Mermin model is shown for three different values of the damping coefficient (γ). Clearly, the higher the value of γ , the broader the plasmon peak. At small- q , below the Landau cut-off (see panel a), the plasmon peak and the single-electron continuum in the Lindhard model are well separated by an energy gap. This energy gap is responsible for the delta-like plasmon peak indicative of a dampless

excitation with an infinite lifetime. In contrast, in the Mermin model, the single-electron continuum is merged with the plasmon peak, resulting in a damped plasmon of a finite lifetime that decreases with γ . At high- q , above the Landau cut-off (see panel b), the Landau damping mechanism is dominant, resulting in the absence of a well-defined plasmon peak. In this case, the excitation spectrum features a single and broad peak that represents the single-electron continuum. Differences between the Lindhard and Mermin models at high- q (above the Landau cut-off) are small.

2.2.4. Plasmon pole model. A simplification of the RPA dielectric function is offered by the so-called plasmon-pole model [70]. In the plasmon-pole model, it is assumed that the excitation spectrum of the HEG is collapsed into a single peak along the plasmon dispersion line. The dielectric function of the plasmon-pole model reads

$$\epsilon_{PP}(E, q; E_{pl}, \gamma) = 1 + \frac{E_{pl}^2}{E_{pl}^2(q) - E_{pl}^2 - E^2 - iE\gamma(q)}, \quad (28)$$

with the corresponding ELF being

$$\text{Im} \left[\frac{-1}{\epsilon_{PP}(E, q; E_{pl}, \gamma)} \right] = \frac{E_{pl}^2 \gamma(q) E}{(E_{pl}^2(q) - E^2)^2 + (E\gamma(q))^2}. \quad (29)$$

The plasmon-pole model of the HEG can be extended to insulators (and semiconductors) by introducing an ‘effective’ bandgap energy (E_g). Then, the dielectric function of the plasmon-pole model becomes

$$\epsilon_{PP}(E, q; E_{pl}, \gamma) = 1 + \frac{E_{pl}^2}{\beta^2(q) + E_g^2 - E^2 - iE\gamma(q)}, \quad (30)$$

where $\beta^2(q) = E_{pl}^2(q) - E_{pl}^2$. At the optical limit equation (30) reduces to the dielectric function equation (19) of the Drude model with $E_b = E_g$. It then follows that the peak in the ELF of equation (30) is shifted from E_{pl} to higher values according to $E_{\text{peak}} \approx \sqrt{E_{pl}^2 + E_g^2}$.

It is trivial to confirm that the plasmon-pole model reduces to the Drude model at the optical limit; thus, the plasmon-pole model is often referred to in the literature as an extended-Drude model. An important difference between the plasmon-pole model and the Lindhard or Mermin models, is that the extension of the plasmon-pole model to $q \neq 0$ is not done automatically, but has to be implemented manually through appropriate dispersion relations for $E_{\text{pl}}(q)$ and/or $\gamma(q)$. Consistent with the RPA, one could use $E_{\text{pl}}^{\text{RPA}}(q)$ from equation (22) and a q -independent damping constant, i.e. $\gamma(q) = \gamma(0)$, that is fixed across the momentum space. An important property of the plasmon-pole model that facilitates the search for suitable dispersion relations is that, owing to its relation to the Drude model, it fulfills the sum rules at all q independent of the particular form of dispersion that is at our disposal.

2.2.5. Local-field corrections. The RPA is a mean-field theory that accounts only for long-range correlation in the motion of the screening electrons. To go beyond RPA, one needs to consider also the short-range interactions among the screening electrons, which arise from the instantaneous Coulomb repulsion and the exchange interaction due to the Pauli principle, the so-called XC effects. As first shown by Hubbard [71] XC effects can be accounted for through the

Thus, the real part of the LFC causes a reduction in the plasmon energy from its RPA value. Likewise, the effect of XC on the dispersion of the plasmon linewidth is

$$\Delta E_{\text{pl}}^{\text{XC}}(q) = \underbrace{\Delta E_{\text{pl}}^{\text{RPA}}(q)}_{=0} + \Delta E_{\text{pl}}^{\text{LFC}}(q), \quad (35)$$

where $\Delta E_{\text{pl}}^{\text{RPA}}(q) = 0$ and

$$\Delta E_{\text{pl}}^{\text{LFC}}(q) = \frac{1}{2} E_{\text{pl}} q^2 \lim_{q \rightarrow 0} \left[\frac{G_2(E = E_{\text{pl}}, q)}{q^2} \right]. \quad (36)$$

Thus, the imaginary part of the LFC gives rise to plasmon damping, which is absent in the RPA. The mechanism of plasmon decay at small q (below the Landau cut-off) is via multi-pair e-h excitations, which arise due to the XC interactions among the excited electrons (which are not allowed within the RPA).

Most calculations of the LFC are carried out in the static limit, which implies that $E \rightarrow 0$ or, more commonly, that the E -dependence of the LFC is ignored. In the static limit, the imaginary part vanishes and the LFC becomes purely real

$$\lim_{E \rightarrow 0} G(E, q) = \lim_{E \rightarrow 0} G_1(E, q) \equiv G(q). \quad (37)$$

In the static limit, the ELF of equation (32) becomes

$$\text{Im} \left[\frac{-1}{\varepsilon_{\text{t-t}}(E, q)} \right]_{\text{SLFC}} = \frac{\text{Im}[\varepsilon_{\text{RPA}}(E, q; E_{\text{pl}})]}{\{1 + [1 - G(q)][\text{Re}[\varepsilon_{\text{RPA}}(E, q; E_{\text{pl}})] - 1]\}^2 + \{[1 - G(q)]\text{Im}[\varepsilon_{\text{RPA}}(E, q; E_{\text{pl}})]\}^2} \quad (38)$$

many-body local-field correction (LFC) yielding the so-called test-charge-test-charge (t-t) dielectric function

$$\varepsilon_{\text{t-t}}(E, q) = 1 + \frac{\varepsilon_{\text{RPA}}(E, q; E_{\text{pl}}) - 1}{1 - G(E, q)[\varepsilon_{\text{RPA}}(E, q; E_{\text{pl}}) - 1]}, \quad (31)$$

where $G(E, q) = G_1(E, q) + iG_2(E, q)$ is the LFC, which, in general, depends upon both the energy- and momentum-transfer. It is trivial to confirm from equation (31) that the RPA corresponds to the condition $G(E, q) = 0$. The t-t dielectric function, $\varepsilon_{\text{t-t}}(E, q)$, describes ‘exactly’ the screening of the HEG to two ‘test’ charges. The corresponding ELF reads

$$\text{Im} \left[\frac{-1}{\varepsilon_{\text{t-t}}(E, q)} \right] = \text{Im} \left\{ \frac{\varepsilon_{\text{RPA}}(E, q; E_{\text{pl}}) - 1}{\varepsilon_{\text{RPA}}(E, q) - G(E, q)[\varepsilon_{\text{RPA}}(E, q; E_{\text{pl}}) - 1]} \right\}. \quad (32)$$

The effect of XC on the dispersion relation of the plasmon energy can be expressed as

$$E_{\text{pl}}^{\text{XC}}(q) = E_{\text{pl}}^{\text{RPA}}(q) + E_{\text{pl}}^{\text{LFC}}(q), \quad (33)$$

where $E_{\text{pl}}^{\text{RPA}}(q)$ is obtained from equation (22) and $E_{\text{pl}}^{\text{LFC}}(q)$ by

$$E_{\text{pl}}^{\text{LFC}}(q) = -\frac{1}{2} E_{\text{pl}} q^2 \lim_{q \rightarrow 0} \left[\frac{G_1(E = E_{\text{pl}}, q)}{q^2} \right]. \quad (34)$$

where SLFC stands for the static-LFC. In the static limit, equation (34) reads

$$E_{\text{pl}}^{\text{SLFC}}(q) = -\frac{1}{2} E_{\text{pl}} q^2 \lim_{q \rightarrow 0} \left[\frac{G(q)}{q^2} \right], \quad (39)$$

where

$$\lim_{q \rightarrow 0} G(q) = A(q/q_F)^2, \quad (40)$$

with A being a positive dimensionless parameter that depends on the electronic density. Thus, in the static limit, the effect of XC on the dispersion relation of the plasmon energy can be written in the following compact form

$$E_{\text{pl}}^{\text{XC}}(q) = E_{\text{pl}} + (\alpha_{\text{RPA}} - \alpha_{\text{SLFC}}) \frac{q^2}{2m}, \quad (41)$$

with α_{RPA} obtained from equation (23) and

$$\alpha_{\text{SLFC}} = \left(\frac{1}{2} \frac{E_{\text{pl}}}{E_F} \right) A. \quad (42)$$

Since, in the static limit, the imaginary part of LFC vanishes, it follows from equation (36) that $\Delta E_{\text{pl}}^{\text{SLFC}}(q) = 0$.

The simplest form of the SLFC has been suggested by Hubbard [71]. In its general form it can be expressed as

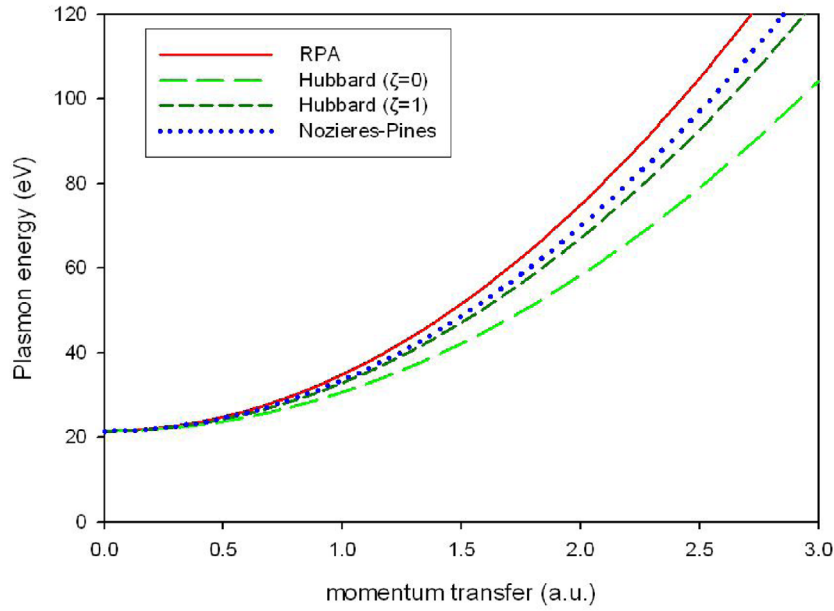


Figure 15. The dispersion relation for the plasmon energy is calculated in the RPA (equation (22)) and with the inclusion of XC effects (equation (41)) using the Hubbard (for two different values of the parameter ζ) and Nozieres–Pines SLFC. The dispersion coefficients α_{RPA} and α_{SLFC} are calculated using the electron density of liquid water ($3.34 \times 10^{23} \text{ e}^- \text{ cm}^{-3}$).

$$G_{\text{Hubbard}}(q) = \frac{q^2}{2(q^2 + q_F^2 + \zeta q_{\text{TF}}^2)}. \quad (43)$$

with $q_{\text{TF}}^2 = \frac{4q_F}{\pi a_0}$ being the Thomas–Fermi inverse screening length. The value of ζ varies between zero and one (with $\zeta = 0$ we recover the original Hubbard SLFC). Inserting equation (43) with $\zeta = 0$ in equation (40) yields $A_{\text{Hubbard}} = \frac{1}{2}$. Another popular form for the SLFC is that of Nozieres and Pines [72] which leads to $A_{\text{N-P}} = \frac{3}{20}$. Most available expressions of the SLFC are either interpolations between its known asymptotic limits or parameterizations of numerical self-consistent calculations and quantum Monte Carlo simulations. It turns out that, for inelastic electron scattering, simple Hubbard-like forms of the SLFC yield similar results to more sophisticated models [73]. In figure 15 we present the effect of XC on the RPA plasmon dispersion using the Hubbard ($\zeta = 0$ and $\zeta = 1$) and Nozieres–Pines SLFCs. Although the various dispersions are still parabolic, as predicted by the RPA, in all cases, the effect of XC is to reduce the plasmon energy from its RPA value (i.e. the plasmon dispersion line is shifted downwards).

On the other hand, since $G(q = 0) = 0$, all models reduce to the RPA values at $q = 0$, which leads to the important conclusion that XC effects vanish at the optical limit. As a result, XC effects cannot explain the experimentally observed finite width of the excitation peaks at $q \approx 0$. Additional damping mechanisms, such as, for example, the collision of excited electrons with nuclei (or impurities), are needed to yield a finite excitation lifetime. Such collisions, which formally describe phonon-assisted damping, can be accounted for by including a finite damping constant γ , as in the Mermin model.

2.3. Optical-data models

Semi-empirical descriptions of the Bethe surface take advantage of the often-available experimental data for the dielectric function at the optical limit ($q \approx 0$). The use of optical data for inelastic scattering calculations in solids was first suggested, independently, by Powell [74, 75], and Howie and Stern [76]. This idea led to the development of optical-data models, whereby the dependence of the dielectric function on energy transfer (at $q = 0$) is determined from experimental data and the dependence on momentum transfer (for extrapolating the optical data to $q \neq 0$) is provided through theoretically motivated so-called extension algorithms [77, 78]. The simplest kind of an extension algorithm is a dispersion relation. Extension algorithms are commonly based on the HEG theory.

Optical-data models currently represent the state-of-the-art for inelastic-scattering calculations in solids for ‘low-energy’ electrons, below $\sim 10 \text{ keV}$ [79, 80]. The most important advantage of optical-data models is that they can be applied to any material (metal, semiconductor or insulator) and they are capable of providing reliable results over a wide energy range. Importantly, the use of experimental data ensures that the excitation properties of the target are considered in a material-specific manner. This requirement is particularly important for low-energy electrons with energies comparable (within a factor of ~ 10) to the excitation energies of the material. Although the use of optical-data models is restricted to materials where experimental optical data are available, methodologies to overcome this restriction have recently been presented using generic optical-data models [81].

The most common source of optical data is measurements of the optical constants of the material as a function of photon energy. The optical constants relate to the dielectric function through the relations

$$\varepsilon(E, 0) = (n(E) + i\kappa(E))^2 \Rightarrow \begin{cases} \text{Re}[\varepsilon(E, 0)] = n(E)^2 - \kappa(E)^2, \\ \text{Im}[\varepsilon(E, 0)] = 2n(E)\kappa(E) \end{cases}, \quad (44)$$

where $n(E)$ and $\kappa(E)$ are the index of refraction and extinction coefficient, respectively, as a function of photon energy. From measurements of the optical constants, and using equation (44), we can then determine the ‘experimental’ ELF at $q = 0$ from

$$\text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E, q=0)}\right] = \text{Im}\left[\frac{-1}{(n(E) + i\kappa(E))^2}\right] = \frac{2n(E)\kappa(E)}{(n(E)^2 - \kappa(E)^2)^2 + (2n(E)\kappa(E))^2}. \quad (45)$$

The optical constants $n(E)$ and $\kappa(E)$ of liquid water for energy transfer in the range 7.6–25.6 eV have been measured at Oak Ridge by Heller *et al* [82]; these 30 year old data have been used as input in most optical-data models for liquid water [83]. A more complete optical data set for liquid water that extends to an energy transfer up to 160 eV has been provided by Hayashi *et al* [61]. In this study, inelastic x-ray scattering spectroscopy is used to measure the GOS of liquid water at nearly zero q . The ELF was then obtained from

$$\text{Im}\left[\frac{-1}{\varepsilon(E, q=0)}\right]_{\text{exp}} = \frac{\pi E_{\text{pl}}^2}{2} \frac{1}{E} \left(\frac{df(E, q=0)}{dE} \right)_{\text{exp}}, \quad (46)$$

where $df(E, q=0)/dE$ is the GOS of liquid water at the optical limit; also known as optical-oscillator-strength. Equation (46) is essentially a generalization of equation (8) to condensed media. Note also that the GOS of equation (46), which corresponds to the condensed phase, is defined *per electron* (also called ‘one-electron’ GOS) and not per atom or molecule as it is used for gas-phase systems.

2.3.1. Extended-Drude model. The most represented optical-data model in the literature of liquid water is the extended-Drude model advanced by Ritchie and co-workers and reviewed in Ritchie *et al* [84]. The extended-Drude model of the dielectric function generalizes the plasmon-pole model of the HEG to real materials, which have a more complex excitation spectrum and, therefore, a single set of model parameters (E_{pl} , γ) that cannot faithfully reproduce their excitation spectrum.

The simplest version of the extended-Drude model [85] describes the ELF of an arbitrary material as a superposition of plasmon-pole ELF's:

$$\text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right]_{\text{Ritchie-Howie}} = \sum_n \frac{A_n}{E_n^2} \text{Im}\left[\frac{-1}{\varepsilon_{\text{pp}}(E, q; E_n, \gamma_n)}\right] = \sum_n \frac{A_n \gamma_n(q) E}{(E_n^2(q) - E^2)^2 + (E \gamma_n(q))^2}. \quad (47)$$

The model parameters $\{A_n, E_n, \gamma_n\}$, related to the height, position, and width, respectively, of the n th excitation peak in the ELF are obtained from fitting the experimental optical data through

$$\text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E, q \approx 0)}\right] \approx \text{Im}\left[\frac{-1}{\varepsilon(E, q=0)}\right]_{\text{Ritchie-Howie}} = \sum_n \frac{A_n \gamma_n E}{(E_n^2 - E^2)^2 + (E \gamma_n)^2}, \quad (48)$$

where $E_n \equiv E_n(q=0)$ and $\gamma_n \equiv \gamma_n(q=0)$. For the model to be self-consistent, the fitting procedure must be carried out under the constraints imposed by the f - and Kramers–Krönig sum rules. The extension of the Ritchie–Howie model to $q \neq 0$ is carried out via the dispersion relations $E_n(q) = f(q; E_n)$ and $\gamma_n(q) = f(q; \gamma_n)$. The most common approach is to assume that the damping constant is independent of q , and fixed across all momenta at the fitting value γ_n , and approximate $E_n(q)$ by the RPA expression of equation (22) to the second order with $\alpha_{\text{RPA}} \approx 1$. More generally, for materials where experimental data at $q \neq 0$ are available, one can treat the quadratic dispersion coefficient in equation (22) as an adjustable parameter and take a q -dependent damping coefficient, $\gamma_n(q)$ [86, 87]

$$E_n(q) = E_n + \alpha_{\text{emp}} \frac{q^2}{2m}, \quad (49)$$

and

$$\gamma_n(q) = \gamma_n(1 + \beta_{\text{emp}} q^2), \quad (50)$$

where the empirical dispersion coefficients α_{emp} and β_{emp} are determined from experiments. In the most used implementation of the Ritchie–Howie model one commonly assumes $\alpha_{\text{emp}} = 1$ and $\beta_{\text{emp}} = 0$.

An established alternative to the Ritchie–Howie model for constructing the ELF of an arbitrary material from its optical data, is the Oak Ridge model of the dielectric function [84], which is integrated in the OREC (Oak Ridge Electron Code) track-structure code. In the OREC model, the ‘effective’ band-gap energy (E_g) of the plasmon-pole insulator model, equation (30), is replaced by a finite number of excitation energies, E_n , with their associated oscillator strength f_n , leading to the expression

$$\varepsilon_{\text{OREC}}(E, q) = 1 + \sum_n f_n(q) \frac{E_{\text{pl}}^2}{E_n^2(q) - E^2 - iE\gamma_n(q)}, \quad (51)$$

where $E_n^2(q) = E_n^2(0) + \beta^2(q)$, in accord with equation (30). The model parameters $\{f_n, E_n, \gamma_n\}$ are obtained from fitting the available optical data for the imaginary part of the dielectric function using the expression

$$\text{Im}[\varepsilon_{\text{exp}}(E, q \approx 0)] \approx \text{Im}[\varepsilon_{\text{OREC}}(E, q=0)] = E_{\text{pl}}^2 \sum_n D(E; E_n), \quad (52)$$

with

$$D(E; E_n) = \frac{f_n E \gamma_n}{(E_n^2 - E^2)^2 + (\gamma_n E)^2}, \quad (53)$$

where D stands for ‘Drude’ -type function. Using the Kramers–Krönig relation (equation (10)) the real part of the dielectric function can also be obtained from the optical data using the expression

$$\begin{aligned} \text{Re}[\varepsilon_{\text{exp}}(E, q \approx 0)] &\approx \text{Re}[\varepsilon_{\text{OREC}}(E, q = 0)] \\ &= 1 + E_{\text{pl}}^2 \sum_n D_{\text{KK}}(E; E_n), \end{aligned} \quad (54)$$

where

$$D_{\text{KK}}(E; E_n) = \frac{f_n(E_n^2 - E^2)}{(E_n^2 - E^2)^2 + (\gamma_n E)^2}. \quad (55)$$

To preserve self-consistency, the fitting of the optical data with equations (50) and (52) must be carried out under the constraints imposed by the sum rules, equations (11)–(13).

A problem with the Ritchie–Howie and OREC models is that both $\text{Im}[-1/\varepsilon(E)]_{\text{Ritchie-Howie}}$ and $\text{Im}[\varepsilon_{\text{OREC}}(E)]$ diminish asymptotically as E^{-3} contrary to the theoretically expected $E^{-4.5}$. To improve upon the asymptotic properties of the OREC model, Ritchie and co-workers [84] have suggested the replacement of the Drude function (D) in equations (52) and (53) by the so-called *derivative* Drude function (denoted here by $\tilde{D}$)

$$\tilde{D}(E; E_n) = \frac{2f_n(E\gamma_n)^3}{[(E_n^2 - E^2)^2 + (E\gamma_n)^2]^2}. \quad (56)$$

The derivative Drude function falls as E^{-5} so it better complies with the expected asymptotic trend ($E^{-4.5}$). Then, the function D_{KK} in equations (54) and (55) is replaced by the Kramers–Krönig pair of the derivative Drude function

$$\tilde{D}_{\text{KK}}(E; E_n) = \frac{f_n(E_n^2 - E^2)[(E_n^2 - E^2)^2 + 3E^2\gamma_n^2]}{[(E_n^2 - E^2)^2 + (E\gamma_n)^2]^2}. \quad (57)$$

Extension of the OREC model to $q \neq 0$ is carried out through the following dispersion relations

$$f_k(q) = f_k[\exp(-a_k q^2) + b_k q^2 \exp(-c_k q^2)], \quad (58)$$

where a_k , b_k , and c_k are empirical parameters,

$$f_n(q) = f_n \left[\left(1 - \sum_k f_k(q) \right) / \left(1 - \sum_k f_k \right) \right], \quad (59)$$

and

$$E_n(q) = E_n + \frac{q^2}{2m}, \quad (60)$$

with n , k running over the ionization shells and discrete excitation levels, respectively, of liquid water. The above dispersion relations provide a reasonable approximation at high q , where $f_k(q \rightarrow \infty) = 0$, $f_n(q \rightarrow \infty) = f_n / \sum_n f_n$, and $E_n(q \rightarrow \infty) = q^2/2m$, enabling the OREC model to reproduce the global characteristics of the Bethe surface of liquid water in a satisfactory manner [88].

In the latest version, called NOREC (NIST-modified OREC), the damping coefficient has become q -dependent through [89]

$$\gamma_n(q) = \gamma_n + R\gamma q, \quad (61)$$

which is a simple representation for introducing momentum-broadening effects due the various dissipative processes that are neglected in the OREC model.

The OREC/NOREC models are based on the optical data of Heller *et al* [82]. Subsequent improvements by Emfietzoglou *et al* [90] and Dingfelder [91] have taken advantage of the more recent optical data of Hayashi *et al* [61]. Specifically, for describing the optical data for the imaginary part of the dielectric function at $q = 0$, a more efficient algorithm has been used, which employs ordinary Drude functions for ionizations and derivative Drude functions for excitations in the following manner

$$\text{Im}[\varepsilon(E, q = 0)] = E_{\text{pl}}^2 \left\{ \sum_n^{\text{ion}} D(E; E_n) + \sum_k^{\text{exc}} \tilde{D}(E; E_k) \right\} \quad (62)$$

where the functions D and $\tilde{D}$ are given, respectively, by equations (53) and (56). Since $\tilde{D}$ is more sharply peaked than D , it is more suitable for fitting the discrete excitation peaks of the absorption spectrum. From equation (62) and the Kramers–Krönig relation (equation (10)) the real part of the dielectric function can be expressed as

$$\text{Re}[\varepsilon(E, q = 0)] = 1 + E_{\text{pl}}^2 \left\{ \sum_n^{\text{ion}} D_{\text{KK}}(E; E_n) + \sum_k^{\text{exc}} \tilde{D}_{\text{KK}}(E; E_k) \right\}. \quad (63)$$

where the functions D_{KK} and $\tilde{D}_{\text{KK}}$ are given, respectively, by equations (55) and (57). In addition, further refinements are added to account for the vanishing contribution of $\text{Im}[\varepsilon_n(E, q)]$ below the binding energies of the corresponding shells, i.e. for the condition $\text{Im}[\varepsilon_n(E, q)] = 0$ for $E < B_n$.

Perhaps the most important improvement to the OREC/NOREC model is related to its extension to $q \neq 0$. From an analysis of more recent experimental data [59, 60] for the ELF of liquid water at $q \neq 0$, Emfietzoglou *et al* [90] proposed new dispersion relations that account for XC effects in a phenomenological manner. The Emfietzoglou–Cucinotta–Nikjoo (ECN) [90] dispersion relations read (with q in atomic units)

$$E_n(q) = E_n + \alpha_{\text{emp}}(q) R\gamma q^2, \quad (64)$$

where $\alpha_{\text{emp}}(q) = 1 - \exp(-cq^d)$ and

$$\gamma_n(q) = \gamma_n + aq + bq^2, \quad (65)$$

with the value of the a , b , c , d coefficients determined empirically through a fit to the experimental data for liquid water at $q \neq 0$. Thus, equations (64) and (65) are material-specific. The effect of the dispersion coefficient $\alpha_{\text{emp}}(q)$ in equation (64) is to reduce $E_n(q)$ at not too large q , thus, shifting the ELF to lower excitation energies in accordance with the effect of α_{SLFC} (equation (41)). Note that the particular form for the q -dependence of $\alpha_{\text{emp}}(q)$ guarantees the correct limiting behavior of equation (64) since $\alpha_{\text{emp}}(q \rightarrow 0) = 0$ and $\alpha_{\text{emp}}(q \rightarrow \infty) = 1$. The linear-quadratic (LQ) dispersion relation for $\gamma_n(q)$ introduces momentum broadening in the ELF at all q (even below the Landau cut-off) as confirmed by the experimental data and predicted by the theory of XC effects [93].

The two main drawbacks of the extended-Drude models is the choice of an appropriate dispersion relation and the procedure to determine the Drude coefficients, which, for materials with a structure-rich spectrum, can become particularly tedious and inefficient.

2.3.2. Penn model. Penn [94] has developed an optical-data model, whereby extension of the optical data to $q \neq 0$ is made via the Lindhard dielectric function as follows

$$\text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right]_{\text{Penn}} = \int_0^\infty A(E') \text{Im}\left[\frac{-1}{\varepsilon_L(E, q; E')}\right] dE', \quad (66)$$

where from equation (24) we obtain

$$A(E) = \frac{2}{\pi E} \text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E, q \approx 0)}\right]. \quad (67)$$

Inserting equation (67) into equation (66) yields the full Penn expression

$$\text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right]_{\text{Penn}} = \frac{2}{\pi} \int_0^\infty \text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E', q \approx 0)}\right] \text{Im}\left[\frac{-1}{\varepsilon_L(E, q; E')}\right] \frac{dE'}{E'}. \quad (68)$$

In practice, the use of equation (68) is particularly cumbersome due to the analytic properties of the Lindhard dielectric function. Moreover, the ELF of equation (68) requires an integration over the excitation energy (E'), thus, being more computationally intensive than the extended-Drude models which involve a summation over a finite number of Drude-like functions. The use of the Lindhard dielectric function (and the Penn model in general) is well justified for the outer-shell electrons, which have a rather small excitation threshold, but not for the inner-shell electrons, where the binding energy far exceeds the plasmon energy of the system. Thus, the Penn model is not recommended for core excitations. As already noted, for materials with a complicated (structure-rich) spectrum the procedure to determine the Drude coefficients in the extended-Drude models can be tedious and inefficient. The Penn model overcomes the problem of a parametric fitting of the optical data, since at each $\{E, q\}$ point of the Bethe surface the ELF is completely determined by the experimental $\text{Im}\left[\frac{-1}{\varepsilon(E)}$ spectrum. Another advantage of the Penn model is that the ELF is automatically extended to $q \neq 0$ through the analytic properties of the Lindhard dielectric function without the need of adopting any particular dispersion relation, as in the extended-Drude models. The Penn model forms the basis of the famous Tanuma–Powell–Penn (TPP) predictive formula for the electron inelastic mean free path in solids [95].

2.3.3. Ashley model. A simplification of the Penn model has been suggested by Ashley [96, 97]. The basis of the Ashley model is the replacement of the Lindhard dielectric function by its plasmon-pole approximation in the zero-damping limit, $\varepsilon_{\text{pp}}(E, q; E_{\text{pl}}, \gamma \rightarrow 0)$. Then, equation (68) becomes

$$\text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right] = \int_0^\infty \text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E', q \approx 0)}\right] \frac{E'}{E} \delta(E - E'(q)) dE'. \quad (69)$$

Using the properties of the delta function, equation (69) is further simplified to

$$\text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right] = \frac{E_0}{E} \text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E_0, q \approx 0)}\right], \quad (70)$$

where E_0 is the positive solution of the dispersion relation $E(q; E_0) = E$. To simplify the numerical work, Ashley suggested using the Lindhard dispersion relation with $\alpha_{\text{RPA}} = 1$. This leads to $E_0 = E - \frac{q^2}{2m}$ that when inserted into equation (70) yields the final expression of the Ashley model

$$\text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right]_{\text{Ashley}} = \frac{E - Q}{E} \text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E - Q, q \approx 0)}\right] \Theta(E - Q), \quad (71)$$

where $Q \equiv \frac{q^2}{2m}$. The main advantage of the Ashley model is its computational simplicity. For example, the integration of the ELF over q can now be done analytically. On the other hand, a drawback of the Ashley model is that it shifts the excitation thresholds to higher energies. This unphysical effect is obviously most problematic at low electron energies and leads to the failure of the model at electron energies comparable to the binding energies.

2.3.4. Liljequist model. Liljequist [98, 99] has proposed the extension of optical data to $q \neq 0$ using a generalization of equation (69) that reads

$$\text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right] = \int_0^\infty \text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E', q \approx 0)}\right] \frac{E'}{E} \delta(E - F(E', q)) dE', \quad (72)$$

where $F(E', q)$ is a generalization of the dispersion relation $E'(q)$ of equation (69), obeying the condition $F(E', q = 0) \equiv E'$, and $\delta(E - F(E', q))$ is called a ‘ δ -oscillator’. The form of $F(E', q)$ in the Liljequist model is based on Bohr’s distinction of inelastic collisions into (i) resonant-like interactions with bound electrons, corresponding to small q , and (ii) binary-like interactions with large q , whereby the target electrons can be considered free (and at rest). Then, $F(E', q)$ takes the form

$$F(E', q) = E' \Theta(E' - Q) + Q \Theta(Q - E'), \quad (73)$$

where $Q \equiv \frac{q^2}{2m}$. Inserting equation (73) into equation (72) we obtain

$$\begin{aligned} \text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right]_{\text{Liljequist}} &= \text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E, q \approx 0)}\right] \Theta(E - Q) \\ &+ \frac{\pi}{2} E_{\text{pl}}^2 \frac{Z_{\text{eff}}(Q)}{Z} \frac{\delta(E - Q)}{E}, \end{aligned} \quad (74)$$

where $Z_{\text{eff}}(Q)$ is the effective number of target electrons that contribute to inelastic collisions with energy transfer up to $Q \equiv \frac{q^2}{2m}$ and given by the ‘partial’ f -sum rule expression

$$\int_0^Q E' \text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E', q \approx 0)}\right] dE' = \frac{\pi}{2} E_{\text{pl}}^2 \frac{Z_{\text{eff}}(Q)}{Z}. \quad (75)$$

The terms in the right-hand-side of equation (74) correspond to the two types of inelastic collisions in the Bohr categorization. Specifically, the first term of equation (74) represents the

soft (or distant) inelastic collisions with small enough q so that $E > Q$; in this case the ELF is approximated by its optical limit. The second term of equation (74) represents the hard (or close) collisions with $E = Q$; in this case the ELF is replaced by a delta function at the binary limit, weighted by the partial f -sum for energy-transfer up to Q . The Liljequist model is not very accurate for small- q collisions (e.g. those that involve valence-electron excitations) due the assumption of a constant ELF in the region $0 < Q < E$, as implied by the first term in the left-hand-side of equation (74). On the other hand, it provides a reasonable approximation for high- q collisions (e.g. those that involve inner-shell ionizations) owing to the use of the f -sum of equation (75) that gives a more realistic account of the contribution of hard collisions (compare, for example, to the high-energy binary collision models). Thus, the Liljequist model is generally considered most effective for inner-shell excitations.

2.3.5. MELF model. Abril and co-workers [100, 101] have proposed an optical-data model, whereby the optical data are extended to $q \neq 0$ using the Mermin energy-loss-function (MELF). Specifically, in the MELF model, the ELF of an arbitrary material is described by a superposition of Mermin-type ELFs using the following expression

$$\text{Im}\left[\frac{-1}{\varepsilon(E, q)}\right]_{\text{MELF}} = \sum_n \frac{A_n}{E_n^2} \text{Im}\left[\frac{-1}{\varepsilon_M(E, q; E_n, \gamma_n)}\right], \quad (76)$$

where the model parameters $\{A_n, E_n, \gamma_n\}$ are obtained from fitting the available optical data taking advantage of the fact that the MELF reduces to the Drude ELF at the limit $q = 0$

$$\begin{aligned} \text{Im}\left[\frac{-1}{\varepsilon_{\text{exp}}(E, q \approx 0)}\right] &\approx \text{Im}\left[\frac{-1}{\varepsilon(E, 0)}\right]_{\text{MELF}} = \sum_n \frac{A_n}{E_n^2} \text{Im}\left[\frac{-1}{\varepsilon_D(E; E_n, \gamma_n)}\right], \\ &= \sum_n \frac{A_n \gamma_n E}{(E_n^2 - E^2)^2 + (E \gamma_n)^2} \end{aligned} \quad (77)$$

Note that the MELF model coincides with the Ritchie–Howie model (see equation (48)) at $q = 0$. Thus, the MELF model also faces the tedious task of a parametric fitting of the optical data to obtain the model coefficients. The advantage of the MELF model is that the extension of the optical data to $q \neq 0$ is provided automatically through the analytic properties of the Mermin dielectric function. Thus, similar to the Penn model, no further approximations are needed with respect to the dispersion relations. However, unlike the Penn model, the form of equation (76) does not involve integrals and, by virtue of the Mermin dielectric function, it accounts for damping at any q .

2.4. Comparison

In figure 16 we compare various optical-data models against the experimental data [59, 60] at three values of momentum-transfer q . In all the models, we have used the same set of optical data [61] to determine the ELF at $q = 0$. Thus, the results depicted in figure 16 essentially compare the different algorithms for extending the optical data to $q \neq 0$. It can be first recognized that, overall, the Penn, ECN, and MELF

models provide a better representation of the experimental data at $q \neq 0$. The OREC and Ashley models exhibit a peak that is clearly too high and narrow compared to the experimental data and the other models. The sharp peak near the excitation threshold in the OREC model is due to the non-dispersive form of the excitations. This peak is not noticeable in the NOREC and ECN models, because it is strongly damped at $q \neq 0$. Note also that the Ashley model tends to displace the excitation threshold at higher energies with increasing q (whereas the band gap of liquid water is q -independent and $\sim 7\text{--}8\text{ eV}$). The NOREC model provides an improvement over the OREC model owing to the adoption of a dispersive damping coefficient, $\gamma(q)$, that induces momentum broadening in the ELF. Nonetheless, the peak in the NOREC model is still too narrow and somewhat displaced to higher excitation energies. The Penn model provides a fair representation of the data, especially at high- q , where lifetime broadening due to the Landau damping mechanism sets in; the Landau cut-off that corresponds to the electronic density of liquid water is $\sim 1\text{ a.u.}$ The ECN model yields a fairly good agreement with the experimental data due to the inclusion of XC effects (in a phenomenological manner) that induce plasmon shift and momentum broadening in the ELF. Thus, the ECN model substantially improves the OREC/NOREC models, which are also based on the extended-Drude model. The ability of a simple model, like the ECN, to faithfully reproduce the overall features of the ELF of liquid water at $q \neq 0$ is indicative of the efficacy of the approach based on optical-data models. This is despite the fact that the evolution of the excitation channels (especially of the inter-band transitions) with increasing q may be varied and complicated as revealed by the peak-and-shoulder structure of the experimental ELF.

The ELF of the optical-data models depicted in figure 16 has been used to calculate differential cross sections (equivalent to differential inverse inelastic mean free path), $d\Lambda(E; T)/dE$, in liquid water based on equation (3). In figure 17 we compare the calculated differential cross sections for two incident electron energies.

At the low incident energy (left panel) the shape and magnitude of the spectrum is different among models, whereas at the high incident energy (right panel) differences become rather small. It follows that the energy loss of low-energy electrons is due to collisions with $q \neq 0$, so the different extension algorithms have a sizeable impact upon the differential cross section. In contrast, the largest contribution to the energy loss of high-energy electrons comes from collisions with $q \approx 0$; thus, the extension algorithm used is inconsequential as long as the same sum-rule-constrained ELF at the optical limit has been employed in the calculations. The explanation of the above observations rests on the following realization. First, we may notice that although equation (3) involves an integration of ELF over q , the factor $\frac{1}{q}$ in the integral makes the differential cross section more sensitive to low- rather than high- q collisions. On the other hand, for $T \gg E$, the low- q limit in the integral ($q_{\min}$) decreases with increasing T (since $q_{\min} \propto 1/v$, with v the incident electron velocity). Thus, as the incident electron energy increases, inelastic collisions with lower q can be probed.

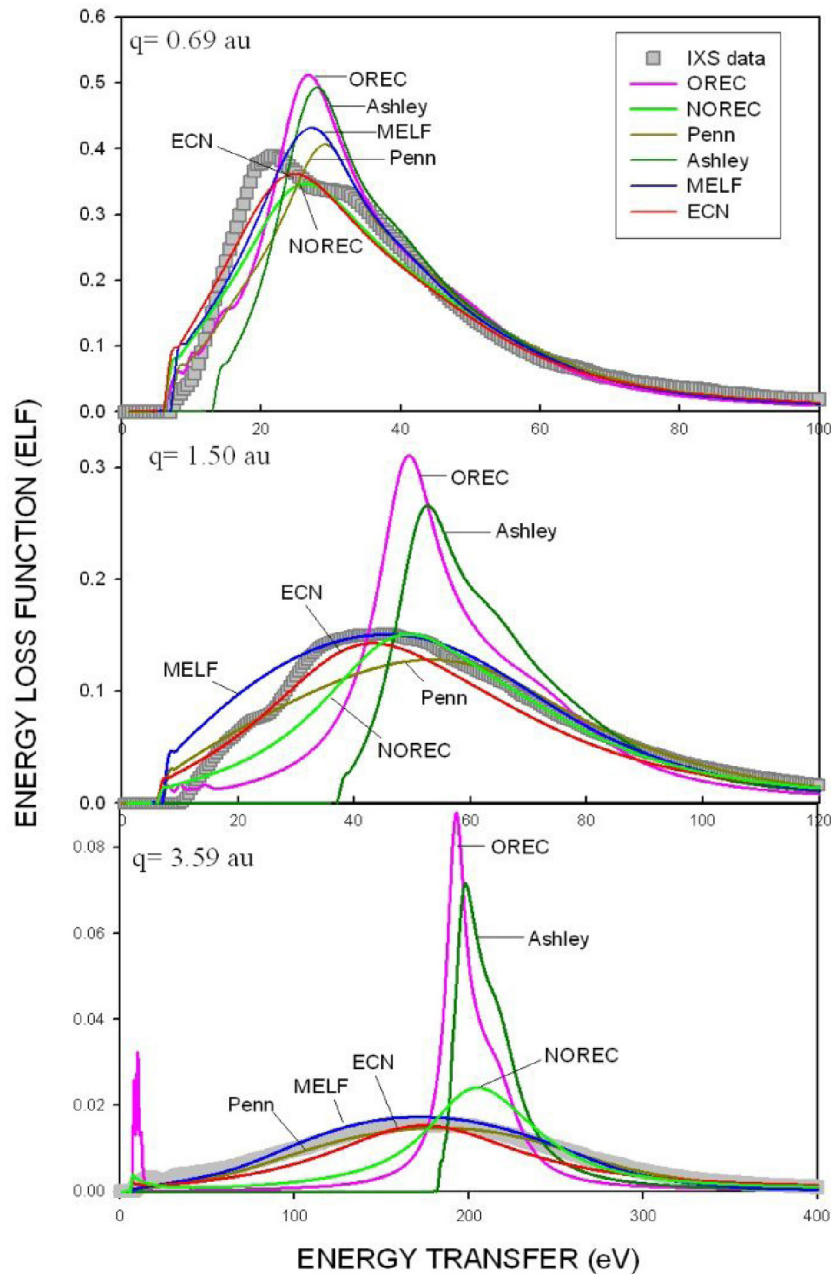


Figure 16. The experimental data for the ELF of liquid water [59, 60] at three different values of momentum transfer ($q = 0.69, 1.50, 3.59$ a.u.) are compared with various model calculations (see text for details on the models).

The electron inelastic mean free path, $\Lambda^{-1}(T)$, in liquid water as a function of incident energy is presented in figure 18. The calculations are based on equation (5) with the ELF determined from the various optical-data models for the dielectric function of liquid water.

The first observation is that all optical-data models gradually agree (to within 10%) above a few hundred eV, while differences increase at low energies and become particularly noticeable below 100 eV. This is in accordance with the discussion on the differential cross section of figure 17. The inelastic mean free path for incident electron energies above a few hundred eV depends mainly upon the ELF at $q \approx 0$. In contrast, the value of the inelastic mean free path at low incident

energies will sensitively depend upon the ELF at $q \neq 0$. Apart from the Ashley model, which results in a steep rise of the inelastic mean free path at low energies, all other optical-data models yield a similar shape with a modest rise at low energies, although in absolute numbers differences among models are quite large (up to a factor of ~ 2). The behavior of the Ashley model at low energies is due to the unphysical shift of the excitation threshold to higher energies (see figure 15). This yields very small probabilities of low-energy excitation, since the low-energy excitation area under the ELF to be integrated in equation (5) is strongly underestimated, thus, resulting in a small cross section or, equivalently, a large mean free path for inelastic collisions. Investigations of the effect of different

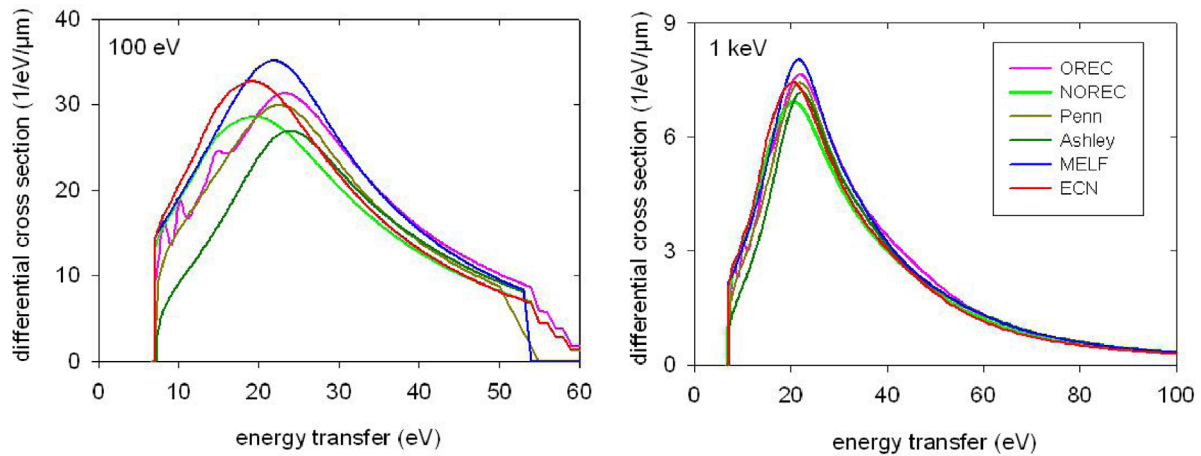


Figure 17. The differential cross section as a function of energy transfer for incident electrons of 100 eV (left panel) and 1 keV (right panel) calculated from equation (3) using different models for the dielectric response function of liquid water (see text for details on the models).

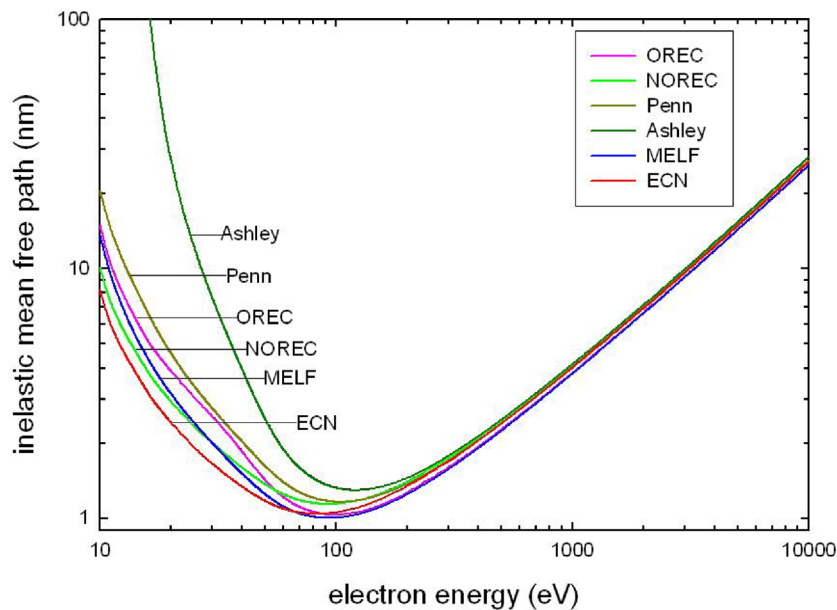


Figure 18. The inelastic mean free path as a function of electron energy calculated from equation (5) using different models for the dielectric response function of liquid water (see text for details on the models).

q -dependencies of the ELF on differential and total inelastic cross sections for various solids have yielded similar conclusions [102–105].

By construction, it is difficult within the simplified approach of optical-data models to include fine details of the excitation spectrum and its evolution across the momentum space. Such features are expected to become important when the incident electron energy becomes comparable to the excitation energies associated with the fine structures in the ELF. Moreover, effects beyond the conventional RPA theory of the dielectric response function (e.g. band-structure and XC effects) are expected to set in at low energies, further increasing the uncertainties in this domain. The inclusion of effects beyond RPA and of *ab initio* calculations of optical spectra seems to hold the most promise in the near future for expanding the range of applicability of optical-data models down to very low energies [73, 92, 93, 106–109].

3. Elastic scattering of low-energy electrons

The radiation damage with which we are concerned here is due to ionizing events, i.e. energy deposition in quanta large enough to exceed an ionization threshold. It might then be thought that elastic scattering events, where no energy (or negligibly little energy) is deposited, should be of little importance. That is, however, not the case. Low-energy electron tracks are highly tortuous, and this is mainly due to elastic scattering. Therefore, elastic scattering is of decisive importance for the spatial distribution of the ionizing events. In section 3.1, we will briefly comment on basic features of electron elastic scattering by atoms and molecules. For a more detailed discussion of theoretical and empirical models of elastic electron scattering used in electron trajectory simulations, see, e.g. [110] and references therein. In section 3.2, we discuss in more detail the not well-understood elastic scattering of very-low-energy electrons in liquid water.

3.1. Electron elastic scattering models used in track simulation

The transport of electrons has, in the context of track structure and radiation damage, so far been calculated by means of the conventional trajectory approach, using either Monte Carlo trajectory simulation or the equivalent transport theory, expressed by the Boltzmann transport equation.

In this approach (i.e. a semi-classical picture, where the electron is treated as a point-size particle moving along a trajectory through the medium) elastic scattering events along the trajectory are usually assumed to occur against single atoms (or, possibly, single molecules), i.e. the possibility of coherent elastic scattering by many atoms or molecules simultaneously is neglected. The elastic scattering, which essentially is the deflection of the electron in the field of a nucleus (screened by its atomic electrons), is then determined by the elastic differential cross section (elastic DCS) per target atom or target molecule, together with the density n of such targets per unit volume. It is convenient to characterize the elastic scattering in terms of the elastic mean free path

$$\lambda_e = \frac{1}{n\sigma_e} \quad (78)$$

where σ_e is the total elastic scattering cross section, i.e. the elastic DCS integrated over all scattering angles, and the transport mean free path,

$$\lambda_{tr} = \frac{\lambda_e}{1 - \langle \cos \theta \rangle} \quad (79)$$

where $\langle \cos \theta \rangle$ is the average of the cosine of the polar scattering angle θ in a scattering event. The transport mean free path may be regarded as the path length over which the electron effectively loses its memory of direction. Under conditions of plural or multiple scattering (rather than single scattering), the transport mean free path (rather than the mean free path or the detailed DCS) is the parameter of primary importance for the spatial distribution of scattering events [111]. Therefore, if—as often happens—a simplified model is used for the DCS, then it is in most Monte Carlo applications imperative that it is adapted to give an accurate transport mean free path [111]. The total elastic mean free path is in fact less important.

We may note that the inverse λ_e^{-1} of the elastic mean free path is sometimes referred to as the macroscopic elastic cross section. A commonly used concept is the transport cross section σ_{tr} , defined by

$$\lambda_{tr} = \frac{1}{n\sigma_{tr}}. \quad (80)$$

The most simple and at least fairly realistic approximation of the DCS for elastic scattering of electrons against atoms may be the so-called screened Rutherford DCS, based on the Wentzel model of an atomic scattering potential with the screening radius

$$R = a_0 \cdot Z^{-1/3} \quad (81)$$

where a_0 is the Bohr radius ($\approx 0.53 \text{ \AA}$) and Z is the atomic number of the target atom. The screened Rutherford DCS is

in general *not* accurate enough, but it can conveniently be used here as a starting point for a discussion. The Wentzel model atomic scattering potential is simply the Coulomb interaction between the atomic nucleus and the incident electron, multiplied by an exponential screening factor $\exp(-r/R)$ which approximately represents the screening of the nuclear Coulomb field due to the cloud of atomic electrons, i.e. (in cgs units)

$$V(r) = -\frac{Ze^2}{r} \exp(-r/R) \quad (82)$$

If the scattering of an electron in this potential is calculated in the Born approximation (i.e. assuming only so-called first order or single scattering of the electron wave by the potential), then the resulting DCS, calculated by means of the Schrödinger equation, may be written as

$$\frac{d\sigma_{SR}}{d\Omega} = \left(\frac{Ze^2}{2mv^2} \right)^2 \frac{1}{\left(\sin^2 \frac{\theta}{2} + \frac{1}{4k^2R^2} \right)^2} \quad (83)$$

where m is the electron mass, v is the electron velocity, k the electron wave number and θ the scattering angle. This is the screened Rutherford cross section. The term $\eta = (4k^2R^2)^{-1}$ is sometimes known as the screening parameter.

For electrons in the energy range 100 keV–20 MeV, the DCS represented by equations (81)–(83) gives a transport cross section that is, if not accurate, at least reasonably close to accurate calculations—the error is in the order of 10% or less [112]. However, at electron energies below 100 keV the error increases rapidly, in particular for high Z atoms. For example, at an electron energy of 100 eV, the calculated transport cross section is a factor ≈ 1.12 too large for $Z = 4$ (Be), a factor ≈ 2.8 too large for $Z = 13$ (Al) and no less than a factor ≈ 66 too large for $Z = 79$ (Au) [112].

The error is mainly due to two factors: first, the simple atomic scattering potential model given by equations (81) and (82) is not accurate enough, and, second, the DCS given by equation (83) is based on the Born approximation. The excessive error at low electron energies and high Z should be due mainly to the second factor, i.e. the use of the Born approximation. As indicated by the numbers exemplified above, the Born approximation typically gives too large a transport cross section, i.e. it exaggerates the elastic scattering, making the transport mean free path too short.

A systematic remedy is, therefore, first to replace equation (82) by an accurate atomic scattering potential $V(r)$ and, second, to go beyond the Born approximation, i.e. apply a higher order quantum calculation of the electron scattering. The first step towards an accurate scattering potential is to calculate the static screening of the nuclear Coulomb field by means of accurately determined atomic electron wavefunctions. Further corrections of this static atomic scattering potential are due to the inclusion of electron exchange effects, polarization effects, and so-called inelastic absorption effects. For a detailed description of the state-of-the-art in calculation of an accurate atomic scattering potential $V(r)$, see [113].

Given an accurate, spherically symmetric potential $V(r)$, an in-principle exact calculation of the quantum scattering DCS can be made by means of partial wave analysis (PWA) [113]. The resulting DCS is, in general, represented numerically, rather than by a closed formula such as equation (83). The Schrödinger equation quantum formalism is usually replaced by the Dirac equation formalism, in which relativistic effects, including effects of electron spin, are automatically included (see, e.g. [113]).

Such accurate elastic DCSs are nowadays used as a standard in detailed Monte Carlo electron trajectory simulation codes. However, the screened Rutherford cross section still enjoys a certain popularity, possibly due to its simplicity and convenience in a simulation context. The use of the screened Rutherford DCS might be justified as a reasonable approximation used for the simulation of plural or multiple scattering, if it has been modified appropriately, i.e. in such a way that (at least) the transport mean free path comes out correctly. One often-used method has been to empirically modify the screening parameter η or the screening radius R . It is also possible to modify the screened Rutherford DCS by means of a simple multiplicative factor [112]. The modification may either be based on PWA analysis of scattering in an accurate potential $V(r)$ (see [112] and references therein), or on empirical data if such data are available.

We have so far discussed the elastic scattering of electrons by single atoms. The electron scattering performed by molecules may in the simplest approximation be treated by means of a simple addition of the atomic contributions to the DCS, where different atoms are then assumed to scatter incoherently, and at higher electron energies this appears to be a good approximation. At lower energies, one may have to take coherent elastic scattering (interference between waves scattered by the different atoms in the molecule) into account (see e.g. [110] and references therein).

We now consider the elastic scattering of electrons in condensed matter. For plural or multiple elastic and inelastic scattering of electrons in the keV–MeV energy range in a gaseous, liquid, amorphous or polycrystalline medium, we may, on the basis of experience, expect the trajectory approach to be adequate. If so, coherent elastic scattering can be neglected, and for the simulation of an electron track it is sufficient that we know the DCS for the elastic scattering of the electron by a single atom. However, we may recall that diffraction patterns do occur under certain scattering conditions, e.g. for zero-loss high energy (~ 100 keV) electron penetration through very thin layers, or for very-low-energy electrons (~ 100 eV) backscattered without energy loss from surfaces. In such cases, it is trivially clear that the trajectory approach fails. An important point is that at very low electron energies the trajectory approach may, as will be discussed below, fail even without the appearance of obvious diffraction effects.

In condensed matter, the scattering potential represented by a single atom is somewhat modified by the presence of its neighboring atoms, since these exert a distorting effect on the atomic electron cloud. Thus, it is likely that the scattering potential V is no longer precisely spherical. In computational practice, the potential is usually assumed to still be spherical

($V = V(r)$), but slightly modified at distances $\approx d/2$, where d is the (average) nearest neighbor distance in the medium. This may have some effect on the total elastic scattering cross section, but little effect on the transport cross section, due to the fact that a considerable contribution to the transport cross section comes from large and medium scattering angle events, where the electron has, classically speaking, passed through the inner parts of the atom. Therefore, the modification of the outer part of the atomic scattering potential $V(r)$ due to the liquid or solid state of the target should, in general, have little effect on the trajectory pattern and the spatial distribution of ionizing events [112].

3.2. Elastic scattering of very-low-energy electrons in liquid water

We now consider the special and rather enigmatic case of very-low-energy electrons (~ 100 eV and below) in liquid water. Here, the theory is very uncertain, and experiments are so far not at all conclusive. Unfortunately, this case is also of major interest in the theory of radiation damage to DNA molecules in a living cell.

At very low electron energies, elastic scattering models used for electrons moving in liquid water and biological matter face (at least) three problems. First, the experimental study of very-low-energy electron transport in liquid water is extremely difficult. Liquid water is not easily compatible with the vacuum required in an electron spectrometer, and the distances of interest (range, transport mean free path) in the scattering medium are microscopically small. Second, on the theoretical side, the validity of the trajectory approach itself may be questioned, due to the increasingly large effects of coherent scattering that may be expected with an increasing electron wavelength. Third, the physical assumptions, which are made when calculating the elastic DCS, may be questioned—it may even be questioned if the conventional scattering cross-section concept is meaningful. One further theoretical difficulty is that elastic electron scattering at very low energies in a condensed medium might possibly be influenced by the polarization properties of the medium.

The seriousness of these problems is underlined by the fact that a considerable discrepancy is found when theoretical calculations of elastic scattering are compared to available experimental data.

3.2.1. Validity of the trajectory approach to electron transport. Let us for a moment disregard inelastic scattering, and consider purely elastic, multiple scattering of an electron in an amorphous or disordered medium (e.g. liquid water). From a quantum mechanical point of view we are here dealing essentially with the multiple elastic scattering and self-interference of an extended wave (the electron wave function) propagating through a medium consisting of scatterers (atoms or molecules) in randomly shifting positions. At any time, the wave is scattered and re-scattered by many scatterers simultaneously. With this picture in mind, we might very well ask ourselves how it can be that a simple trajectory treatment of the electron transport is at all possible.

In response to this question, the limits of validity of the trajectory approach have during the past decade been extensively studied by means of a comparison of conventional trajectory simulation with the corresponding *exact* quantum calculation. This is practically possible and has been performed for the case of a particle scattered in a medium consisting of point (point-size) scatterers; this case offers a simple as well as an exact solution to the quantum multiple scattering problem. The results so far, summarized in two recent papers [114, 115], indicate that trajectory simulation gives a good approximation of the *average* quantum multiple elastic scattering in a medium with randomly varying scatterer positions, provided that the de Broglie wavelength λ of the scattered particle is short, compared to d_{nn} , where $d_{nn} = n^{-1/3}$ is a measure of the average distance between nearest neighbor scatterers. For a point scatterer medium, it is found that if inelastic scattering is absent, or at least rare, and if local order (locally ordered structure) in the medium is neglected, then the trajectory method is a good approximation (in the sense that the error in spatial and angular distributions is generally less than 5%) if $\lambda < d_{nn}$. The presence of local order (such as found, e.g. in liquids) will somewhat increase the error of the trajectory method, shifting the limit of validity to a higher electron energy [116]. On the other hand, the presence of inelastic scattering will, assuming that it destroys coherence, decrease the error, thus shifting the limit of validity of the trajectory method to lower energies [114]. Intuitively, this may be understood by observing that if coherent scattering is always destroyed in a scattering event, then coherence effects should of course disappear entirely. It is not trivially clear that this would lead to the validity of the trajectory method, but calculations using the point scatterer model indicate that it does [114].

For clarity, it should be remarked that the trajectories calculated for very-low-energy electrons do not represent the motion of wave-packets. A low-energy electron wave-packet may be larger than the entire simulated trajectory [115]. Rather, the trajectories are an approximate representation of the multiple wave-scattering that occurs within a propagating wave-packet. For this reason, the validity of the trajectory method, in this context, has been referred to as ‘circumstantial’ [115], in order to point out the difference from the ‘explicit’ validity of the trajectory picture, which may be expected when it is used e.g. for the transport of heavy ions through a medium.

For electrons moving in a point-scatterer model of liquid water ($n \approx 0.0332 \text{ \AA}^{-3}$, $d_{nn} \approx 3.11 \text{ \AA}$), a wavelength $\lambda = d_{nn}$ corresponds to an electron energy $\approx 15 \text{ eV}$. This is the estimated limit of validity (5% error) of the trajectory approach if scattering is predominantly elastic ($\lambda_i \gg \lambda_e$) and local order is absent [116]. The presence of local order similar to that in liquid water shifts the 5% limit to somewhere in the range 50–100 eV, while the 10% error limit is then in the range 20–30 eV [116]. A demonstration performed with a point-scatterer model of liquid water, with local order as well as inelastic scattering included [114], shows that if inelastic scattering (mean free path λ_i) is sufficiently frequent ($\lambda_i \sim \lambda_e$), and coherence is assumed to be destroyed in each inelastic event, then the trajectory method may be a fair approximation even at electron energies as low as a few eV [114].

The results obtained for a medium of point scatterers have been tentatively extrapolated to a medium consisting of atoms or molecules [115]. The basis for this extrapolation is that the error of the trajectory method appears to increase in proportion to amplitude of the multiple wave-scattering within the medium. This suggests that the error should be roughly proportional to the square root of the elastic scattering cross section. In general, this means that the error is expected to be somewhat larger than that for the point-scatterer model [115]. Unfortunately, a full treatment of multiple wave-scattering in a cluster of finite-size scatterers is computationally very demanding.

It should finally be observed that even if the trajectory approach, in general, may be a valid approximation down to a very low electron energy, coherent scattering effects (e.g. interference patterns in the energy deposition) may still occur under certain conditions [116, 117]. It can be argued that the coherence length of an electron moving in liquid water should be in the order of the total mean free path [117]. Therefore, coherent scattering effects should be possible if the electrons are scattered by objects immersed in liquid water at a well-defined distance from each other (e.g. different parts of a DNA molecule) provided that the distance is at most in the order of the total mean free path [117]. Indeed, the possibility of coherent scattering of a low-energy electron within the DNA molecule has previously been shown by detailed and explicit theoretical calculations [118].

3.2.2. Theoretical and experimental elastic scattering cross sections. For the remaining discussion, we will, tentatively assume that the conventional trajectory approach is a valid approximation. The next question is then: How reliable is our present knowledge of the elastic DCS in liquid water for electrons with very low energies?

At present, there seems to be essentially four data sets of interest. They may be listed as follows:

- (1) Experimental and theoretical elastic cross sections for electron scattering in *gaseous* water, i.e. for electron scattering by free water molecules (here referred to as ‘gas-phase data’). A comprehensive review has been given by Itikawa and Mason [119]. Gas-phase data from [119–124] are shown in figure 19, together with data based on additivity, i.e. cross sections obtained by the simple addition of oxygen and hydrogen atomic cross sections [125]. The relative uncertainties in the experimental data are typically between $\pm 15\%$ – $\pm 30\%$.
- (2) Theoretical elastic cross sections based on x-ray-determined molecular electron density in liquid water [126], also indicated in figure 19.
- (3) Experimental elastic cross sections derived from electron backscattering in thin amorphous ice layers on cold metal backing [127], also shown in figure 19. Here, the estimated relative uncertainty in experimental data is between $\pm 30\%$ – $\pm 45\%$ [127].
- (4) Experimentally determined effective attenuation length (EAL), derived from photoelectron escape from x-ray-irradiated liquid water [128–130]. The relative uncertainty

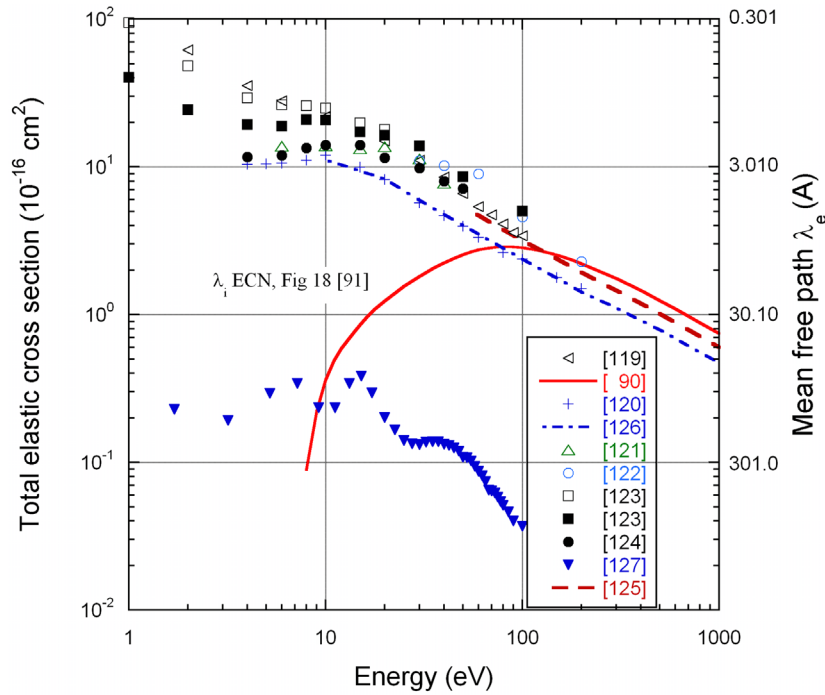


Figure 19. Total elastic scattering cross section σ_e and corresponding elastic mean free path $\lambda_e = 1/n\sigma_e$ in liquid water, according to cross sections for free water molecules (gas-phase data [119–125]), according to the calculation of Aouchiche *et al* [126] for liquid water, and according to the measurements of electron scattering in amorphous ice by Michaud *et al* [127]. The inelastic scattering total cross section (and inelastic mean free path) obtained with the ECN model [90] is indicated for comparison (see section 2 and figure 18). Khakoo *et al* [123] is shown with two data sets, one where the Born dipole extrapolation has been used for small scattering angles (open squares), and one without this extrapolation (filled squares).

in the reported EAL data appears to be about $\pm 25\%$. It should be noted that the uncertainties in data sets (3) and (4) appear to be largely due to the calibration methods and models, which are used for analyzing the raw experimental data.

Total elastic cross sections are shown in figure 19 for the data sets (1), (2) and (3). Also shown, by means of the numbers indicated on the right-hand vertical axis, is the conversion from total elastic cross section to elastic mean free path in liquid water, obtained by using equation (78) with n set equal to the number of molecules per unit volume in liquid water ($n = 0.0332 \text{ \AA}^{-3}$).

Two striking features in figure 19 should be noted. First, the difference between data sets (1) and (2), on the one hand, and data set (3), on the other, is dramatic nearly two orders of magnitude. In addition, there is no sign of convergence of the two data sets when the electron energy approaches 1 keV. This absence of any convergence is unexpected and problematic, since condensed phase effects are expected to be less important at higher electron energies. It should also be noted here that the gas-phase elastic scattering data may include a very small rotational energy loss, while the elastic scattering in amorphous ice is well resolved from inelastic events—those inelastic events that involve the smallest energy losses are the electron–phonon interactions [127].

The second striking feature appears for electron energies below 20–50 eV. Here, most of the data set (1) (gas-phase data) predicts, by equation (78), an elastic mean free path

which is *smaller* than the average distance $d_{nn} = n^{-1/3} \approx 3.1 \text{ \AA}$ between nearest neighbor water molecules in liquid water. As will be discussed, this should raise some serious doubts as regards the validity of gas-phase data when applied to liquid water at such low electron energies. The gas-phase elastic cross sections are nevertheless frequently used in track simulation codes, apparently since they are available, comparatively well determined and straightforward to apply.

Aouchiche *et al* [126] have tried to replace the gas phase data by a physically more realistic alternative by replacing the scattering potential of a free water molecule with a potential derived from x-ray scattering in liquid water, also adding polarization–correlation effects according to the theory presented by Salvat [113]. The resulting elastic scattering cross sections are, in general, (see figure 19) somewhat smaller (about 10%–15%) than the experimental elastic cross sections obtained for the water in the gas phase (although close to those measured by Danjo and Ishimura [120]).

We now turn to the experimental data that are available for water in condensed phase. The Michaud *et al* [127] cross sections are based on measurements of the energy-resolved spectrum of electrons backscattered from thin layers of amorphous ice on a cold metal backing. It is reasonable to assume that amorphous ice should have electron-scattering properties that are at least similar to liquid water. The cross sections are obtained from the measured data by means of an analytic electron transport model, which involves several approximations and physical assumptions. The results include inelastic as well as elastic scattering cross sections [127]—only the result for

elastic scattering is shown in figure 19. We may, as mentioned, note the dramatic difference from the elastic-scattering data sets (1) and (2).

It is interesting to note that an independent support for the possibility of a large elastic mean free path comes from a different type of experiment, namely x-ray-induced photoelectron escape from liquid water, first described by Ottosson *et al* [128]. In these experiments, here presented as data set (4), a microjet of liquid water shoots out into vacuum, where it is irradiated by x-rays from an x-ray synchrotron. Depending on the x-ray energy, photoelectrons of varying energy are generated in the water and escape from the surface of the microjet. Some escape in directions into an electrostatic energy analyzer. The analysis of data is, like the analysis performed by Michaud *et al* [127], complex and model-dependent. In the Ottosson *et al* measurement [128], calibration was obtained by means of an elaborate process that involved a molecular dynamics simulation of the positions of Na and I ions in a NaI solution in water. An apparently simpler calibration method was later used by Suzuki *et al* [130]. From the experimental data the so-called EAL was derived. The EAL may be defined as the distance from the point of origin of the photoelectron to the liquid surface for which the probability of escape without energy loss has decreased by a factor $1/e$. If the photoelectron had started in a direction perpendicularly towards the surface, and if, in addition, elastic scattering had been negligible, then we would simply expect $EAL = \lambda_i$. In reality, photoelectrons are emitted from the photo-effect site with a certain angular distribution relative to the incident x-ray beam, which means that we, for purely geometrical reasons, should expect $EAL < \lambda_i$. If there is also strong elastic scattering ($\lambda_e \ll \lambda_i$), then we may expect $EAL \ll \lambda_i$ due to ‘wrinkling’ of the electron path. The EAL thus depends on both inelastic scattering and elastic scattering, and a derivation of the elastic scattering properties from the EAL requires knowledge of the inelastic scattering as well as the initial angular distribution of the photoelectron emission.

An $EAL \sim 20 \text{ \AA}$ in the energy range 10–100 eV is reported by [130]; a similar value $\sim 20\text{--}30 \text{ \AA}$ is reported by [128] at energies of about 100 eV, while a somewhat lower value $\approx 10\text{--}12 \text{ \AA}$ is reported in [129]. We may compare this with the inelastic mean free path $\lambda_i \approx 10\text{--}15 \text{ \AA}$ at $E = 100 \text{ eV}$ in figure 18 and with the elastic mean free path $\lambda_e \approx 5\text{--}15 \text{ \AA}$ for the gas-phase data at the same electron energy in figure 19; the amorphous ice data give $\lambda_e \approx 700 \text{ \AA}$. Ottosson *et al* [128] argue qualitatively on the basis of their results that the elastic scattering in liquid water should be weak in the considered energy range. However, the experimental results seem too uncertain and mutually inconsistent to allow any definite conclusions. It may be hoped that future studies of the AEL in liquid water, combined with inelastic scattering data such as provided by the calculations described in section 2, might yield experimental information about elastic as well as inelastic electron scattering in liquid water at very low electron energies.

3.2.3. Critical discussion of elastic mean free paths derived from gas-phase scattering data at energies below 50 eV. As mentioned, the cross sections obtained in data sets (1) and (2) for electron energies below about 50 eV correspond to mean

free path values less than $3 \text{ \AA}$, i.e. less than d_{nn} (figure 19). It has previously been proposed that such a short mean free path is unrealistic [131]. We may, following [132], consider the fact that the standard sampling method for simulating the step length between two successive scattering events generates a Poisson distribution of events within any path length interval. If the mean free path is of a magnitude similar to d_{nn} , it follows that several successive elastic scattering events could occur, with a substantial probability, within a distance corresponding to the passage through a single water molecule [132]. This picture is, however, not compatible with meaning of the quantum elastic scattering cross section, which in principle is the cross section for finding the electron, after the scattering event, and being far away from the scattering molecule, in a quantum state different from the initial one. Thus, in the calculation of the cross section, the interaction between the electron and the molecule is considered as a single event. Alternative ways of performing the sampling of step length are discussed in [133].

An implicit assumption in the standard method for sampling step length between scattering events in a Monte Carlo trajectory simulation (as well as in analytic transport theory, e.g. the Boltzmann transport equation) is that the medium can be regarded as a homogeneous continuum. This is clearly a physically reasonable assumption only if $\lambda_e \gg d_{nn}$, i.e. if $\sigma_e \ll d_{nn}^2$. This objection, as well as the previous one, suggests that the usual application of gas-phase elastic cross sections, as well as the elastic cross sections predicted by Aouchiche *et al* [126], to the elastic scattering of very-low-energy electrons in condensed matter might be inadequate for their purpose. When discussing a possible remedy, the following arguments should be considered.

- (a) In the standard definition and calculation of the cross section, it is assumed that the source is placed at a distance d_{ST} very far from the target, more precisely, it is required that $d_{ST} \gg R$, where R is the range of the scattering potential. Then, the incident particle may be assumed to be initially (well outside the range R) in a plane-wave state $\exp(i\mathbf{k} \cdot \mathbf{r})$, which is the assumption made in the standard quantum calculation of the scattering cross section. Now, in a condensed medium, R is typically comparable to d_{nn} . If the calculated cross section σ_e is large enough to make the mean free path $\lambda_e = (n \cdot \sigma_e)^{-1}$ comparable to or smaller than d_{nn} , then d_{ST} is evidently in the order of R . The initial wave function that hits the target molecule is then very far from a plane wave, and the appropriateness of the standard cross section may, therefore, be questioned. One fortunate exception from this may be noted: for a point scatterer we have $R = 0$, so the standard cross section is always valid (since certainly $d_{ST} \gg R$).
- (b) As regards proposals for a remedy, Michaud *et al* [127] suggest that the large elastic mean free path obtained in the amorphous ice experiments could be understood in terms of coherence effects and screening as described by the inverse of the squared magnitude of the dielectric response function ϵ —similar to the screening of inelastic interaction.

- (c) An alternative suggestion is that very-low-energy electrons in liquid water should be treated as polarons, i.e. quasiparticles made up of the electron and a surrounding cloud of polarized medium (virtual phonons). Ganachaud and Mokrani [131] proposed, in their work on secondary electron emission from insulators, that very-low-energy electron transport might be polaronic in nature. The polaronic nature of a low-energy electron in liquid water is also strongly suggested by the fact that it may end up as a hydrated electron; indeed, the polaronic nature of hydrated electron diffusion in liquid water has been demonstrated [134]. Polarons typically have an effective mass larger than the mass of the bare electron, and one may speculate on whether this might associate polaron formation with a weaker elastic scattering. A recent application of the Ganachaud–Mokrani model is presented by Dapor [135].
- (d) Continued experimental studies of the EAL in liquid water might be useful since they may, as mentioned before, provide information about the elastic scattering. One might, therefore, hope that more reliable data on the EAL could be used as a basis for an empirical model of the transport of very-low-energy electrons in liquid water.

3.2.4. Critical discussion of elastic mean free paths derived from amorphous ice measurements. The work of Michaud *et al* [127] includes a transport theoretical analysis in order to extract the elastic and inelastic mean free path values from the observed experimental data. This so-called two-beam analysis contains certain assumptions and approximations. We will discuss two of them.

First, the two-beam analysis is based on a trajectory picture. It has recently been shown [136] that coherent scattering (in other words, electron diffraction in the amorphous ice layer) should have influenced the results. Such effects are not accounted for by the two-beam analysis. However, the actual effect of diffraction on the results may have been moderate [136].

Second, it has been pointed out by Michaud *et al* [127] that a forward scattering component may be missing in the two-beam analysis, making the derived cross section smaller than the real one. It has previously been suggested that this might be quantitatively discussed in terms of the transport mean free path or equivalently the transport cross section [116]. For this, we may recall that two different plural or multiple scattering processes will produce similar results under multiple scattering, provided that their transport mean free path is the same [111]. For the elastic scattering at very low electron energies, Michaud *et al* [127] assumed a model with explicitly equal forward and backward scattering in each scattering event. Evidently, the average $\langle \cos \theta \rangle$ is then zero, making the transport and elastic mean free path identical (equation (79)). Since the procedure used by Michaud *et al* essentially was to fit this model to experimental data, it may be suggested that the elastic scattering datum that Michaud *et al* [127] obtained from their analysis at very low electron energies may have been an estimate of the elastic transport cross section rather than the total elastic cross section.

Table 6. Tentative correction of the total elastic cross section reported by Michaud *et al* [127]. The correction factor is $2 \times (\sigma_e/\sigma_{tr})_{\text{gas}}$, rounded off to the nearest integer value.

Electron energy (eV)	Correction factor
1	34
4	15
7	7
10	5
15	5
20	5
40	6
100	6

We will here conclude this argument with a simple quantitative estimate. Assuming, for the present discussion, that the elastic cross section values obtained by Michaud *et al* [127] at very low electron energies should be interpreted as the transport cross section, we wish to estimate the real total elastic cross section. We then need some information about the average of $\langle \cos \theta \rangle$ in the real scattering process. As a possibly reasonable guess, we assume the angular distribution to be the same as for scattering from a free water molecule at the same energy. Using equations (79) and (80) and gas-phase data from [119], this leads to a very simple correction formula, namely

$$\sigma_e^{(\text{corrected})} = \sigma_e^{(\text{Michaud et al})} \cdot (\sigma_e/\sigma_{tr})_{\text{gas}}. \quad (84)$$

We may here also add that Meesungnoen *et al* [137], when applying the amorphous ice data to simulation, empirically increased the elastic and inelastic scattering cross sections by a factor 2. They attributed this factor to a larger structural disorder in the liquid than in the solid. Combining this correction with equation (84) and using gas-phase data from [119], we get a total correction factor shown in table 6.

In table 6, the large correction factors for energies below 10 eV might be unrealistic as they arise due to small angle scattering of very-low-energy electrons by the dipole moment of free water molecules [119]. The final tentative correction is otherwise generally ~ 5 , which means that the Michaud *et al* [127] data set in figure 19 is translated upwards so that the elastic mean free path in the energy range 1–20 eV would be roughly 30 Å. The problematic absence of any sign of convergence with gas-phase elastic scattering data at energies above 100 eV still remains.

3.2.5. Conclusions as regards very-low-energy electron elastic scattering in liquid water. Studies of trajectory simulation compared to an exact quantum treatment of multiple scattering in a point-scatterer medium indicate that the trajectory method might be a reasonable approximation down to quite low electron energies—a few tens of eV or perhaps, if inelastic scattering predominates, even a few eV. Coherent scattering effects may appear under special circumstances.

Fundamental considerations about the multiple scattering process as well as available experimental data indicate that the elastic mean free path obtained from molecular gas-phase cross sections is smaller than the real elastic mean free path in liquid water at very low electron energies (< 50 eV). On the

other hand, it seems not unlikely that the elastic mean free path previously derived from amorphous ice data is larger.

A more sophisticated and presumably more complex quantum treatment may be required for the elastic scattering of very-low-energy electrons in liquid water, perhaps involving multiple wave scattering as well as consideration of the polaronic nature of the electron. In view of the complexity of this theoretical approach, new experiments seem to be highly motivated to provide an empirical basis for improved scattering models.

4. Interaction cross sections for ions

Inelastic interactions of ions arise from electromagnetic and nuclear interactions. Electromagnetic interactions are responsible for the energy loss of ions and production of secondary electrons in ion tracks. Nuclear interactions cause changes in the direction of incident ions and can produce nuclear fragments and neutrons in high-energy ion beams. In general, the occurrence probability is much higher for electromagnetic interactions. Therefore, electromagnetic interactions are usually followed event-by-event by MCTS simulations. Nuclear non-elastic interactions are not considered directly in MCTS simulations, because a large number of interaction channels and secondary particles are involved, making the simulation procedures too complicated. General-purpose Monte Carlo radiation transport codes can be used for generating energy spectra of secondary particles induced by nuclear interactions, which can be coupled to MCTS simulations for investigations of energy deposition at the cellular and subcellular levels.

4.1. Electromagnetic interactions of ions

4.1.1. General aspects. When ions traverse a medium, they interact with bound electrons and atomic nuclei of the medium. Interactions between ions and bound electrons are considered to be electromagnetic interactions including ionization, excitation and charge exchange (or charge transfer interactions). Ionization and excitation occur when the energy and momentum of ions are higher than the associated energy threshold of an electronic transition. Both interactions are important for high and intermediate energy ions (>1 MeV u^{-1}). In general, high-energy ions are fully stripped, i.e. the ionic charge is equal to the nuclear charge. As ions slow down to a speed comparable to the speed of bound electrons (~ 1 MeV u^{-1}), they can capture target electrons and become ‘dressed’ or ‘accompanied’ by the captured electrons. These electrons can be emitted in subsequent interactions. The process is known as projectile electron loss or projectile ionization. The ionic charge state corresponds to the difference between the nuclear charge and the number of accompanying electrons. After a large number of collisions (e.g. in a thick target), population and depopulation of each ionic charge state are equilibrated and the average ionic charge corresponds to the equilibrium charge of ions. Figure 20 shows the ratios of the equilibrium charge to the nuclear charge of

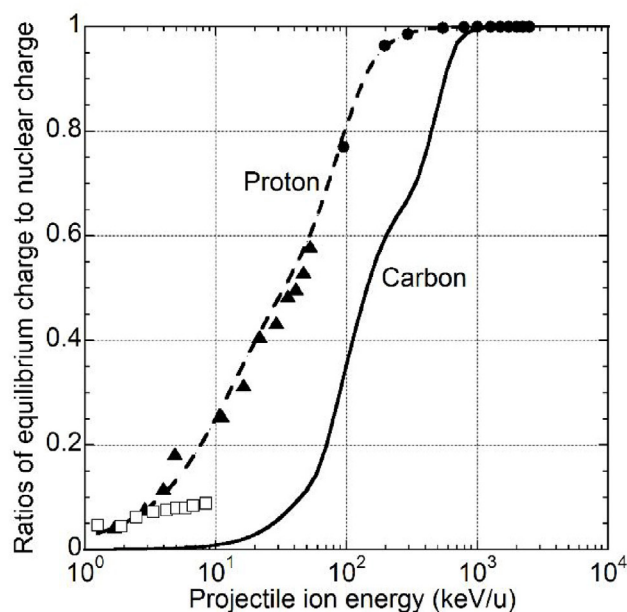


Figure 20. The ratios of the equilibrium charge to the nuclear charge of protons and carbon ions. Lines are calculated based on the cross sections as used in the KURBUC codes [177], circles and triangles are experimental data for protons [172, 173], and squares are calculated from the experimental data for C^0 and C^+ [150].

protons and carbon ions. Low-energy ions are screened by accompanying electrons and the equilibrium charge is lower than the nuclear charge, while in the MeV energy range both nuclei are completely bare.

For very-low-energy ions (<1 keV u^{-1}), electromagnetic interactions need to be described in terms of chemical reactions. However, ions below 1 keV u^{-1} scatter causing very few ionizations and excitations, have a very short range of less than 100 nm in water [138, 139, 140], and more importantly, cross-sections are not available. Therefore, they are usually assumed to be locally absorbed in the track-structure description.

Electromagnetic interactions play an important role in ion track-structure calculations. Ionization and excitation are most important for ion energy loss, while charge exchange determines the ionic charge, which characterizes the strength of ion–atom interaction. Moreover, due to projectile ionization, high-energy electrons can be emitted in the medium and contribute to energy deposition and the spatial distribution of ion tracks.

4.1.2. Multiple electronic interactions. For a collision system involving many electrons (e.g. dressed projectiles), multiple electronic interactions can take place when the collision energy and electronic transition allow. Examples of multiple electronic interactions are multiple target or projectile ionization (direct ionization), multiple electron capture, simultaneous target and projectile ionization (loss ionization) and simultaneous electron capture and target ionization (transfer ionization). Although information on these interactions for water and biomolecular targets is very scarce, available data from ion–atom collisions can shed light on the contribution of these processes in ion track structures.

While multiple electronic interactions were found to be unimportant for proton impact, they play a non-negligible role in heavier ion track structures [141]. In general, multiple electronic interactions become increasingly important as the projectile charge increases [141] and as the impact energy decreases [142]. It was found that the occurrence of double-electronic interactions relative to single-target ionization maximizes for projectile energy of ca. 100 keV u⁻¹ [143]. Transfer ionization was found to be the major multiple electronic interaction in the low-energy region, while double-target ionization and loss ionization are non-negligible in the intermediate energy region [144]. For high-energy projectiles, loss ionization tends to outweigh pure projectile electron ionization. In addition, inner-shell ionization by high-energy ions can take place. If electrons from valence orbitals fill the core vacancies, Auger electron emission can occur as a relaxation process of the target.

4.1.3. Two-center effect. Secondary electrons are the major cause of energy deposition in ion tracks. In ion-atom/molecule collisions, electron emission due to electromagnetic interactions can be investigated in terms of ‘centers’ [142]. A center refers to a Coulombic interaction between a nucleus (target nucleus or projectile nucleus) and active (interacting) electrons during and immediately after the collision. Zero-center electron emission is characterized by weak perturbation of projectile and target nuclei to the collision system, while a binary collision between a projectile nucleus and an active electron leads to single-center electron emission. Due to the long-range Coulomb potential of low- and intermediate-energy ions, target and projectile nuclei simultaneously interact with active electrons, giving rise to two-center electron emission. In this case, outgoing electrons are influenced by the electromagnetic fields of both nuclei. Two-center electron emission includes electron capture to the projectile continuum (ECC) and projectile electron loss to the continuum (ELC). The two-center effect is characterized by the enhancement of forward electron emission and emission of electrons with a similar speed to the projectile, i.e. the electron kinetic energy is close to T/λ , where T is kinetic energy per atomic mass unit of the ion and λ is the proton-to-electron mass ratio ($\lambda \approx 1836$). The two-center effect is expected to be important when the projectile speed is sufficiently low ($v \leq 3$ atomic units) [145].

4.2. Nuclear interactions

Nuclear interactions between projectile and target nuclei can be categorized as nuclear elastic and nuclear non-elastic interactions [146]. If kinetic energy of the collision system is conserved, the interaction is considered elastic, and in another case it is non-elastic. Nuclear elastic scattering has a high occurrence probability for low-energy ions. It leads to the deflection of projectile ions from the original path, but energy loss due to elastic interactions is relatively low compared to energy loss due to electromagnetic interactions.

In high-energy proton beams, nuclear non-elastic interactions are responsible for the removal of primary protons from the incident beam. Cross sections for nuclear non-elastic interactions between hydrogen and oxygen nuclei maximize at proton energies of ~25–30 MeV and reach a plateau at energies above 100 MeV [146]. Secondary particles induced by nuclear interactions include secondary protons, neutrons, photons and heavy recoils. Among dose contributions by secondaries, those by secondary protons are most important (~10% of total dose) and mainly present in the entrance region of proton beams. In the Bragg peak region, dose is primarily deposited by primary protons [147].

For high-energy carbon ion beams, nuclear non-elastic scattering is more crucial than for proton cases, due to the higher probability of nuclear non-elastic scattering. Nuclear non-elastic interactions can lead to projectile and target fragmentation. Secondary particles produced by nuclear fragmentation can traverse the tissue beyond the position of the Bragg peak, causing the so-called fragmentation tail [148, 149].

4.3. Theoretical models for inelastic interaction cross sections for ion impact

Cross sections for electromagnetic interactions are important input parameters for MCTS simulations. Most of experimental cross sections are available for electron and proton impact, while semi-empirical formulas and *ab initio* calculations are usually applied for heavy ion impact. Theoretical cross-section models are either based on the classical theory or the quantum mechanical theory. The classical theory is valid when the characteristic dimension of the electron orbit exceeds the de Broglie wavelength, which is associated with the relative motion of the collision system [58], i.e. the Bohr criterion must be fulfilled as follows [151]

$$\kappa = 2zv_0/v > 1 \text{ or } T < z^2 100 \text{ keV u}^{-1}, \quad (85)$$

where z and v are the charge and velocity of the projectile, v_0 is the Bohr velocity and T is the kinetic energy of the projectile ($v \approx 0.2\sqrt{T(\text{keV u}^{-1})}$ atomic units).

Examples of the classical calculations are the Rutherford formula, the binary encounter approximation (BEA) and the classical trajectory Monte Carlo (CTMC) method. The Rutherford formula and the BEA describe head-on collisions between a projectile and an active electron (projectile-center case), while the CTMC method is based on trajectory simulations of all collision partners including a projectile nucleus, a target nucleus and active electrons (two-center case).

Quantum mechanical calculations are either perturbative or exact in terms of the Hamiltonian. Commonly used perturbative calculations are the first Born approximation (FBA) [58] and distorted-wave methods [152, 153]. The FBA treats the projectile field as perturbation to the collision system, while the initial- and final-state wave functions of the active electron are influenced only by the target field (target-center) [58].

Table 7. Approximate regions of applicability of the theoretical models for ion-impact cross sections.

Model	Energy range of applicability	Center concept
Rutherford formula	$T/\lambda > 10B$	Projectile center
Binary encounter approximation	$T/\lambda > 10B$	Projectile center
Classical trajectory Monte Carlo	$T < z^2 100 \text{ keV u}^{-1}$	Target and projectile centers
First Born approximation	$T > 500 \text{ keV u}^{-1}$	Target center
Bethe theory	$T > 1 \text{ MeV u}^{-1}$	Target center
Distorted-wave calculations	$T > \text{a few hundreds of keV u}^{-1}$	Target and projectile centers
AO and MO close-coupling methods	$T < 10 \text{ keV u}^{-1}$	Target and projectile centers

Note: λ is the proton–electron mass ratio and B is the electron binding energy.

In distorted wave calculations, the long-range Coulomb potential of the projectile is included and emitted electrons are treated as moving in the projectile continuum [152, 153]. In non-perturbative calculations, such as atomic and molecular orbital (AO and MO) close coupling methods, the Hamiltonian is the exact solution of a many-body problem [154].

Table 7 shows the approximate region of applicability and the center concept of each theoretical model commonly used for calculating inelastic interaction cross sections for ion impact. The Rutherford formula and the BEA are the only models that depend explicitly on the electron binding energy and the ion's speed (correlated with T/λ) due to the assumption of head-on collisions of these models. The description of each model is given in the following sub sections.

4.3.1. Rutherford formula. The Rutherford formula for ion impact is derived classically from a head-on collision between a projectile ion and a stationary target electron (impulse approximation). It is valid for hard collisions in which electron emission energy is much larger than electron binding energy ($\varepsilon \gg B$) and $T/\lambda > 10B$ [155]. The Rutherford formula for single differential cross sections (SDCS) is given by

$$\frac{d\sigma^{\text{Ruth}}}{dE} = \frac{4\pi a_0^2 z^2}{T/\lambda} \left(\frac{R}{E} \right)^2, \quad (86)$$

where σ is the interaction cross section, E is energy transfer in the collision ($E = \varepsilon + B$), a_0 is the Bohr radius ($0.529 \cdot 10^{-10}$ m) and R is the Rydberg energy (13.6 eV).

4.3.2. Binary encounter approximation (BEA). The BEA is valid for hard collisions, in which secondary electrons are emitted into intermediate angles relative to the projectile's path. Small angle scattering is associated with distant collisions, which are better described by a quantum mechanical framework. Backward scattering is not possible in the BEA, due to a constraint of the classical kinematics, where energy spectra of outgoing electrons have a maximum at

$$\varepsilon \approx 4(T/\lambda) \cos^2 \theta, \quad (87)$$

where θ is the electron emission angle relative to the incident direction of the projectile. The equality of equation (87) is obtained when the outgoing electron is initially free and at rest and recoil of the target nucleus is negligible.

The SDCS in electron energy for ion impact derived within the BEA is given by [145, 155]

$$\frac{d\sigma}{d\varepsilon} = \frac{d\sigma^{\text{Ruth}}}{dE} \left(1 + \frac{4U}{3E} \right) \quad \text{for } E_{\min} \leq E \leq E_-, \quad (88a)$$

$$\frac{d\sigma}{d\varepsilon} = \frac{d\sigma^{\text{Ruth}}}{dE} \frac{U}{6E} \left[\left(\frac{4T/\lambda}{U} \right)^{3/2} + (1 - \sqrt{1 + E/U})^3 \right] \quad \text{for } E_- \leq E \leq E_+ \text{ and} \quad (88b)$$

$$d\sigma/d\varepsilon = 0 \quad \text{for } E > E_+, \quad (88c)$$

where $E_{\pm} = 4T/\lambda \pm 4\sqrt{(T/\lambda)U}$, $E_{\min} = B$ and $T > U$, U is the kinetic energy of the bound electron, and $T > U$.

4.3.3. The first Born approximation (FBA) and the Bethe theory. The FBA is a first-order quantum mechanical calculation. It is valid for sufficiently fast charged particles. In the FBA, a projectile is characterized by a plane wave in the entrance and exit channels. The initial state of an active electron is given by a stationary wave function, which describes electron binding to the target nucleus. The final-state wave function of the outgoing electron describes the electron as moving in the target continuum. Therefore, the FBA is a target-center calculation. The low-energy limit for applications of the FBA is given by the Sommerfeld criterion as follows [151]

$$zv_0/v < 1 \quad \text{or} \quad T > z^2 25 \text{ keV u}^{-1}, \quad (89)$$

where the accuracy of the FBA is not necessarily ruled out outside this energy range.

If an interaction is Coulombic and the associated momentum transfer of the interaction is averaged over all atomic orientations, the SDCS calculated using the FBA is reduced to the basic formula of the Bethe theory according to [58]

$$\frac{d\sigma}{dE} = \int \frac{4\pi a_0^2 z^2}{T/\lambda} \frac{R^2}{E} \frac{df(E, q)}{dE} d[\ln(qa_0)^2], \quad (90)$$

where $df(E, q)/dE$ is the GOS of the medium and q is the momentum transfer. For soft collisions in which the projectile initial momentum and energy are much larger than the momentum- and energy-transfers of the interaction ($p \gg q$ and $T/\lambda \gg E$), the GOS is reduced to the dipole oscillator strength, df/dE , and the cross-section in the dipole approximation is given by [58]

$$\frac{d\sigma}{dE} = \frac{4\pi a_0^2 z^2}{T/(\lambda R)} \left[\frac{R}{E} \frac{df}{dE} \ln \left(\frac{4c(E)T}{\lambda R} \right) + O \left(\frac{\lambda E}{T} \right) \right], \quad (91)$$

where $c(E)$ is a function of energy transfer that depends on the particular q -dependence of GOS and $O(\lambda E/T)$ is the next higher order term in $\lambda E/T$. The first term in the bracket is identical for any charged particles with the same speed, while $O(\lambda E/T)$ is slightly mass dependent [58]. If $O(\lambda E/T)$ is negligibly small, the inelastic scattering cross section is scaled with the square of the effective charge of the projectile according to

$$\sigma_1 = z_1^2 \sigma_2 / z_2^2, \quad (92)$$

where σ_1 and σ_2 are interaction cross sections for two charged particles with the effective charges z_1 and z_2 , respectively. The charge scaling method is a commonly used approach for scaling experimental data for electron and proton impact to interaction cross sections for heavy ion impact. A similar approach that scales the ion cross section with the effective charge has also been reported [178]

The Bethe theory, based on equations (91)–(92) and equations (95)–(96) have facilitated formulations of semi-empirical expressions for target ionization and target excitation by ion impact. For example, the SDCS in electron energy for target ionization by proton impact can be calculated using the Rudd formula according to [145]

$$\frac{d\sigma}{dw} = 4\pi a_0^2 N(R/B)^2 \frac{F_1 + F_2 w}{B(1+w)^3 \{1 + \exp[\alpha(w - w_c)/v]\}}, \quad (93)$$

where $d\sigma/dw$ approaches the Bethe cross section, equation (91), at high energies, w is the reduced electron energy ($w = \varepsilon/B$), N is the number of electrons in the orbital, ε is the outgoing electron energy, R is the Rydberg energy, B is the orbital binding energy, $v = \sqrt{T/(\lambda B)}$, w_c is the cutoff energy for a proper dependence of electron energy, α is a dimensionless parameter near unity related to the size of the target, and F_1 and F_2 are the fitting parameters.

For target excitation, a semi-empirical formula for total cross sections (TCS) for proton impact on water vapor, has been formulated by Miller and Green based on the experimental data for electron impact [156] according to

$$\sigma_{\text{exc}}(T) = \frac{\sigma_0(Za)^\Omega(T - W)^\nu}{J^{\Omega+\nu} + T^{\Omega+\nu}}, \quad (94)$$

where $\sigma_0 = 10^{-16} \text{ cm}^2$, Z_t is the nuclear charge of the target, W is the threshold energy for a given electronic transition, a and Ω are the fitting parameters for a high-energy limit, and J and ν are the fitting parameters for a low-energy limit.

4.3.4. Classical trajectory Monte Carlo (CTMC) method. The CTMC method is based on simultaneous trajectory simulations involving collision partners [157]. The three-body CTMC method is the simplest CTMC approach for heavy ion–atom/molecule collisions. The collision partners are a screened target nucleus, a screened projectile nucleus and an active (interacting) electron. The nucleus to which the active electron is bound is called a ‘core’. For dressed projectiles, the active electron is either bound to the target nucleus or projectile nucleus. The target core is associated with target ionization and electron capture, while the projectile core is relevant

for projectile ionization. Both target and projectile nuclei are assumed to be screened by their bound non-interacting electrons (spectator electrons).

Classical calculations are valid in the energy range where molecular effects (low-energy effects) and dipole interactions (high-energy effects) are not important. Although the lower energy limit of the CTMC method has not been well described, this approach has been used successfully for ion projectiles with energies as low as 1 keV u^{-1} [158]. Furthermore, the three-body CTMC method together with statistical methods has been used for generating interaction cross sections for carbon projectiles of energies of 1 keV u^{-1} – 10 MeV u^{-1} in water vapor [159], i.e. the inputs of the MCTS code Kurbuc_Carbon [25].

The CTMC method is attractive for low- and intermediate-energy ion impacts, because it takes into account two-center effects. Interactions between each collision partner are described with the classical Newtonian mechanics according to

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = - \sum_{j \neq i} \nabla_i \frac{Z_i(r_{ij}) Z_j(r_{ij})}{|\mathbf{r}_i - \mathbf{r}_j|} \text{ and} \quad (95)$$

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i, \quad (96)$$

where m_i is the mass of the collision partner i , $\mathbf{r}_i$ its position vector in the Cartesian coordinate system (x, y, z), $\mathbf{v}_i$ its velocity vector and $Z_i(r_{ij})$ its effective charge with respect to the distance r_{ij} to another collision partner j .

The effective charge can be calculated using a model potential that describes a nucleus as being completely screened by its bound or accompanying electrons at large distances and completely bare at infinitesimal distances. For example, cross sections for carbon projectiles as used in Kurbuc_Carbon were calculated based on the model potential of Garvey *et al* [160] according to

$$V_i(r_{ij}) = \frac{Z_i(r_{ij})}{r_{ij}} = \frac{Z - N[1 - \Omega(r_{ij})]}{r_{ij}}, \quad (97a)$$

$$\Omega(r_{ij}) = \frac{1}{(\eta/\varsigma)(e^{r_{ij}^\varsigma} - 1) + 1}, \quad (97b)$$

$$\eta = \eta_0 + \eta_1(Z - N - 1) \text{ and } \varsigma = \varsigma_0 + \varsigma_1(Z - N - 1), \quad (97c)$$

where Z is the nuclear charge of a nucleus i , N is the number of spectator electrons bound to i and $\Omega(r_{ij})$ is the screening function parameterized with the screening parameters η and ς . The screening parameters as used in equations (97a)–(97c) have been generalized for any atomic and ionic species of $Z \leq 54$ [160]. For Kurbuc_Carbon, the water molecule was described as a spherically symmetric atom with electron binding energies of a water molecule and the nuclear charge 10 [159].

Equations (95)–(96) can be solved conveniently in the relative coordinate system ($\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3$), where $\mathbf{R}_1 = \mathbf{r}_p - \mathbf{r}_t$, $\mathbf{R}_2 = \mathbf{r}_e - \mathbf{r}_t$, and $\mathbf{R}_3 = \mathbf{R}_2 - \mathbf{R}_1$, and the indices p , t , and e denote the projectile nucleus, the target nucleus and the

active electron, respectively. In this way, 12 differential equations, including $\mathbf{R}_1$, $\mathbf{R}_2$ and their time derivatives, are to be integrated over time using the Runge–Kutta method with an adaptive step size. Time integration or the so-called trajectory simulation should be carried out until the interaction probability between both nuclei is negligibly small.

Prior to a trajectory simulation, the initial distance between the active electron and the core, r_{ec} , needs to be determined. For example, this parameter can be randomly sampled from a classical microcanonical distribution of the following form [161]

$$f(r_{ec}, p_{ec}) \propto \delta \left(-E_{ec} - \frac{p_{ec}^2}{2\mu_{ec}} + \frac{Z_c(r_{ec})}{r_{ec}} \right), \quad (98)$$

where E_{ec} is the electron binding energy of an investigated core, p_{ec} is the electron momentum relative to the core, μ_{ec} is the reduced mass of the active electron-core system and $Z_c(r_{ec})$ is the effective charge of the core with respect to the position of the active electron. r_{ec} is constrained to $0 < r_{ec} < r_{max}$, where r_{max} is the positive root of $-E_{ec} + \frac{Z_c(r_{ec})}{r_{ec}}$. The relationship between the binding energy and the momentum of the active electron relative to the core is given by

$$E_{ec} = \frac{p_{ec}^2}{2\mu_{ec}} - \frac{Z_c(r_{ec})}{r_{ec}}. \quad (99)$$

The initial directions of the position and momentum vectors of the active electron can be randomly selected. Furthermore, it should be ensured that the initial distance between the target nucleus and projectile nucleus is large enough so that the starting position does not affect the calculated result.

Trajectory simulations are performed for different impact parameters, b . At the maximum impact parameter, b_{max} , the interaction probability between both nuclei should be negligibly small. At the end of each trajectory simulation, electron energies relative to the target and projectile nuclei (E_{et} and E_{ep} , respectively) are calculated according to equation (99) and used for the selection of the interaction type. If E_{et} and $E_{ep} > 0$ the interaction is associated with target or projectile ionization, while if $E_{et} \leq 0$ and $E_{ep} > 0$ it corresponds to electron capture.

The interaction probability obtained using the described selection procedure is considered to be the one-electron transition probability. To take into account the many-electron problem of heavy ion–atom/molecule collisions, statistical methods such as the independent particle model (IPM) [162] and the independent event model (IEVM) [163] can also be applied to the three-body CTMC solution. The statistical approach was found to improve the agreement between the calculated and measured cross sections for low- and intermediate-energy carbon projectiles in water [159].

Finally, the TCS, single-differential cross section and double-differential cross section, respectively, are calculated from

$$\sigma_R = 2\pi \sum_i b_i P_R(b_i) \Delta b, \quad (100a)$$

$$\frac{d\sigma_R}{d\varepsilon} = 2\pi \sum_i \frac{b_i P_R(b_i, \varepsilon) \Delta b}{\Delta \varepsilon} \text{ and} \quad (100b)$$

$$\frac{d^2\sigma_R}{d\varepsilon d\Omega} = 2\pi \sum_i \frac{b_i P_R(b_i, \varepsilon, \theta) \Delta b}{2\pi \sin \theta \Delta \theta \Delta \varepsilon}, \quad (100c)$$

where $P_R(b_i)$ is the many-electron interaction probability of the interaction R at the impact parameter b_i , $P_R(b_i, \varepsilon)$ is the corresponding probability for electron emission energy ε and $P_R(b_i, \varepsilon, \theta)$ is the corresponding probability for electron emission energy ε and emission angle θ . i refers to the i th impact parameter of simulated ion trajectories. In general, the increment of b_i is 0.1 atomic unit and the maximum b_i ranges from 0–25 atomic units, depending on the ion energy.

4.3.5. Continuum distorted wave (CDW) and continuum distorted wave-eikonal initial state (CDW-EIS) methods. The CDW [152] and CDW-EIS [153, 164] methods are valid for impact energies above the maximum in the ionization cross section (a few hundreds of keV u^{-1}). Both methods describe the initial and final-state wave functions of the active electron as being distorted by the electric field of the projectile ion. Both methods use the same final-state wave function. However, for the entrance channel, the CDW method uses the exact form of a distorted wave, which was found to be improperly normalized [153]. Therefore, in the CDW-EIS approach, the initial-state distorted-wave function takes the asymptotic approximation, the so-called eikonal approximation. This modification should be justified, because soft collisions dominate the total ionization cross section [142] and the normalization condition of the eikonal wave function is fulfilled. Interaction cross sections calculated using the CDW and CDW-EIS methods converge to each other for fast projectiles and converge to FBA cross sections at asymptotic projectile energies. For ion impact on water and biomolecular targets, distorted wave calculations have been investigated for bare ions [165, 166].

4.3.6. Atomic orbital (AO) and molecular orbital (MO) close-coupling methods. In the AO and MO close-coupling methods [154], electronic states of the collision system are described as a linear combination of basis wave functions, which considers strong interactions between the active electron and the target and projectile nuclei. The close-coupling calculations are used for calculating cross sections for low-energy ions in the energy range from a few tens of eV–10 keV u^{-1} . As the number of required basis wave functions increases with impact energy, the close-coupling method becomes computationally impractical for higher impact energies.

4.4. Interaction cross sections used in KURBUC

KURBUC is a suite of MCTS codes for simulating electrons and ions in water as a substitute for tissue-equivalent media [32]. For ion simulations, the KURBUC codes can be used for simulating ion track segments or the full-slowing-down of ions by taking into account all charge states of simulated ions. Kur_buc_Proton is used for simulating H^+ and H^0 of initial energies of

Table 8. Inelastic scattering cross sections for H^+ and H^0 as used in Kurbuc_Proton [24].

Interactions	Sources of interaction cross sections
Single target ionization $H^q + H_2O \rightarrow H^q + H_2O^+ + e^-$	TCS - H^+ ($T < 1$ MeV): experimental data [168] - H^0 ($T < 1$ MeV): the empirical formula of Green and McNeal [169] with the asymptotic form that complies with the Bethe theory, equation (92) - H^0 and H^+ ($T \geq 1$ MeV): the charge-scaling method, equation (92), based on the experimental data for electron impact [18] Single differential cross sections, $d\sigma/d\varepsilon$: - $T < 1$ MeV: the Rudd model [145], equation (93) - $T \geq 1$ MeV: the BEA, equation (88) Double differential cross sections, $d^2\sigma/(d\varepsilon d\Omega)$ - $T < 1$ MeV: experimental data [170, 171] - $T \geq 1$ MeV: the classical kinematics, equation (87)
Target excitation $H^q + H_2O \rightarrow H^q + H_2O^*$	The empirical formula of Miller and Green [156] as described in equation (94)
Single-electron capture (σ_{10}) $H^+ + H_2O \rightarrow H^0 + H_2O^+$	Fitting to the experimental data [172, 173] using the formula of Miller and Green [156]
Projectile ionization $H^0 + H_2O \rightarrow H^+ + H_2O + e^-$	TCS: the empirical formula of Miller and Green [156] Electron emission energy: T/λ Electron emission angle: 0° relative to the incident direction of the projectile

Table 9. Inelastic interaction cross sections for C^{q+} as used in Kurbuc_Carbon [25].

Interactions	Sources of interaction cross sections
Single-target ionization (SI) (q from 0–6) $C^{q+} + H_2O \rightarrow C^{q+} + H_2O^+ + e^-$	The CTMC method with the IEVM [159]
Single-electron capture (SC) (q from 1–6) $C^{q+} + H_2O \rightarrow C^{(q-1)+} + H_2O^+$	The CTMC method with the IPM [159]
Single-electron loss from the projectile (SL) (q from 0–5) $C^{q+} + H_2O \rightarrow C^{(q+1)+} + H_2O + e^-$	TCS: The CTMC method with the IEVM [159] Single-differential cross sections, $d\sigma/d\varepsilon$: - $T < 1$ MeV: The CTMC method with the IEVM for C^0 – C^{3+} [159] and $\varepsilon = T/\lambda$ for C^{4+} and C^{5+} - $T \geq 1$ MeV: $\varepsilon = T/\lambda$ for all carbon charge states Electron emission angle: 0° relative to the incident direction of the projectile
Target excitation (Exc) (q from 0–6) $C^{q+} + H_2O \rightarrow C^{q+} + H_2O^*$	Scaling the Miller and Green formula, equation (94), with the square of the carbon charge state according to equation (92)
Double-target ionization (DI) (q from 0–6) $C^{q+} + H_2O \rightarrow C^{q+} + H_2O^{2+} + 2e^-$	The CTMC method with the IEVM [159]
Transfer ionization (TI) (q from 1–6) $C^{q+} + H_2O \rightarrow C^{(q-1)+} + H_2O^{2+} + e^-$	A mixture of the CTMC method with the IPM and IEVM [159]
Loss ionization (LI) (q from 0–5) $C^{q+} + H_2O \rightarrow C^{(q+1)+} + H_2O^+ + 2e^-$	The CTMC method with the IEVM [159]

1 keV–300 MeV [20, 24, 167] and Kurbuc_Carbon is applicable for simulating carbon projectiles of the charge states 0–6 in the energy range of 1 keV u^{-1} – 10 MeV u^{-1} [25].

4.4.1. Inelastic scattering cross sections for ions. KURBUC uses inelastic scattering cross sections for ion impact on the water molecule. Those cross sections are either based on available experimental data or well-validated theoretical models. Details of the cross sections used in Kurbuc_Proton and Kurbuc_Carbon are given in tables 8 and 9, respectively.

The TCS for hydrogen and carbon projectiles that are inputted in the codes are presented in figures 21 and 22, respectively. Kurbuc_Proton considers single-electronic interactions, while Kurbuc_Carbon also includes double-electron interactions in the simulation.

4.4.2. Elastic scattering cross sections for ions. Elastic scattering for low-energy ions and neutrals can be described classically when the de Broglie wave length of the incident atom (λ_p) is negligibly small compared with the collision diameter

and screening length of the collision [174]. KURBUC uses the classical mechanics trajectory methods [174] for elastic scattering cross sections for ion impact. In this approach, the collision diameter (b) or the closest distance in a head-on collision is given by

$$b = Z_p Z_t e^2 / T_{CM}, \quad (101)$$

where T_{CM} is kinetic energy in the center-of-mass system. The screened Coulomb potential was used for describing nuclear elastic interactions between the projectile nucleus and the target nucleus, according to

$$V(r) = \frac{Z_p Z_t e^2}{r} F_s(r/r_s), \quad (102)$$

where $F_s(r/r_s)$ is the screening function of Molière with [175]

$$F_s(r/r_s) = \sum_{i=1}^3 C_i e^{-b_i r/r_s} = 0.10e^{-6r/r_s} + 0.55e^{-1.2r/r_s} + 0.35e^{-0.3r/r_s}, \quad (103)$$

and r_s is the screening length, written as [174]

$$r_s = \frac{0.88534a_0}{\sqrt{Z_p^{2/3} + Z_t^{2/3}}}. \quad (104)$$

Z_p and Z_t are the nuclear charges of the projectile and the target, respectively.

To calculate the angular distribution of the projectile, the scattering angle in the center-of-mass system (θ_{CM}) was calculated using

$$\theta_{CM}(p) = \pi - 2 \int_{r_{min}}^{\infty} \frac{p dr}{r^2 \sqrt{1 - V(r)/T_{CM} - p^2/r^2}}, \quad (105)$$

where p is the impact parameter of the collision, r is the distance between the target nucleus and the projectile nucleus, and r_{min} is the closest distance between the two nuclei. r_{min} corresponds to the root of the square root term. Scattering angles were calculated for a set of p/r_s and b/r_s for large and small angles. In the case of small angles, a straight-line trajectory of the incident particle was assumed to simplify the exact integration. The angular distribution of the projectile or the SDCS in scattering angles were obtained by numerical differentiation of the following form

$$\frac{d\sigma}{d\Omega_{CM}} = -\frac{p}{\sin \theta_{CM}} \frac{dp}{d\theta_{CM}}. \quad (106)$$

The TCS for elastic scattering (σ_{el}) were computed by integrating the SDCS over solid angles, according to

$$\sigma_{el} = 2\pi \int_{\theta_{cut}}^{\pi} \frac{d\sigma}{d\Omega_{CM}} \sin \theta d\theta. \quad (107)$$

θ_{cut} was introduced to avoid an infinite value of the integral. The recommended value of θ_{cut} of the ICRU report [176] was used, namely

$$\theta_{cut} \approx \frac{1}{137} Z_t^{1/3} \frac{m_e c^2}{Pc}, \quad (108)$$

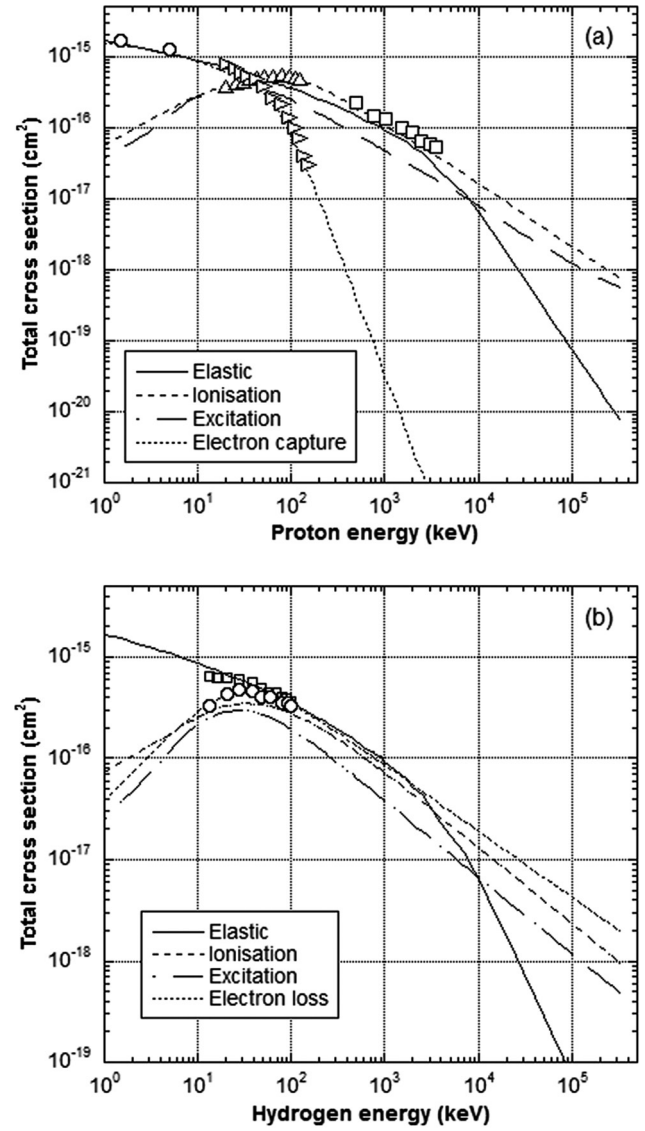


Figure 21. TCS for H^+ (a) and H^0 (b) in water. Lines are the cross sections as used in Kurbuc_Proton and symbols are the experimental data for: (panel a) electron capture (circles [179]) and right-pointing triangles [180] and ionization (triangles [180] and squares [181]); (panel b) projectile ionization (circles) and target ionization (squares) [181] (Reprinted with permission from [24], copyright 2010, Taylor & Francis.

where P is the momentum of the projectile.

In the simulation, scattering angles in the center-of-mass system are converted to the scattering angle in the projectile frame using

$$\theta = \arctan\left(\frac{m_t \sin \theta_{CM}}{m_p + m_t \cos \theta_{CM}}\right), \quad (109)$$

where m_p and m_t are molar masses of the projectile and the target, respectively.

For protons with energies larger than 10 MeV, the Mott scattering formula of the following form was used instead of the classical mechanics trajectory calculation, because it gives more realistic track structures of protons at high energies [176]

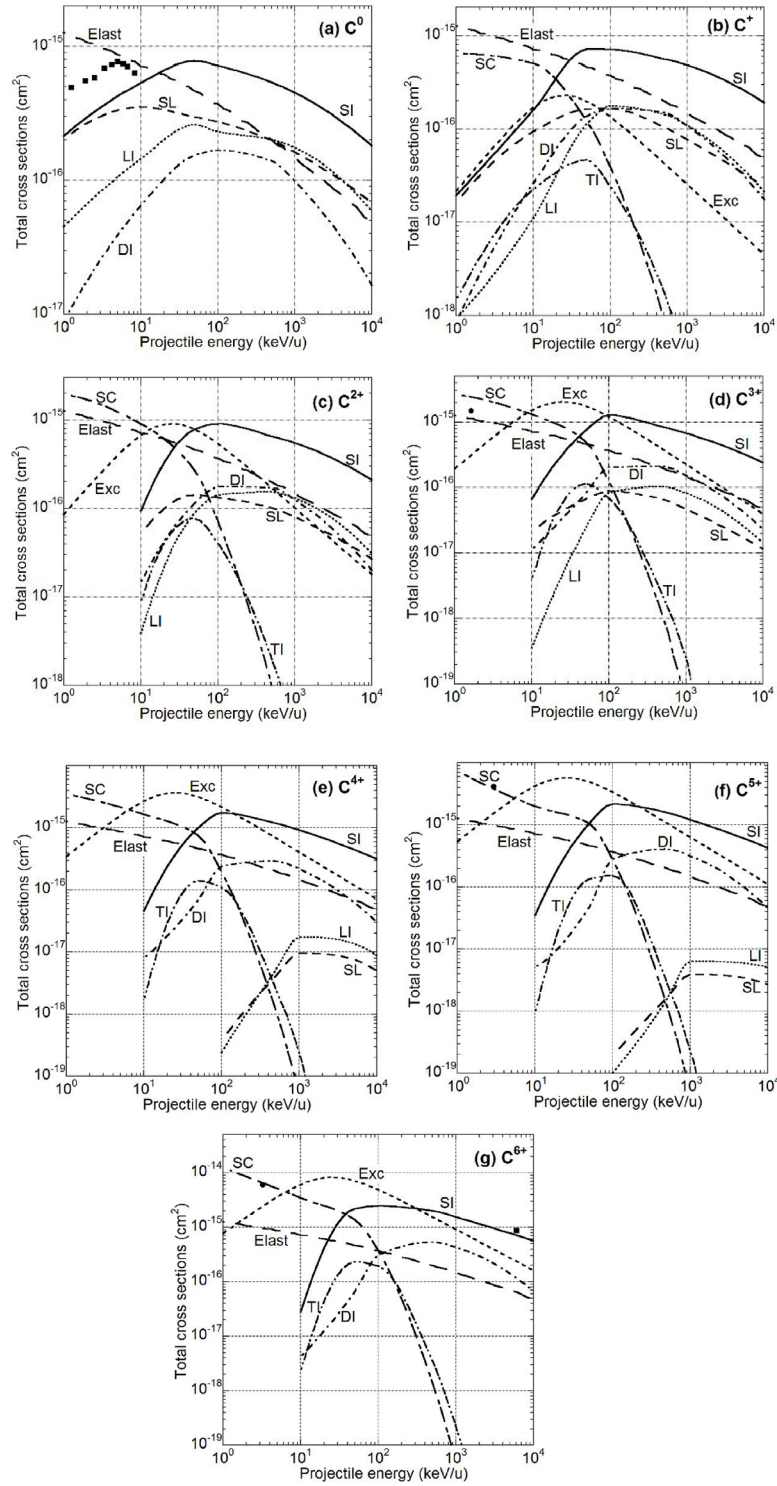


Figure 22. TCS for carbon projectiles of the charge states 0–6 in water (panels a–g, respectively). The abbreviations of the interaction types are given in table 9. Lines are the cross sections as used in Kurbuc_Carbon and symbols are experimental data for: single-target ionization (squares in panels a [150] and g [182]) and single electron capture (circles in panels d [183], f [184] and g [183]). From [25], © Institute of Physics and Engineering in Medicine. Reproduced by permission of IOP Publishing Ltd. Copyright 2013, all rights reserved.

$$\frac{d\sigma}{d\Omega_{CM}} = \frac{1}{4} Z_t^2 Z_p^2 r_e^2 \left(\frac{m_e c}{P\beta} \right)^2 \frac{1 - \beta^2 \sin^2(\theta_{CM}/2)}{\sin^2(\theta_{CM}/2)}. \quad (110)$$

Figure 23 shows the TCS for elastic scattering for protons and neutron hydrogen that are used in Kurbuc_Proton [20, 24],

compared with the cross sections used in Geant4-DNA [185]. Both codes use the same classical approach for projectiles of energies below 1 MeV but different cut-off values for scattering angles. The TCS for elastic scattering for carbon projectiles that are used in Kurbuc_Carbon are given in figure 22.

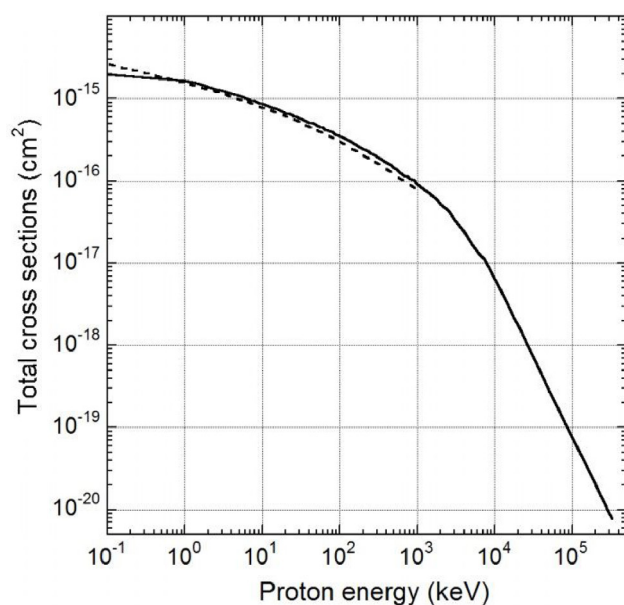


Figure 23. TCS for elastic scattering for proton and neutral hydrogen impacts in water that are used in Kurbus_Carbon (solid line) [20, 24] and Geant4-DNA (dotted line) [185].

5. Modeling cellular response to radiation damage

In order to model cellular response to DNA damage, biological models are required. Traditionally, semi-empirical or phenomenological models were used to model the cellular response such as clonogenic cell survival. In this approach, the experimental survival curves were fitted to exponential formula and the parameters in the model were described as a function of dose and cellular-dependent parameters. The most famous phenomenological model is the LQ model [186, 187]. The LQ model was successfully applied to conventional radiotherapy [188]. Although phenomenological models had a successful history in radiotherapy, they were not capable of radiation cancer-risk assessment. The second approach to model radiation response is the mechanistic model. In this approach, the known biological processes that are effective in defining a specific cellular response are considered. The mechanistic approach requires detailed information of the specific response. The first response to radiation damage is cell signaling (intracellular and extracellular signaling) and repairs [189, 190]. If the damage is not repaired, cells activate either cell-cycle arrest or cell-death processes to avoid mutation. If the repair is not correctly performed, it might lead to mutation. The persistence of mutation in the DNA might initiate a series of events that might eventually cause heritable disease or cancer [191]. DNA repair is the key component of cellular response to radiation. Only recently, with the advances in the biological experiments that describe the DNA repair processes, has it been possible to model DNA repair response. In this chapter, we will discuss the achievement in mechanistic models describing DNA repair.

The damage to the DNA is in the form of base lesion (BL), SSB and DSB. BLs and SSBs are repaired with base excision repair (BER) pathway. The BER pathway is a fast and simple repair process. DSBs are repaired by multiple different repair

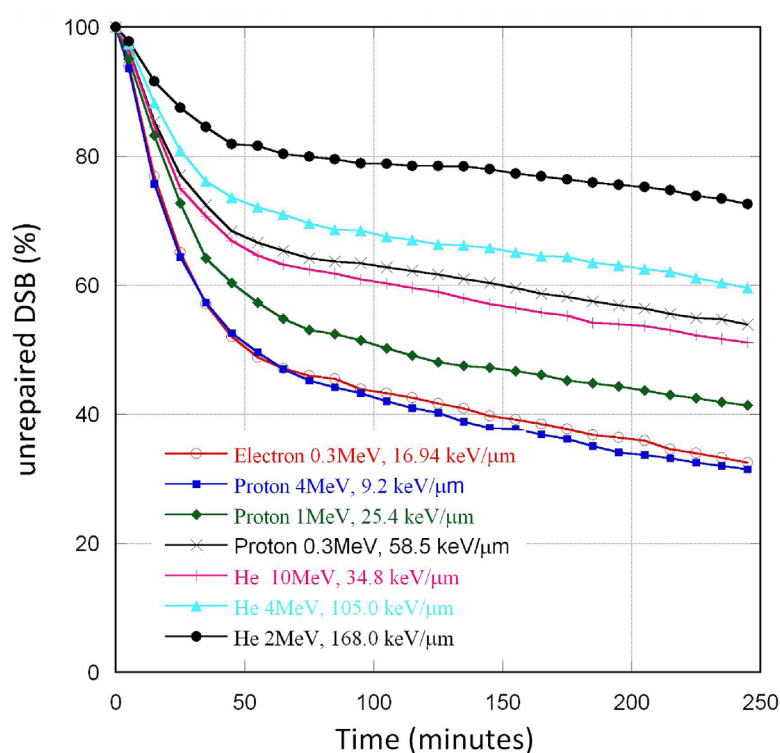
pathways, namely nonhomologous end-joining (NHEJ), homologous recombination (HR), single-strand annealing (SSA) and microhomology-mediated end-joining (MMEJ). The NHEJ pathway is the main repair process in mammalian cells that is active through the whole cell cycle. HR is a complicated and slow repair process that is involved in the repair of the damage that is induced in the G2/late S phases of the cell cycle. SSA and MMEJ pathways in mammalian cells are not well studied, but it is suggested that these pathways are backup to the NHEJ process.

5.1. DSB repair model

DSB repair kinetics can be measured using two experimental methods. The first method is pulsed-field gel electrophoresis (PFGE) (reviewed in [192, 193]) and γ -H2AX foci counting as a function of time (reviewed in [194]). PFGE measurements require high doses of radiation (>20 Gy), and γ -H2AX foci experiments are performed at more relevant and lower doses. PFGE is however more accurate for measuring repair kinetics in comparison to γ -H2AX foci experiment. The main reason for the greater accuracy of PFGE is the delay of the foci signal (depending on the cell line, it could take about 15 min for the maximum signal). During this delay, a relatively large number of DSBs is repaired. The γ -H2AX foci cover nearly 2 Mbp of the chromosome that is larger than the size of the DSB that is defined to 10bp. The large size of the foci could, therefore, cover more than one DSB and underestimate the number of DSBs. The underestimation is dependent on beam quality (LET). The higher the LET, the more condensed is the number of DSBs along the track of the particle, and, therefore, the number of DSBs is more underestimated. Neither γ -H2AX foci nor PFGE experiments are able to measure the complexity of DSB. The complexity of DSB is defined as a DSB that is in close proximity (within 10bp) of another SSB or DSB. The experimental methods are not sensitive enough to measure the complexity of DSB within 10bp resolution. However, γ -H2AX foci appear to be larger for more than one DSB in close proximity. The repair kinetic measurements with both methods show at least two distinct repair components known as slow and fast DSB repair processes. The simplest method to characterize the fast and slow component of DSB repair is to use a bi-exponential [195–198] or other exponential fits [199] to the experimental repair kinetics. Although bi-exponential models characterize the DSB repair curves, they do not provide any mechanistic information of the repair processes. The alternative approach to study the two- or multi-exponential components of DSB repair is the mechanistic approach. Using the mechanistic method, it was suggested that the reason for the fast and slow repair component is the switch from a fast to a slow repair process that mainly happens due to the complexity of the DSB. The first mechanistic approach discussed in this chapter is the stochastic method developed by Friedland *et al* [200–203]. In this model, DNA damage is calculated using the PRTRAC Monte Carlo simulation code. The PARTRAC simulation considers water ionization and excitations and the production of water radicals. The damage in the DNA is calculated by overlapping the energy deposition induced by the track with a multi-scale chromatin model in the nucleus. In the repair model, only the NHEJ repair pathway is modeled.

Table 10. Yield of DSB by complexity for electrons, protons and He-4 particles^a.

Particles	DSB yield #10 ⁻¹¹ /Gy/Da	DSB (%)	DSB+ (%)	DSB++ (%)	DSBC (%) ^b
Electrons					
0.3 keV	2.3	2.4	0.9	0.1	28
1.5 keV	1.8	1.7	0.6	0.1	29
4.5 keV	2.9	1.5	0.6	0.0	29
Protons					
0.3 MeV	3.5	4.8	3.6	1.3	51
0.5 MeV	3.2	3.9	2.3	0.6	46
0.75 MeV	2.9	3.3	1.6	0.3	37
1.0 MeV	2.6	2.7	1.1	0.3	37
2.0 MeV	2.6	2	0.7	0.1	28
4.0 MeV	1.8	1.6	0.5	0.1	26
He-4 particles					
2.0 MeV	4.0	4.8	6.2	5.7	73
3.0 MeV	3.6	5.3	5.7	3.7	64
4.0 MeV	3.7	4.3	4.1	2.2	60
6.0 MeV	3.7	3.5	2.7	1.3	56
8.0 MeV	3.4	3.9	2.4	0.7	47
10.0 MeV	2.9	3.2	1.9	0.4	45

^a DNA damage data from [213, 214].^b DSBC: complexity of DSB is defined as a percentage of (DSB++ DSB+)/(DSB + DSB++ DSB+).**Figure 24.** DSB repair kinetics for electrons, protons and He-4 particles using the DSB repair model [204].

The repair model considers the spatial movement of the DSB ends and the kinetics of the main proteins involved in the NHEJ DSB repair pathway. The repair model in PARTRAC considers different scenarios for repair and has successfully predicted the repair kinetics of the DSB induced by ⁶⁰Co gamma irradiation and Ni irradiation [203].

The second mechanistic approach to calculate the repair kinetic of DSB [53, 55, 193–210], the kinetics of γ -H2AX foci formation and disappearance [208, 211, 212] and BER

kinetics [56] is a biochemical kinetic method. In this method, the repair proteins involved in the modification of the damages ends are identified and a schematic model is formed. The schematic model is translated into a system of differential equations by explaining every step of repair with a separate equation [205, 206, 208]. In order to solve the system of differential equations, protein rate constants and concentrations are required. The system of differential equations was nonlinear due to the mathematical approach used to solve

the problem. The rate constants and concentrations of the repair proteins are not easy to measure. The first assumption to solve the system of equations was that the concentrations of repair proteins in cells that are not deficient in repair are abundant. In order to calculate the rate constants, the system of equations was solved by estimating the rate constants for cells irradiated with photons. Appendix C provides the list of equations and the rate constants, table C1, used to solve the equations. To verify that the rate constants selected for photons are applicable to other radiation species such as electrons, ultrasoft x-rays, protons, and other light ions, MCTS simulation and the inverse transform sampling method (ITS) were employed [53, 55, 206]. The MCTS simulation was used to compute the damage spectra induced by electrons, ultrasoft x-rays, protons, and other light ions. The repair kinetics for each DSB was separately calculated using the ITS method. The complex-type DSBs undergo the same repair pathway that is different from simple-type DSBs and DSBs that fall in the heterochromatin. However, the stochastic nature of the repair kinetics is conserved by the ITS method such that two identical complex DSBs may be repaired with different kinetics. The repair kinetics of the DSBs induced by electrons, ultrasoft x-rays, protons, and other light ions were calculated using the ITS method. The calculated repair kinetics could predict the experimental repair kinetics [53, 55, 206]. In addition, for early-acting proteins such as DNA-PKcs and Ku80, the initial concentration kinetics were verified by comparing the calculated concentration kinetics to published measurements that used green fluorescent tagged foci [205].

In the repair model, it was assumed that the complexity of DSBs and distribution of DSBs in the heterochromatin played the main role in selecting the repair pathway [230–232]. The simple-type DSBs were repaired with the NHEJ pathway in a fast process. The DSBs in the heterochromatin were repaired by employing proteins such as Artemis, ATM, and KAP-1 that delayed the repair process. The complex-type DSBs require resection that delayed the repair process and was the main reason for the slow repair component of DSB repair kinetics. The complex DSBs in the Late S and G2 phases of the cell cycle were repaired with HR and in the Early S and G1 phases of the cell cycle were repaired with backup NHEJ pathways. Table 10 lists the DSB yield and complexity induced by electrons, protons, and Helium-4 particles [213, 214]. The repair kinetics of the DSBs induced by electrons, protons, and Helium-4 particles were calculated using the G1 and Early S phase repair model. Figure 24 plots the computed repair kinetics. It is shown that the repair kinetics of 0.3 keV electrons and 4 MeV protons is similar (figure 24) because the DSB complexity of 0.3 keV electrons and 4 MeV protons is 28% and 26%, respectively. Accordingly, the repair kinetics of 0.3 MeV protons is similar to the repair kinetics of 10 MeV He-4 particles, because the DSB complexity is respectively, 51% and 45%.

5.2. Conclusion

The mechanistic repair model presented here has been successful in describing the repair kinetics induced by radiation of different quality. In the model, we have considered two main

reasons for the slow repair component. The first reason is the complexity of DSB and the second is the DSBs that fall in the heterochromatin part of the nucleus. Radiation quality mainly affects the complexity of the DSB and the complexity of the DSB increases with LET. The DSB repair model successfully predicted the repair kinetics of DSB measured for different radiation quality. The agreement confirms the assumption that the slow repair component that is LET-dependent is mainly due to the complex-type DSB.

One direction for future work is mechanistic modeling of deletions, mutations and chromosome aberrations [215, 216] that arise from the repair of DNA damage. For this purpose, the kinetic model (temporal model) should be integrated with a spatial model. The cell-nucleus model is required to note the location of the damage in the nucleus [42, 217]. The main challenge for the spatial model is chromosome dynamics that happen after DNA damage. Another direction for future research is to study cell signaling, including cell-death kinetics. Key proteins such as ATM and ATR are involved in the activation of different cellular responses such as cell death or cell-cycle arrest. These proteins are involved in post-translational processes. The main challenge for modeling cell signaling is the large number of proteins involved in the process and variations in the response of different cell lines.

6. Concluding remarks

The work summarized in the pages above is a reflection of what has taken some 30 years of research activity. Douglas Lea, 70 years ago, mapped the road for radiation biology for the future generation. We have contributed to those aims and objectives in a small way. The physics of energy loss and scattering of low-energy electrons in the condensed phase is undoubtedly one of the most important research areas towards understanding radiation track structure and its relation to biological damage. Our work in modeling electron interactions at energies below 1 keV has been able to reduce the uncertainties to an acceptable level, despite the very limited experimental information for condensed-phase water. Another significant achievement on the physics part has been the extension of ion track-structure in the Bragg peak. This allows a near-complete description of the full-slowng-down of protons (1 keV–300 MeV), alpha-particles (1 keV u^{-1} –2 MeV u^{-1}), and carbon ions (1 keV u^{-1} –10 MeV u^{-1}).

The elastic scattering of low-energy electrons in liquid water is, however, still not well understood, and further basic experimental and theoretical work is required.

In this paper, we did not review extensively the chemistry of water radicals as most relevant experiments had been performed many decades ago, most models agree with each other, and a large number of reviews have been published.

The work on the DNA damage response and repair has become very popular and important over the past decade, and we have seen a surge in activity and the publication of a large number of highly exciting work. However, not a single fully mechanistic description of DNA damage induced by ionizing particles has ever been published, by taking into account all the chemistry it entails. Our

own work by-passes all the initial chemistry in the processes, after establishing whether energy deposition in the DNA volume leads to damage or not. It is a remarkable fact that the work of Roger Martin (1971) [48] and David Charlton and John Humm (1988) [47] in determining the energetics of an SSB by experiment and modeling, respectively, have endured since the early 1980s. It is also important that the Monte Carlo track structure simulation method still remains the only tool for producing accurate quantitative spectra of DNA damage for electrons and heavy ions.

Although the concept of complex DNA DSB was introduced in the literature in the mid 1980s by a number of workers, we first quantified the concept by producing frequencies of simple and complex DNA damage produced by electrons and particulate ions, and then produced for, the first time, a complete model of DBS repair, including the NHEJ, MMEJ, SSA and BER, showing that the complex DSBs were responsible for the slow repair kinetics.

This is where the work of the authors in the the field of radiation biophysics stands, but the way forward is very challenging and will require that we deal with issues such as, for example,

the localization of energy loss for very-low-energy electrons in condensed media, and the cell-signaling network arriving at the point where the cell makes the decision to repair, continue and divide with mis-repair in the genome, or die. To achieve these goals is not a ‘low-hanging fruit’ and requires the close collaboration of physicists, mathematicians, and molecular biologists.

Finally, we remind the reader of what we referred to, in the abstract; the role of physics in helping to elucidate the cancer problem. You only need to look at the statistics of cancer death for the 1950s and compare it to the 2004 records (for the USA), which show almost no change (Source: NCHS—National Vital Statistics System). We believe physics, in line with the scheme we described in figure 1, can bring unique insights into solving the cancer problem.

Acknowledgments

The work of the Radiation Biophysics Group (RBG) was supported by the Swedish Radiation Safety Authority (SSM), and the Karolinska Institutet.

Appendix A

Number and types of events generated in a single track of 2 MeV u^{-1} 4He₂:

csda range = 72.1460 μ m

shortest path-length = 72.1460 μ m

center of the sphere = 0.0000 $\times 10^0$ μ m

sphere diameter = 1.4500 $\times 10^2$ μ m

No. of primary interactions = 196619

Interaction	Type	No. of events	%
He ²⁺	Elastic	60401	30.72
He ²⁺	Ioniz	85844	43.66
He ²⁺	Excit	10185	5.18
He ²⁺	2e ⁻ capt	29	0.01
He ²⁺	1e ⁻ capt	1155	0.59
He ⁺	Elastic	9587	4.88
He ⁺	Ioniz	8392	4.27
He ⁺	Excit	2442	1.24
He ⁺	1e ⁻ capt	1223	0.62
He ⁺	1e ⁻ loss	1140	0.58
He ⁰	Elastic	9243	4.70
He ⁰	Ioniz	4220	2.15
He ⁰	Excit	1506	0.77
He ⁰	1e ⁻ loss	1209	0.61
He ⁰	2e ⁻ loss	43	0.02

Interaction	Type	No. of events	%
2nd e ⁻	Ioniz 12.62	72495	18.92
2nd e ⁻	Ioniz 14.75	53744	14.03
2nd e ⁻	Ioniz 18.51	29869	7.80
2nd e ⁻	Ioniz 32.4	5858	1.53
2nd e ⁻	Ioniz 539.7	39	0.01
2nd e ⁻	Excit A1B1	73917	19.29
2nd e ⁻	Excit B1A1	26455	6.90

2nd e ⁻	Excit RydAB	9885	2.58
2nd e ⁻	Excit RydCD	14 155	3.69
2nd e ⁻	Excit DB	46 492	12.13
2nd e ⁻	Excit Lyman	13 039	3.40
2nd e ⁻	Excit Balmer	2542	0.66
2nd e ⁻	Excit OH*	34 653	9.04

No. of 2nd-electron interactions = 383 143

No. of He ionization = 98 456

No. of He excitation = 14 135

No. of He capture = 2407

No. of He loss = 2392

No. of He elastic = 79 231

2nd e⁻ ionization = 162 005

2nd e⁻ excitation = 221 138

2nd e⁻ elastic = 43 109 603

Total ionization = 260 461

Total excitation = 235 271

Total elastic = 43 188 834

Total no. of interactions by 2nd electrons (elastic not included) = 579 762

Lower cut-off energy for 2nd e⁻ = 10.0 eV

Energy deposition	keV	%
Primary e ⁻	2.4352×10^3	3.0678×10^0
Secondary e ⁻	5.5029×10^3	6.9322×10^1
Dose (mGy)		
He ion	2nd-e	Total
2.22456×10^{-1}	5.28290×10^{-1}	7.50746×10^{-1}
Number of track in the sphere		
He ion (primary)	2nd e ⁻	Total
1	53 527	53 528.0

Electron fluence within the sphere (channel width = 10.0 eV)

0	14 983	8379	5259	3736	2779	2208	1811	1574	1227
950	894	770	684	616	532	470	378	359	323
310	252	273	228	194	189	205	179	170	131
139	127	108	94	113	105	101	89	82	59
69	75	61	60	62	62	52	56	51	50
63	214	48	52	38	56	46	36	30	43
28	23	37	22	26	33	23	30	24	17
22	16	27	20	21	23	18	23	16	13
15	17	11	20	11	14	14	17	18	11
14	15	9	10	13	14	11	10	8	12
12	5	9	9	15	7	6	8	16	9
4	4	7	4	9	7	7	3	7	16
6	3	5	5	6	7	2	8	7	7
3	7	8	5	7	0	3	4	2	6
6	4	6	8	9	2	2	5	4	1
2	2	1	2	2	7	3	4	2	5
6	3	6	1	4	2	6	1	1	0
2	2	1	4	2	3	3	1	2	1
3	6	0	0	2	0	2	1	1	5
0	1	2	6	2	2	2	1	1	3
1	4	6	3	1	2	0	0	2	2
1	1	2	2	2	0	2	2	0	1
0	2	0	1	1	0	1	0	1	3
0	0	4	0	0	2	2	2	0	1
0	0	0	2	1	1	1	1	3	1

Appendix B

Table B1. Microdosimetry parameters for electrons and heavy ions in small cylinders and spheres of various height and diameters

Proton energy	1 MeV			2 MeV			5 MeV		
LET (keV μm^{-1})	26.8			16.5			7.8		
Cylinder	2.3 nm $\times$ 2.3 nm	10 nm $\times$ 5 nm	30 nm $\times$ 30 nm	2.3 nm $\times$ 2.3 nm	10 nm $\times$ 5 nm	30 nm $\times$ 30 nm	2.3 nm $\times$ 2.3 nm	10 nm $\times$ 5 nm	30 nm $\times$ 30 nm
$\bar{y}_F$ (keV μm^{-1})	1.81×10^1	1.71×10^1	1.95×10^1	1.64×10^1	1.28×10^1	1.23×10^1	1.61×10^1	9.77×10^1	7.71×10^1
$\bar{y}_D$ (keV μm^{-1})	3.37×10^1	3.44×10^1	3.37×10^1	2.92×10^1	2.73×10^1	2.24×10^1	2.75×10^1	2.19×10^1	1.60×10^1
$\bar{z}_F$ (cGy)	4.64×10^7	3.47×10^6	2.95×10^5	4.20×10^7	2.61×10^6	1.85×10^5	3.89×10^7	1.99×10^6	1.07×10^5
$\bar{z}_D$ (cGy)	8.65×10^7	7.01×10^6	5.09×10^5	7.51×10^7	5.56×10^6	3.39×10^5	7.07×10^7	4.47×10^6	2.41×10^5
Proton energy	10 MeV			50 MeV			250 MeV		
LET (keV μm^{-1})	4.4			1.2			0.4		
Cylinder	2.3 nm $\times$ 2.3 nm	10 nm $\times$ 5 nm	30 nm $\times$ 30 nm	2.3 nm $\times$ 2.3 nm	10 nm $\times$ 5 nm	30 nm $\times$ 30 nm	2.3 nm $\times$ 2.3 nm	10 nm $\times$ 5 nm	30 nm $\times$ 30 nm
$\bar{y}_F$ (keV μm^{-1})	1.47×10^1	8.39×10^1	4.93×10^1	1.22×10^1	7.08×10^1	2.89×10^1	1.43×10^1	7.57×10^1	2.71×10^1
$\bar{y}_D$ (keV μm^{-1})	2.55×10^1	1.97×10^1	1.34×10^1	1.88×10^1	1.84×10^1	9.95×10^1	2.58×10^1	1.95×10^1	1.14×10^1
$\bar{z}_F$ (cGy)	3.77×10^7	1.71×10^6	7.46×10^4	3.14×10^7	1.45×10^6	4.38×10^4	3.66×10^7	1.55×10^6	4.12×10^4
$\bar{z}_D$ (cGy)	6.82×10^7	4.02×10^6	2.02×10^5	4.84×10^7	3.76×10^6	1.50×10^5	6.63×10^7	3.98×10^6	1.71×10^5

Table B2. Microdosimetry parameters for heavy ions in a wall-less proportional counter (1 μm diameter)^a.

Ion	E (MeV amu^{-1})	LET (keV μm^{-1})	$\bar{y}_F$ (keV μm^{-1})	$\bar{y}_D$ (keV μm^{-1})	$\bar{z}_F$ cGy	$\bar{z}_D$ cGy	Quality factor Q
Proton	0.1	20.2	20.2	22.8	4.1	4.6	6.8
Proton	0.2	10.1	9.7	11.3	2.0	2.3	3.4
Proton	0.3	60.7	58.7	67.3	11.9	13.7	19.8
Proton	0.5	44.4	42.8	49.6	8.7	10.1	14.8
Proton	0.7	35.3	34.4	40.0	7.0	8.1	12.0
Proton	1.0	26.7	25.4	30.3	5.2	6.2	9.1
Proton	2.0	16.4	15.1	18.8	3.1	3.8	5.6
Proton	5.0	8.5	7.3	9.5	1.5	1.9	2.9
Proton	10.0	5.3	4.5	5.8	0.9	1.2	1.7
Proton	20.0	2.9	2.4	3.9	0.5	0.8	1.2
Proton	50.0	1.5	1.1	2.5	0.2	0.5	0.8
Proton	100.0	0.9	0.6	1.8	0.1	0.4	0.5
Proton	200.0	0.7	0.3	0.5	0.1	0.1	0.2
Proton	1000.0	0.2	0.2	0.7	0.0	0.2	0.2
Alpha	0.8	126.5	113.9	1143.3	23.1	29.1	28.5
Alpha	1.2	91.4	79.2	102.8	16.1	220.9	26.8
Alpha	2.5	55.9	44.3	62.2	9.0	12.6	18.3
Alpha	5.0	33.7	25.5	35.6	5.2	7.2	10.7
Alpha	12.5	17.0	12.9	17.1	2.6	3.5	5.1
Alpha	25.0	10.0	7.4	9.6	1.5	2.0	2.9
Alpha	250.0	1.9	1.2	2.1	0.2	0.4	0.6
Carbon	3.0	443.0	232.4	464.3	47.2	94.3	12.9
Carbon	4.0	375.3	205.1	442.4	41.7	89.9	13.6
Carbon	5.0	299.1	157.0	315.9	31.9	64.1	17.6
Carbon	6.0	271.8	139.9	263.1	28.4	53.4	20.3

(Continued)

Table B2. (Continued).

Ion	E (MeV amu ⁻¹)	LET (keV μm^{-1})	$\bar{y}_F$ (keV μm^{-1})	$\bar{y}_D$ (keV μm^{-1})	$\bar{z}_F$ cGy	$\bar{z}_D$ cGy	Quality factor Q
Carbon	7.0	238.3	121.6	231.9	24.7	47.1	22.2
Carbon	8.0	204.5	103.8	196.0	21.1	39.8	24.4
Carbon	9.0	198.8	100.4	190.2	20.4	38.6	24.9
Carbon	10.0	181.6	99.4	179.9	20.2	36.5	25.7
Carbon	11.0	156.2	76.1	147.8	15.4	30.0	26.8
Carbon	12.0	139.2	73.6	129.9	15.0	26.4	27.3
Carbon	13.0	148.4	72.3	135.0	14.7	27.4	27.5
Carbon	15.0	128.3	64.2	117.0	13.0	23.8	27.3
Carbon	16.0	371.5	191.7	456.6	38.9	92.7	13.2
Carbon	17.0	120.9	55.3	103.4	11.2	21.0	26.3
Carbon	18.0	112.1	55.1	102.4	11.2	20.8	25.9
Carbon	19.0	110.4	63.2	102.7	12.8	20.9	26.5
Carbon	20.0	103.3	55.1	95.5	11.2	19.4	25.3
Carbon	25.0	82.4	47.4	77.5	9.6	15.7	22.2
Carbon	30.0	78.6	40.8	66.2	8.3	13.4	10.4
Carbon	50.0	54.4	31.0	47.2	6.3	9.6	14.1
Carbon	70.0	40.6	23.5	33.0	4.8	6.7	9.9
Carbon	80.0	36.6	21.2	29.6	4.3	6.0	8.9
Carbon	100.0	31.0	18.5	25.1	3.8	5.1	7.5
Carbon	250.0	13.7	10.5	13.9	2.1	2.8	4.2
Carbon	300.0	13.3	9.6	12.3	2.0	2.5	3.7
Oxygen	5.0	538.8	264.8	547.5	53.8	111.2	10.9
Oxygen	10.0	322.8	155.4	306.4	31.6	62.2	18.0
Oxygen	50.0	94.2	47.5	77.6	9.7	15.8	22.1
Oxygen	100.0	54.9	228.9	53.7	5.9	8.9	13.1
Oxygen	250.0	34.7	17.2	22.6	3.5	4.6	6.8
Oxygen	500.0	21.8	11.9	15.6	22.4	3.2	4.7
Oxygen	1000.0	15.3	9.6	12.2	1.9	2.5	3.7
Neon	10.0	502.3	218.9	478.5	44.5	97.1	12.2
Neon	20.0	298.6	127.5	262.8	25.9	53.4	20.2
Neon	30.0	222.4	88.2	180.7	17.9	36.7	25.5
Neon	40.0	167.4	80.4	147.8	16.3	30.0	27.8
Neon	100.0	90.2	44.1	68.9	8.9	14.0	20.0
Neon	200.0	50.3	28.8	40.2	5.8	8.2	12.0
Neon	1000.0	26.8	14.2	19.6	2.9	4.0	5.9
Silicon	10.0	302.9	41.4	153.3	8.4	31.1	19.6
Silicon	50.0	300.2	119.9	240.2	24.3	48.8	21.6
Silicon	100.0	171.2	72.9	134.4	14.8	27.3	27.6
Silicon	200.0	101.4	45.4	79.8	9.2	16.2	22.5
Silicon	300.0	75.6	40.2	62.3	8.2	12.7	18.4
Silicon	400.0	15.3	9.8	48.7	2.0	9.9	14.5
Silicon	500.0	89.6	31.3	48.0	6.4	9.7	14.3
Silicon	600.0	57.2	28.6	43.6	5.8	8.9	13.0
Silicon	700.0	54.7	30.5	41.9	6.2	8.5	12.5
Silicon	1000.0	50.0	25.2	35.2	5.1	7.2	10.6
Iron	100.0	619.0	231.8	466.5	47.1	94.7	12.5
Iron	150.0	440.4	155.3	325.7	31.5	66.1	17.1
Iron	200.0	313.1	137.2	283.0	27.9	57.5	19.2
Iron	360.0	213.7	96.4	186.8	19.6	37.9	25.1
Iron	540.0	200.4	86.5	163.3	17.6	33.2	26.6
Iron	700.0	189.7	69.1	130.7	14.0	26.5	27.4
Iron	790.0	194.4	72.2	127.4	14.7	25.9	27.8
Iron	1000.0	158.8	66.3	121.4	13.5	24.7	27.4
Iron	2000.0	132.8	60.8	102.6	12.3	20.8	26.4
Iron	3000.0	140.1	60.0	103.9	12.2	21.1	26.4

^a Data from Monograph 2006, H Nikjoo, Microdosimetry parameters for heavy ions in a tissue equivalent proportional counter (TEPC) (unpublished).

Table B3. References to monographs for microdosimetry parameters^a.

1. Monograph 2013, Taleei R, Charlton D E, Nikjoo H (2013), Energy depositions by monoenergetic α -particles (0.3–20.0 MeV u^{-1}) in cylindrical targets (1–100 nm diameters). Karolinska Institutet (220 pages)
2. Monograph 2010a, Hultqvist M, Charlton D E, Nikjoo H (2010), Energy depositions by monoenergetic carbon ions (10–300 MeV u^{-1}) in cylindrical targets, Karolinska Institutet (120 pages)
3. Monograph 2010b, Liamsuwan T, Uehara S, Charlton D E, Nikjoo H (2010), Energy deposition by monoenergetic protons (0.3–300 MeV) in cylindrical targets. Karolinska Institutet (212 pages)
4. Monograph 2003, Kylonen J E, Girard P, Uehara S, Lindborg L, Charlton D E, Nikjoo H (2003) Energy deposition by heavy ions in cylindrical targets, MRC Genome Stability Unit, Harwell (200 pages)
5. Monograph 2001b, Nikjoo H, Nuta O, Boyden-Pratt N, Uehara S, Goodhead D T, Charlton D E (2002) Energy deposition by alpha-particles in cylindrical targets, MRC Genome Stability Unit, Harwell (220 pages)
6. Monograph 2001a, Nikjoo H, Boyden-Pratt N, Uehara S, Goodhead D T, Charlton D E (2001) Energy deposition by monoenergetic protons (2 keV–1 MeV) in cylindrical targets. MRC Genome Stability Unit, Harwell (202 pages)
7. Monograph 1994, Nikjoo H, Goodhead D T, Charlton D E, and Paretzke H G (1994) Energy deposition by monoenergetic electrons (100 eV–100 keV) in cylindrical targets. MRC Radiobiology Unit, Harwell (250 pages)
8. Monograph 1988, Nikjoo H, Goodhead D T, Charlton D E, and Paretzke H G (1988) Energy deposition by C, Al, Ti, and Cu x-rays in cylindrical volumes. MRC Radiobiology Unit, Harwell (150 pages)
9. Monograph 1985, Charlton D E, Goodhead D T, Wilson W E, Paretzke H G (1985) Energy deposition in cylindrical volumes: (a) Protons—energy 0.3–4.0 MeV, (b) Alpha-particles—energy 1.2–20 MeV. MRC Radiobiology Unit, Harwell (125 pages)

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Appendix C

The system of differential equations that describe the repair step-by-step based on the model described in [204]. Y stands for the concentrations at each step of repair, D stands for dose, K stands for the rate constants and the $[]$ defines the protein concentrations.

$$\frac{dY_1}{dt} = \alpha \frac{dD}{dt} - V_1$$

$$V_1 = K_1[Ku70, Ku80]Y_1$$

$$\frac{dY_2}{dt} = V_1 - V_2$$

$$V_2 = K_2[DNA - PKcs]Y_2$$

$$\frac{dY_3}{dt} = V_2 - V_3$$

$$V_3 = K_3Y_3$$

$$\frac{dY_4}{dt} = V_3 - V_4$$

$$V_4 = K_4Y_4$$

$$\frac{dY_5}{dt} = V_4 - V_5 - V_8 - V_{12}$$

$$V_5 = K_5[\text{Polymerase } \mu - \lambda]Y_5$$

$$V_8 = K_8[\text{Artemis, ATM}]Y_5$$

$$V_{12} = K_{12}[\text{MRN}]Y_5$$

$$\frac{dY_6}{dt} = V_5 - V_6$$

$$V_6 = K_6[\text{XLF, XRCC 4, Lig IV}]Y_6$$

$$\frac{dY_7}{dt} = V_6 - V_7$$

$$V_7 = K_7Y_7$$

$$\frac{dY_9}{dt} = V_8 - V_9$$

$$V_9 = K_9[\text{Polymerase } \mu - \lambda]Y_9$$

$$\frac{dY_{10}}{dt} = V_9 - V_{10}$$

$$V_{10} = K_{10}[\text{XLF, XRCC 4, Lig IV}]Y_{10}$$

$$\frac{dY_{11}}{dt} = V_{10} - V_{11}$$

$$V_{11} = K_{11}Y_{11}$$

$$\frac{dY_{13}}{dt} = V_{12} - V_{13}$$

$$V_{13} = K_{13}[\text{PARP 1}]Y_{13}$$

$$\frac{dY_{14}}{dt} = V_{13} - V_{14}$$

$$V_{14} = K_{14}[\text{PARP 1}]Y_{14}$$

$$\frac{dY_{15}}{dt} = V_{14} - V_{15}$$

$$V_{15} = K_{15}[\text{Polymerase } \beta]Y_{15}$$

$$\frac{dY_{16}}{dt} = V_{15} - V_{16}$$

Table C1. Repair-rate constants used in model calculations.

Rate constants (h ⁻¹)	G1/Late S model
k_1	350
k_2	500
K_3	50
K_4	20
k_5	25
k_6	18
K_7	3
k_8	9
k_9	2
k_{10}	0.8
K_{11}	0.5
k_{12}	3
k_{13}	1
k_{14}	0.7
k_{15}	0.75
k_{16}	0.5
K_{17}	0.15

$$V_{16} = K_{16}[\text{XRCC 1, Lig III}]Y_{16}$$

$$\frac{dY_{17}}{dt} = V_{16} - V_{17}$$

$$V_{17} = K_{17}Y_{17}$$

In order to solve the system of equations, the parameters have been scaled with a scaling factor C that is big enough to ensure that the sum of the total concentration of repair complexes and proteins remains constant. For this purpose, it is assumed that the sum of the total concentration of repair complexes and proteins (Y_i and E_i) is constant and equal to C .

$$C = [E_i] + \sum_{i=1}^{17} Y_i = cte$$

$$y_i = \frac{Y_i}{C}$$

$$k_i = CK_i$$

$$c_i = \frac{\sum_{i=1}^{17} y_i}{C}$$

$$c'_i = \frac{\sum_{i=1}^{17} y_i}{C}$$

$$c''_i = \frac{\sum_{i=1}^{11} y_i}{C}$$

The following equations are derived, considering the scaling factor C and substituting the new parameters in the model.

$$\frac{dy_1}{dt} = \frac{\alpha}{C} \frac{dD}{dt} - v_1$$

$$\frac{dy_2}{dt} = v_1 - v_2$$

$$\frac{dy_3}{dt} = v_2 - v_3$$

$$\frac{dy_4}{dt} = v_3 - v_4$$

$$\frac{dy_5}{dt} = v_4 - v_5 - v_8 - v_{12}$$

$$\frac{dy_6}{dt} = v_5 - v_6$$

$$\frac{dy_7}{dt} = v_6 - v_7$$

$$\frac{dy_9}{dt} = v_8 - v_9$$

$$\frac{dy_{10}}{dt} = v_9 - v_{10}$$

$$\frac{dy_{11}}{dt} = v_{10} - v_{11}$$

$$\frac{dy_{13}}{dt} = v_{12} - v_{13}$$

$$\frac{dy_{14}}{dt} = v_{13} - v_{14}$$

$$\frac{dy_{15}}{dt} = v_{14} - v_{15}$$

$$\frac{dy_{16}}{dt} = v_{15} - v_{16}$$

$$\frac{dy_{17}}{dt} = v_{16} - v_{17}$$

where

$$v_1 = k_1(1 - c_2)y_1$$

$$v_2 = k_2(1 - c_3)y_2$$

$$v_3 = K_3y_3$$

$$v_4 = K_4y_4$$

$$v_5 = k_5(1 - c'_6)y_5$$

$$v_6 = k_6(1 - c'_7)y_6$$

$$v_7 = K_7y_5$$

$$v_8 = k_8(1 - c''_9)y_5$$

$$v_9 = k_9(1 - c_9'')y_9$$

$$v_{10} = k_{10}(1 - c_{11}'')y_{10}$$

$$v_{11} = K_{11}y_{11}$$

$$v_{12} = k_{12}(1 - c_{13})y_5$$

$$v_{13} = k_{13}(1 - c_{14})y_{13}$$

$$v_{14} = k_{14}(1 - c_{15})y_{14}$$

$$v_{15} = k_{15}(1 - c_{16})y_{15}$$

$$v_{16} = k_{16}(1 - c_{17})y_{16}$$

$$v_{17} = K_{17}y_{17}$$

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